Change-marked copy: additions relative to the d8b2aa5 40-slide writing baseline are blue.

# Supplementary Information

**Learning conditional language in continuous image space**

Imagized Language Model Project

## Supplementary Note 1: experimental question and scope

The target is an independent visual language model whose runtime consumes writing images and generates writing images. Encoded text and local language models may prepare data offline. They cannot supply the answer during model inference. The present experiment isolates canonical Chinese continuation as one prerequisite. It does not cover the complete instruction-following or historical-script objective.

The repository retains the earlier 96-page research record in its legacy image-native publication folder. That record documents prior experiments without treating every rejected architecture as a contribution to the compact article. This supplement specifies the directly conditional block test, the subsequent exact-raster density and the distinction between ranking and generation.

## Supplementary Note 2: conditional uncertainty

Let $H$ denote a visible history and $X$ the next writing image. Under squared pixel loss, the optimum $f^*(H) = \mathbb{E}[X \mid H]$ can mix valid glyphs. Under independent Bernoulli loss, each marginal equals $\Pr(X_p = 1 \mid H)$, which likewise does not specify joint stroke identity. An image distribution is consequently required for stochastic writing.

Selection and sampling have different expected identity accuracy. If a model exactly recovers the true next-form distribution $p(x \mid H)$, deterministic top-1 identity has conditional accuracy $\max_x p(x \mid H)$. Independently sampling one future and comparing it to one observed continuation has expected accuracy $\sum_x p(x \mid H)^2$. Neither quantity is a general measure of semantic correctness. Thus the reader's 19.97% top-1 cannot be used as the expected accuracy of a correct stochastic writer. Paired context controls, sampled probabilities, direct readability and meaningful longer continuations are needed together.

## Supplementary Note 3: raster transform and gradients

For signed ink image $S = 2X - 1$ and a $32 \times 32$ orthonormal basis $B$, the visual coordinate is

$$\phi(S) = \frac{\operatorname{vec}(BSB^{\mathsf{T}})}{\|\operatorname{vec}(BSB^{\mathsf{T}})\|_2}. \tag{1}$$

For binary observed images this matches the reader's field. For soft predicted images it must be computed directly on $S$. Applying the reader's input threshold inside this training objective would cut all pixel gradients. The pretraining inspection found and corrected that defect. A regression test confirms a finite, nonzero gradient from the field cosine alone to the predicted pixels and equality with the reader transform on binary inputs.

The transform is a full spatial coordinate, not a learned visual vocabulary. Its background contribution also limits its interpretation as a perceptual identity metric. A high cosine can accompany an incorrect glyph.

## Supplementary Note 4: directly conditional architecture

The frozen reader has 16,277,249 parameters. Its final 384-dimensional hidden state and 1,024-dimensional anchor form a continuous condition. A 5,899,970- parameter U-Net receives a complete $32 \times 128$ future strip. Three spatial scales have 64, 128 and 256 channels. Each residual scale receives the projected condition and a sinusoidal time embedding. Additional input channels carry the first-cell predicted plan, a horizontal coordinate and the cell noise levels.

At the bottleneck, four pooled cell vectors enter four-head attention and a 512-wide MLP. A zero-initialized scalar gates the spatial mixer residual. The output channel predicts velocity over four visible cells. The total runtime contains 22,177,219 parameters. The prior checkpoint only initializes matching U-Net weights. An epsilon predictor is not thereby a qualified velocity predictor, so all copied layers are fine-tuned under the new loss.

## Supplementary Note 5: data and optimization

The corpus manifest SHA-256 is

```
76048753b52735d14c98f0d9e4eb8d751401fe8810e82eaaaebff517f6866c03.
```

Source identifiers define deterministic splits before rendering. Original and OpenCC-simplified views inherit the same split. Data generation samples 65-cell segments with a saved dataset seed. Only the first 64 cells are used in this test. A

training example has 60 context and four target cells. A symbolic character identifier is never an input or target to a model method.

On odd updates, the four future positions receive one level from each of 0–24, 25–59, 60–89 and 90–99 under a random permutation. On even updates, one uniformly sampled level from 0–99 applies to all four future cells. The latter pretraining amendment supplies synchronous noise patterns like those encountered during sampling. It was frozen before optimization.

Ten percent of rows drop the condition. All-dropped batches have zero order loss. The foreground loss weights pixels by $1+2X$. The complete objective is

$$L = L_v + 0.10L_{fg} + 0.10L_{dice} + 0.10L_{field} + 0.10L_{repr} + 0.25L_{order}, \tag{2}$$

$$L_{order} = \mathbb{E}_{t\geq 60}\,\mathrm{softplus}\left[0.02 - (c_{ordered} - c_{shuffled})\right]. \tag{3}$$

Only rows with an available condition enter the order loss. The reader remains frozen. Noise, drop masks and data offsets derive from saved RNG states. The run uses seed 20265310, dataset seed 20265311, AdamW with $\beta = (0.9, 0.95)$, weight decay 0.01, gradient norm clip 1, EMA decay 0.999, batch eight and 8,000 updates. The full run consumes 64,000 context segments and 256,000 supervised future cells. These are sampled segments rather than 64,000 independent source records.

## Supplementary Note 6: locked audit and uncertainty

The audit seed is 20265320. It uses 512 fixed development windows for denoising, 256 for direct generation and 64 for two-block visible feedback. The first 60 cells form the context and the remaining four context-window cells form the first target block. The four subsequent continuation cells form the second block. Denoising probes use levels 75 and 95, with identical noise under all conditions. Prefix shuffling preserves the final four visible cells. The zero condition removes all reader state and plan channels.

The 1,024-image bank is rendered after every denoising and generation call. Every primary output is the first sample. Nearest-bank identity is evaluator metadata, not an image replacement or glyph selected by the model. Per-position coverage records how many true targets appear in the bank. All out-of-bank targets receive zero exact-identity credit. The rendered output remains the actual model image even when a bank identity is assigned.

Bootstrap uncertainty resamples source identifiers, retaining all correlated windows from a record together, with 2,000 resamples. Paired differences keep the same starting noise. The intervals do not include training-seed variation. There is one primary trained seed and no sealed generalization result.

## Supplementary Table 1: direct visual block audit

| Condition | First identity (%) | Block identity (%) | Pixel F1 | Isolated ink |
|---|---|---|---|---|
| Ordered | 1.172 | 0.586 | 0.3198 | 0.0050 |
| Shuffled | 1.172 | 0.488 | 0.3176 | 0.0053 |
| Zero condition | 0.000 | 0.195 | 0.3018 | 0.0053 |

All 256 prefixes contribute one four-cell sample. Target coverage in the Han bank is 60.94, 58.98, 64.45 and 68.36% by position. No out-of-bank target receives identity credit.

## Supplementary Note 7: numerical and compute receipts

Training takes 361.6 seconds and peaks at 0.543 GiB of allocated CUDA memory on GPU 0, an RTX 4090 D. These figures describe the bounded experiment. They exclude prior foundation training, data preparation, idle GPU memory, operating-system memory and evaluation. They are not an estimate of the cost of a broadly capable language model. The independent reader and prior training costs remain in their original run reports.

The continuous versus interrupted four-step smoke check compares every denoiser and EMA tensor and the saved training generator. All match exactly. Production checkpoints save optimizer groups, CPU/CUDA/Python/NumPy RNGs and the deterministic data offset. Foundation, data, source and protocol hashes are checked before scoring. No other project's GPU service is part of this run.

## Supplementary Note 8: limitations and next evidence

The post-result motor diagnostic uses the first 64 fixed prefixes and paired noise. Replacing the first plan with the exact target raster gives F1 0.4586. Replacing both plan and visual anchor gives F1 0.4594 and identity 5/64, compared with F1 0.3775 and identity 1/64 for the predicted condition. An exact inverse-transform control has F1 1.0. Only 60.94% of these targets occur in the Han identity bank, so its 60.94% identity is the available ceiling for perfect copies. The

target-supplied test was designed after the locked audit and does not change its decision. It isolates a remaining motor limitation.

Reliable autonomous continuation, calibrated conditional sequence likelihood, held-out short answers, font transfer, independent seed replication and capability-matched text baselines remain outstanding. No selected gallery can substitute for these tests. Future teacher-generated exercises must retain source provenance and separate prompt/answer splits. A deployed teacher or an image-bank lookup would change the model contract.

The older guided model's terminal-signal observation is a mathematical limit, not a proven failure explanation. Correct conditional diffusion also starts from a context-independent noise law. Direct interventions on trained conditional scores and matched initialization are needed to distinguish weak condition learning from a defective sampling trajectory.

## Supplementary Note 9: normalized raster density

The exact-density model (V54) freezes the same 16,277,249-parameter reader. A row transformer has width 192, four layers, six heads, MLP ratio three and zero dropout. Its inputs are shifted earlier rows, a projection of the aligned 32-pixel plan row, spatial row positions and a projected continuous condition (hidden384 plus field1024). A MADE with two 256-wide SiLU layers masks pixels within each row. Its output mask is strict: pixel $c$ cannot see pixel $c$ or any later pixel. Earlier rows may be observed in full. Gradients with respect to forbidden pixels vanish even with nonzero randomized output weights. Enumeration of all 16 four-bit outcomes gives total probability one.

The inverse reader field, scaled by 32 and clipped to $[-2,2]$, supplies an aligned logit at scale six. Residual output weights start at zero. Thus a supplied binary image has greedy reconstruction exactly equal to itself at initialization. This copy path does not learn language and is not a lookup bank. It applies to arbitrary supplied binary rasters. The experiment measures whether optimization retains that path while learning natural continuation.

Each update samples eight streams and 16 causal positions per stream, four from each quarter, including the last position. Of the resulting 128 target images, 25% in expectation receive their actual image field and a zero hidden state as explicit motor exercises. The remainder receive frozen predicted states. Antialiased target images are thresholded at 0.5 before unweighted Bernoulli NLL; soft labels are rejected by the loss. There is no additional foreground, contrastive or guidance objective. Seeds are 20265410 for training and 20265411 for data. AdamW uses learning rate $2\times10^{-4}$, $\beta=(0.9,0.95)$, weight decay 0.01, 400 warmup updates, cosine decay to 10%, norm clip one, BF16 and EMA 0.999. The run makes 1,024,000 selected-image presentations over 8,000 updates; these are not independent records.

Training completes in 207.8 s with 0.400 GiB peak allocated CUDA memory. The runtime has 18,711,841 parameters, of which 2,434,592 are new density parameters. The 52.16-second audit peaks at 0.231 GiB allocated CUDA memory. Neither timing includes earlier reader training. Source hashes, the corpus hash, optimizer, all RNGs and the dataset offset are saved. An interrupted-versus-continuous smoke run matches every model and EMA tensor and the training RNG exactly. The production checkpoint SHA-256 is

`f2c3fab74981541243a963c8cffab6bbbd8c7f9d051512b228347db04a57a9ee.`

## Supplementary Note 10: exact-density audit and decision

Audit seed 20265420 selects 512 development windows from 169 source records for exact target NLL. Pixel autoregression is teacher forced only for this likelihood calculation. The paired earlier-history shuffle preserves the last four visible cells. Ordered NLL is 15.219 nats per image and shuffled NLL is 15.254. Their advantage is 0.0355, with interval [-0.1014, 0.1693] from 2,000 source-record draws (bootstrap seed 20265519). The zero-condition NLL is 47.930 nats; zeroing is an out-of-distribution control, not an optimized unconditional model.

Generation uses 256 prefixes with paired Bernoulli uniforms. Within each image, 32 row-transformer and 1,024 masked-head evaluations produce pixels without rejection or reranking. All generated images precede evaluator-bank construction. The bank combines all distinct training-form rasters with every target raster from generation, motor and feedback tests. Pixel deduplication gives 9,932 entries and full target coverage, including punctuation. Its nearest identity remains a proxy and can credit a malformed image. It differs from the 1,024-Han bank used in earlier experiments.

### Supplementary Table 2: exact raster density audit

| Condition | Identity (%) | Exact image (%) | Pixel F1 | Ink ratio |
|---|---|---|---|---|
| Ordered sample | 1.562 | 0.781 | 0.35033 | 0.945 |
| Shuffled sample | 1.172 | 0.391 | 0.35124 | 0.945 |
| Zero-condition sample | 0.000 | 0.000 | 0.19715 | 0.435 |
| Ordered greedy | 5.469 | 5.469 | 0.21454 | 0.330 |
| Supplied image, sample | 100.000 | 81.250 | 0.99949 | 1.000 |
| Supplied image, greedy | 100.000 | 100.000 | 1.00000 | 1.000 |
| Deranged image, greedy | 0.391 | 0.391 | 0.35769 | 1.000 |

Each row uses 256 images. The evaluator bank contains 9,932 distinct rasters and covers every target. Supplied-image rows reveal the intended image and are motor controls, not language results.

The paired ordered-minus-shuffled identity difference is 0.391 percentage points with interval [0, 1.210], using 134 source records. Four-step visible feedback on 64 prefixes reaches mean identity 4.297% versus 3.125% shuffled. Ordered step-wise identity is 4.688, 4.688, 4.688 and 3.125%; shuffled values are 4.688, 1.563, 3.125 and 3.125%. These secondary measurements do not repair the failed primary context effect.

Four of eight preregistered gates pass: runtime boundary, motor copying, positive feedback difference, and finite/resource checks. Ordered likelihood, ordered sampled identity, sampled quality and isolated-ink surface gates fail. The primary sample has isolated-ink fraction 0.01404, above the 0.01 limit. The decision remains `reject`. No sealed evaluation or independent training-seed replication is reported.

### Supplementary Note 11: independent inference export

An inference-only checkpoint embeds the reader, row model, masks and configuration. It loads without original foundation checkpoints. An aligned 32-pixel image grid is the native prompt interface; optional host text rendering does not pass symbolic identities into the network. Eight-cell greedy output from the exported checkpoint is pixel-identical to the training checkpoint's output with the same prompt. The native motor CLI also matches the binarized input exactly. The export is a 75,225,847-byte research artifact, not a qualified conversational model. General instruction following, open-script language generation and etymology remain unestablished.

### Supplementary Note 12: exploratory context attribution

A plan written after the locked result specifies a read-only likelihood diagnostic on the same 512 windows. It leaves the checkpoint, primary audit and rejection unchanged. Ordered and earlier-shuffled NLL reproduce the saved per-example values within $7.63 \times 10^{-6}$ nat. To derange the complete condition, windows are sorted by source record and the index list is shifted by half its length. Every donor source differs from its target source. Both hidden state and predicted field come from that donor. No image bank or optimization enters this intervention.

Complete-condition derangement raises NLL from 15.219 to 20.630 nats/image. The paired ordered advantage is 5.411 nats, with source-record bootstrap 95% interval [4.514, 6.416]. Earlier-history shuffling, which retains the final four cells, still gives the unresolved 0.0355-nat difference. The density therefore uses the visual condition; its earlier-history ordering effect remains weak. Short-suffix or order-insensitive features are compatible explanations, not an established exclusive mechanism. The 512-image answer-supplied NLL is 0.190 nat; deranging that supplied image raises NLL to 1,954.872 nats. These answer-revealing controls describe the motor path.

The first eight image rows contain 8.644 of 15.219 ordered nats. Whole-context derangement adds 2.365 nats in that band and 1.530, 1.127 and 0.390 in the remaining eight-row bands (Fig. S1). Earlier-shuffle effects vary in sign across bands. No band-wise significance claim or multiple-comparison-adjusted result is made. These model-specific likelihood interventions are not calibrated mutual-information estimates. They do not establish autonomous coherent writing. The next mechanism must retain this context sensitivity while selecting globally coherent future forms.

### Supplementary Note 13: matched joint likelihood learning

The joint experiment (V55) retains the exact V54 network and probability law. Both arms initialize from its EMA, with new optimizers and EMA copies. The frozen control trains only the 2,434,592 density parameters. The joint candidate also trains reader parameters through the exact image likelihood, for 18,711,840 participating parameters. The unused scalar contrastive temperature is fixed. Runtime size stays 18,711,841 in both. Reader dropout remains disabled in both arms, independently of gradient flow. The deployed student continues to receive and generate image tensors, without a vocabulary or external model.

Each arm makes 8,000 updates with eight streams and 16 selected causal positions per stream, using 25% supplied-image motor exercises. Training seed 20265510 and dataset seed 20265511 give matched new draws from the same public corpus. Both final training-generator states match exactly. The frozen reader is bit-identical to initialization; the

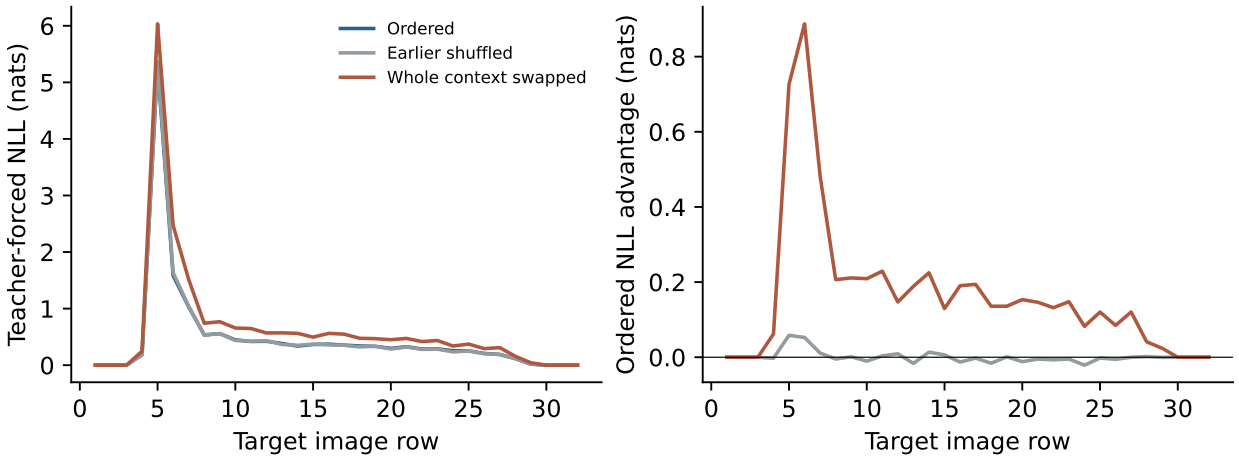



**Fig. S1 | Exploratory row-wise attribution of raster likelihood.** Left, teacher-forced NLL by target-image row for ordered, earlier-shuffled and completely deranged conditions on 512 windows. Right, ordered advantage relative to each control. Earlier target pixels are legitimately observed by the image density, so low later-row NLL cannot itself establish language information from the prompt. Curves are descriptive means; whole-image uncertainty is reported in Supplementary Note 12.

joint reader changes. Reader likelihood gradients are finite and nonzero only in the joint arm. A four-step interrupted smoke reproduces every model and EMA tensor, plus CPU/CUDA/training RNG states. Reader gradients cannot pass through answer-supplied motor rows because those replace the natural state with a zero hidden vector and a fixed observed field.

The density learning rate is $2 \times 10^{-4}$ in both arms; joint reader rate is $2 \times 10^{-5}$. Other optimizer, warmup, cosine, clipping, BF16 and EMA settings match the earlier density experiment. The control takes 218.0 s at 0.400 GiB allocated CUDA memory; the joint arm takes 342.4 s at 0.648 GiB. Both run sequentially on physical GPU 0. These costs exclude V42/V54 pretraining and data preparation.

Both checkpoints finish before either development result opens. The same 512 NLL windows, 256 generation/motor prefixes and 64 four-step feedback streams use audit seed 20265420. Joint ordered NLL is 7.026 nats per image versus 7.579 for the matched frozen control. The paired gain is 0.553 nat, interval [0.450, 0.665]. The joint-minus-control sample identity gain is 1.172 percentage points, interval [-0.397, 2.893]. All intervals use 2,000 source-record bootstrap draws. Joint earlier-history advantage is 0.185 nat [0.128, 0.244], versus 0.119 [0.052, 0.186] for the continued frozen arm. Extra training accounts for part of the likelihood progress; the matched gain isolates the reader update's additional benefit.

## Supplementary Table 3: matched learning audit

| Arm / decoding | Identity (%) | Exact image (%) | Pixel F1 | Isolated ink |
|---|---|---|---|---|
| Frozen / ordered | 2.344 | 1.953 | 0.3643 | 0.0070 |
| Frozen / shuffled | 2.344 | 1.953 | 0.3638 | 0.0071 |
| Frozen / greedy | 6.641 | 6.641 | 0.2296 | 0.0028 |
| Joint / ordered | 3.516 | 3.516 | 0.3620 | 0.0074 |
| Joint / shuffled | 3.125 | 3.125 | 0.3625 | 0.0065 |
| Joint / greedy | 7.422 | 7.422 | 0.2230 | 0.0024 |

Both arms use 256 paired prefixes and the same post-generation evaluator bank. No sampled output is selected or replaced.

Both arms preserve exact greedy copying on 256/256 targets. Stochastic-copy F1 is 0.99993 frozen and 0.99994 joint; exact-image match is 96.875% and 97.266%, respectively. Joint four-step feedback identity is 3.125% ordered and 1.563% shuffled; control values are 2.344% and 2.734%. The joint arm passes six of eight unchanged gates, failing sampled quality and the ordered generation advantage. The control passes five, additionally failing feedback. The matched decision remains `reject`. Neither the primary development reuse nor the paired intervals establish independent seed or sealed-partition generalization.

## Supplementary Note 14: form validity after joint learning

A post-result diagnostic compares saved images with the same 9,932 distinct canonical rasters used for evaluation. Exact membership asks whether an output is a known form; exact target match asks whether it equals the observed

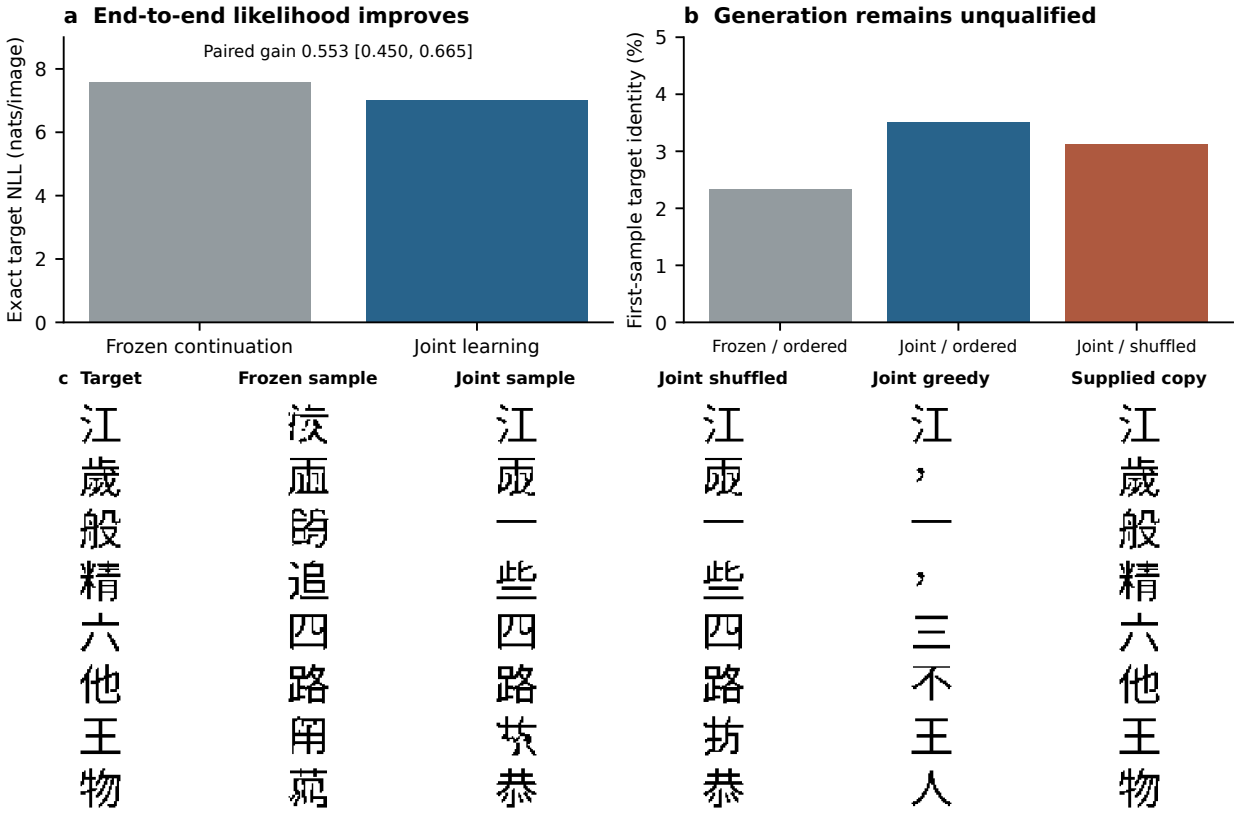



**Fig. S2 | Matched end-to-end learning of exact raster probability. a**, Joint reader–density training improves target NLL over the matched frozen-reader continuation; annotation gives the paired source-record interval. **b**, First-sample identity on 256 prefixes remains low under ordered and earlier-shuffled conditions. **c**, First eight fixed audit rows: target, frozen sample, joint sample, joint shuffled sample, joint greedy output and answer-supplied copying. All outputs are raw model rasters, without selection, bank replacement or OCR repair. The image panels expose errors that the likelihood improvement alone cannot resolve.

next image. Neither uses nearest-neighbour replacement. The bank is confined to evaluation and never enters decoding. These descriptive measurements reuse development outputs and do not change any original decision.

Among 256 first samples, exact known-form membership rises from 9.375% in V54 to 33.594% in the continued frozen arm and 43.750% in the joint arm. The paired joint-minus-frozen gain is 10.156 percentage points, with 95% source-record interval [4.219, 16.240]. Exact observed-target matches are 0.781, 1.953 and 3.516%, respectively. In four-step feedback, joint form membership is 46.875% versus 38.672% frozen; the paired gain is 8.203 points [1.230, 14.683]. The first-sample joint greedy output has 98.438% membership but only 7.422% exact target match and 34 distinct forms, compared with 78 distinct stochastic outputs. Form coherence and linguistic choice remain separate requirements (Fig. S3).

## Supplementary Note 15: raw-pixel beam decoding

The V56 decoder changes search alone. V55 joint weights and the complete binary-image law remain fixed. A beam of eight prefixes expands the next binary pixel to ink or background, adds its exact conditional log probability and retains the eight best scores. The row model observes preceding rows; the masked within-row head observes preceding pixels. Parent gathering keeps row history and row conditions aligned. Stable ties prefer ink and then the earlier parent. Width one reproduces the original greedy rule. No persistent glyph inventory, target identity, image replacement, rejection filter or external model enters the decoder. Beam eight is approximate mode search, not exact global maximization over all $2^{1024}$ images.

A two-image numerical preflight found that the original high matmul precision gave incremental/parallel score differences above the frozen $10^{-3}$-nat tolerance. An amendment recorded before production sets both beam and greedy controls to highest FP32 matmul precision. It changes no weights, windows, beam width, tolerance or quality gate. Production maximum score discrepancy is $2.10 \times 10^{-5}$ nat and the width-one equality check passes.

The fixed audit uses 2,048 development windows, seed 20265620, 64 visible cells and eight future targets. Exact next-image matches are 238/2,048 for ordered beam, 202/2,048 for earlier-shuffled beam and 169/2,048 for ordered greedy: 11.621, 9.863 and 8.252%, respectively. Shuffling preserves the final four visible cells. The paired beam-

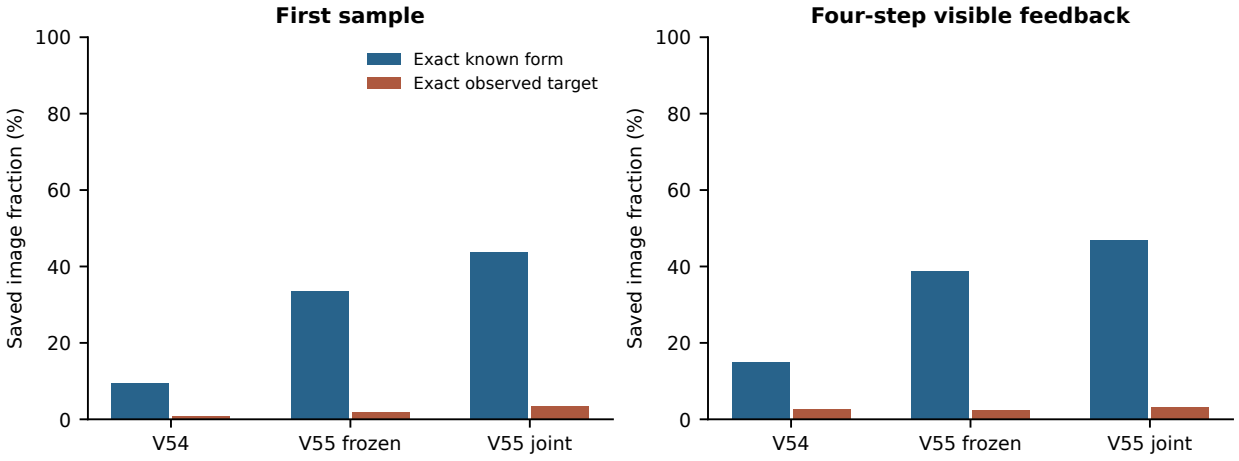



**Fig. S3 | Exact canonical form does not imply correct continuation.** Saved first samples and four-step feedback images are compared with a fixed canonical raster bank. Blue bars require exact pixel equality with any known form; brown bars require equality with the observed future image. All three checkpoints use the same bank. The paired uncertainty is stated in Note 14; the bars are descriptive fractions, not independent replications.

minus-greedy gain is 3.369 points [2.351, 4.412]; ordered-minus-shuffled is 1.758 points [0.966, 2.552]. Intervals use 2,000 source-record bootstrap draws, seed 20265719, with 188 source records. All ordered and shuffled beam outputs exactly match a known canonical form. The post-generation bank has 9,934 distinct rasters and covers every target; it is descriptive, not the search space. Beam emits 127 distinct first forms, versus 116 greedy. Its mean self NLL is 3.108 versus 3.508 nats/image; this scores the selected output, not the true target, and is not a training gain.

Visible feedback uses the first 128 prefixes for eight steps. Exact observed image matches are 68/1,024 ordered beam, 51/1,024 shuffled beam and 50/1,024 greedy. The beam-minus-greedy difference is 1.758 points [0, 3.770], with 89 source records. Adjacent image repetition is 228/896 transitions (25.446%) for beam, versus one transition (0.112%) in the observed streams. Shuffled beam repeats 32.366% and greedy 17.746%. Mean distinct images per eight-step stream are 4.711 beam and 7.789 observed. Thus search improves immediate conditional choice while longer output remains repetitive (Fig. S4). Five of six gates pass; feedback repetition fails and the decision is `reject_decoder`.

All 256 supplied motor images still copy exactly. The audit takes 269.3 s with 0.154 GiB peak allocated CUDA memory on GPU 0, excluding previous training and data preparation. No new training or sealed evaluation is performed. This bounded search result does not establish question answering, instruction following, historical-script generation or capability-matched efficiency against a text language model.

## Supplementary Note 16: grounded image-answer learning

The V57 model retains the V55 joint reader and exact raster writer, extending the allowed causal context from 64 to 512 cells. Parameter-free rotary positions require no weight-shape change; this extension does not itself establish trained long-context competence. Two $384 \times 384$ projections produce source-position attention from the final causal query and observed prompt states. A linear copy gate and a linear stop head give 295,682 new parameters, for 19,007,523 runtime parameters. The contrastive temperature remains unused and fixed. Both reader and density continue learning.

For observed source images $X_i$, attention $a_{ti}$ and copy gate $\lambda_t$, the continued image distribution is

$$q_t(Y) = \lambda_t \sum_i a_{ti} \mathbf{1}[Y = X_i] + (1 - \lambda_t) r_t(Y). \tag{4}$$

Here $r_t$ is the normalized exact raster density. Duplicate source images contribute summed mass. Equality uses lossless bit packing of observed binary pixels; there is no persistent symbolic identity or writing bank. A query cannot attend to source positions later than itself, including when natural training sequences are packed together. With stop hazard $s_t$, $p_t(\text{stop}) = s_t$ and $p_t(Y, \text{continue}) = (1 - s_t) q_t(Y)$. Training includes the terminal stop event after a reference answer. Reference length determines supervision masks, not a deployed input. Runtime receipts retain an inherited Bernoulli-density description; it describes $r_t$ alone, not the complete mixture/event law.

CMRC 2018 supplies natural questions over Wikipedia paragraphs [12]. Pinned source revision and original annotation offsets are checked during data preparation. Whitespace is removed consistently, unsupported font forms are excluded, full rendered prompts are limited to 480 cells and retained answers to 16. No answer-centred crop is used. Answers must match the source substring and not occur verbatim in the question. The result contains 3,211 training questions over 1,134 paragraphs and 1,321 development questions over 421 paragraphs. Article-title and context-hash overlap are zero across splits. The data retain CC-BY-SA-4.0 attribution. Canonical rendering and thresholding match V55.

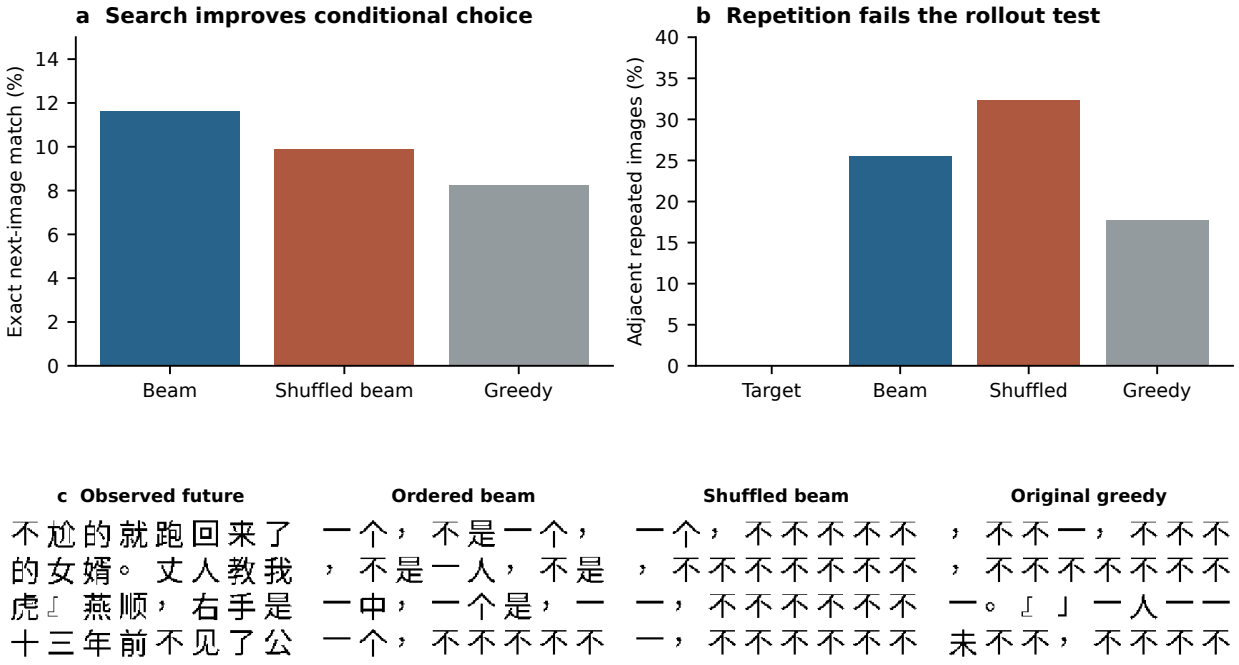



**Fig. S4 | Pixel search improves next-image choice but fails rollout.** **a**, Exact target-image matches on 2,048 fixed prefixes, with unchanged weights across ordered beam, earlier-shuffled beam and greedy controls. **b**, Adjacent raster repetition over eight-step feedback, compared with observed future streams. **c**, First four fixed audit streams: observed future and raw generated images. No examples are selected or repaired. Paired intervals are reported in Note 15; the feedback failure remains the decoder's qualification decision.

No symbolic label is passed to a student method.

Both arms initialize from the same V55 EMA, train for 4,000 updates with batch four and use identical data RNGs. Three of four updates draw uniform training questions with replacement; the fourth retains 16 causal positions per natural stream, including 25% supplied-image motor conditions. Motor conditions use zero hidden states and exact observed fields, bypass copying and omit the stop loss. The no-copy arm fixes $\lambda = 0$; its source-attention heads are frozen. The first retained answer supervises event likelihood, while evaluation accepts every retained reference. Loss is mean exact event NLL over valid image and stop events, without the earlier image-only $1/1024$ rescaling.

AdamW rates are $3 \times 10^{-5}$ for the reader, $10^{-4}$ for the density and $3 \times 10^{-4}$ for new heads; betas are (0.9,0.95), weight decay 0.01, warmup 200 updates, cosine decay to 10%, norm clip one and EMA 0.999. BF16 training uses deterministic CUDA algorithms, math SDPA and fixed cuBLAS workspace configuration; reader dropout stays disabled. A four-update smoke matches interrupted two-plus-two training exactly for all model/EMA tensors and data, training, CPU and CUDA RNG states. A masked-copy backward regression test covers matches occurring only at forbidden future positions. Copy and no-copy training take 278.572 and 269.126 s, with 1.053 and 1.049 GiB peak allocated CUDA memory on GPU 0. These costs exclude previous foundation training and preparation.

Both final EMA checkpoints complete before development opens. Primary IDs are 256 samples without replacement from sorted development records, seed 20265720. The decoder stops at hazard $\geq 0.5$; otherwise it selects the copy component at gate $\geq 0.5$, sums attention over equal observed rasters and emits the largest-mass image. The free component uses pixel beam eight. This is hierarchical greedy decoding, not global mixture-mode search. The 17-cell safety cap is recorded separately from stopping. No answer location, reference length or candidate answer list is available at inference.

Exact answer match requires every output image, its order and learned stop to match a retained reference. Raster F1 measures multiset overlap of exact rasters, maximizing over references. It can credit reordered images and is not the official word-segmented CMRC metric. Primary copy exact match is 10/256 (3.906%), versus 1/256 with question pixels blanked and 0/256 no-copy. Copy F1 is 15.718%, versus 5.553% and 6.582%. Its paired gain over blanking is 10.165 points [7.043, 13.371]; over no-copy it is 9.136 [5.879, 12.483]. Intervals use 2,000 source-article draws, seed 20265819, retaining related questions together (198 primary articles). The blanking intervention removes question ink while retaining passage and framing; it is not an in-distribution alternative natural question.

All 393 shared-paragraph/disjoint-answer question pairs are also tested. Only one has both answers exactly correct

(0.254%). Mean primary answer length is 10.457 cells, and 58/256 cases reach the safety cap. Every primary copy-arm image uses copying, with no free-branch emission. Teacher-forced mean event NLL across all 1,321 development questions is 1.329 copy versus 12.025 no-copy; mean sequence NLL is 8.377 versus 91.477. All 256 V55 supplied motor images remain exact. The full audit takes 1,117.1 s and peaks at 0.140 GiB allocated CUDA memory, including the slower 2,106-image free-writing control. Four of seven gates pass: boundary, clean split/resources, motor and question-dependent F1. Exact answers, absolute F1 and paired-question correctness fail. The decision is `reject_reading_model`.

A post-result, zero-model-call diagnostic on the saved primary outputs finds 43/256 correct first images. Only 10/43 of those become exact complete answers; 14/256 outputs contain a complete reference prefix followed by unwanted additional images. Mean aligned correct prefix length is 0.750 cells; adjacent repetition is 2.230%. These descriptive observations do not alter the locked decision. They motivate testing answer-level structure alongside better evidence selection. All results remain limited to canonical, answerable, short-paragraph reading; independent seed replication, raw page segmentation, historical writing and general language generation are unestablished.

## Supplementary Note 17: complete observed image blocks

V58 continues the completed V57 copy EMA with a normalized distribution over all contiguous observed prompt blocks $X_{i:j}$, where $1 \le j-i+1 \le 16$. For causal states $h_i$ and final prompt state $h_L$, define

$$s_{ij} = \frac{(W_q^s h_L)^\top W_k^s h_i + (W_q^e h_L)^\top W_k^e h_j}{\sqrt{384}} + b_{j-i+1}, \qquad \pi_{ij} = \frac{e^{s_{ij}}}{\sum_{uv} e^{s_{uv}}}.$$

The answer-image law is

$$p(Y \mid X) = \sum_{ij:X_{i:j}=Y} \pi_{ij}.$$

Equality is lossless raster equality, computed from binary pixels; indices address only the current prompt. There is no persistent glyph inventory. Training minimizes exact negative log block mass, divided by reference cell count plus one boundary event across the batch. Annotated source offsets do not enter the loss. Mode decoding sums mass over equal sequences and chooses the most probable sequence, with first-source tie breaking. No reference length, answer list, OCR or output repair enters generation. The predicted end boundary terminates a nonempty block; the 16-cell support is fixed before training. The retained copy/free/stop continuation path remains separate.

Four new 384-by-384 projections and sixteen length biases bring the runtime to 19,597,363 parameters. Projection pairs initialize from V57 copy attention; the matched glyph control constructs and freezes the same new heads. Both start from identical V57 copy EMA weights and make 4,000 updates, batch four, using the previous three-to-one QA/natural-motor schedule, optimizer rates and deterministic settings. Seeds are 20265810 (training) and 20265811 (data). A four-update uninterrupted versus two-plus-two resumed GPU run matches all model/EMA tensors and data, training, CPU and CUDA RNG states exactly. Block training takes 210.995 s, 0.967 GiB peak allocated memory; the glyph control takes 302.237 s, 1.056 GiB. Earlier foundation costs are excluded.

Both final checkpoints precede development inspection. Primary IDs are the same 256 as V57, explicitly reused development, not a sealed final test. Exact answers are 25/256 (9.766%) block, 3 blank-question and 21 matched glyph control. F1 is 27.033%, 9.169% and 22.140%. Article-grouped bootstrap uses 2,000 draws, seed 20265919. Block-minus-control F1 is 4.893 points [0.469, 9.047]; exact match is 1.563 [-1.509, 4.652]. Normal-minus-blank F1 is 17.864 [13.438, 22.289]. Only 4/393 pairs have both answers correct; the paired gate fails and the registered decision remains `reject_reading_model`. All 256 supplied motor images remain exact. Mean conditional block sequence NLL is 6.505, versus 9.552 for glyph-event sequence likelihood; these laws have different supports and must not be treated as interchangeable language perplexities. The audit takes 98.951 s at 0.142 GiB allocated memory. The 78,772,107-byte inference-only export reproduces native prompt-PNG answers and events exactly. Raw first audit failures appear in Figure 5.

## Supplementary Note 18: frozen visual foundation probe

V59 pins MIXAR-477M stage-2 revision `c5d5bcfbad2f290296fd83810558f396c6a6333a`, reusing its shared Apache-2.0 checkpoint. Author-reported multilingual pretraining and its four-H200 compute are external work, not part of single-4090 training here. Both it and the frozen V58 reader consume identical Noto CJK 26-pixel glyphs inside binary 32-pixel cells. This differs from MIXAR's native rendering. An odd-length prompt gets a blank final cell for two-cell pairing; readout candidates exclude that extra cell. Both encoders supply states after each complete pair, repeated over its two source cells.

A fixed CPU-QR semi-orthogonal projection maps MIXAR's 896 states to 384 coordinates, seed 20265900; student states already have width 384. Both append 128 projected normalized-DCT coordinates, seed 20265901, plus two within-pair phase channels. Semantic and fine-image parts are separately RMS-normalized. The 514-dimensional

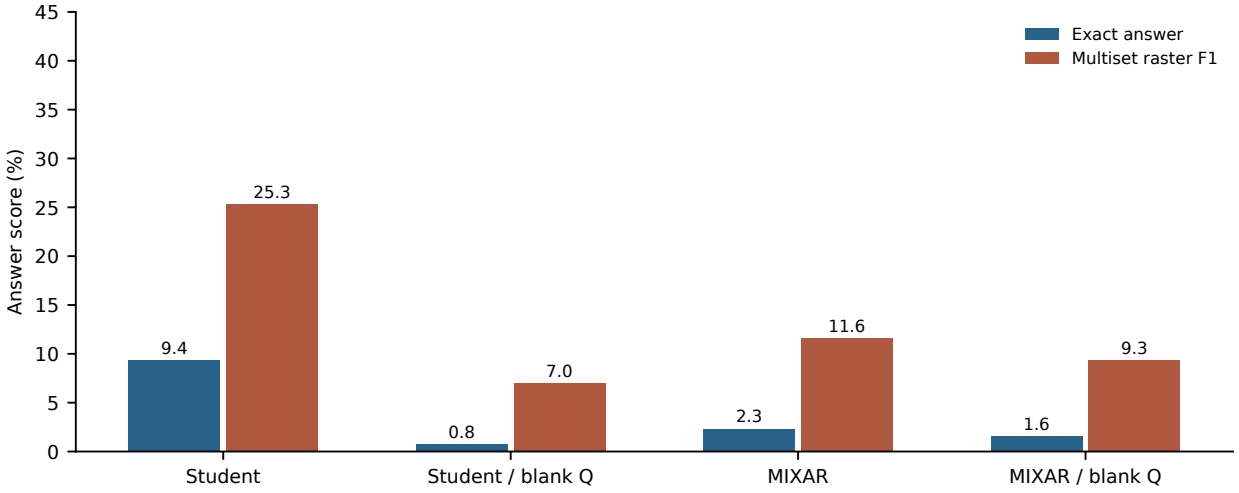



**Fig. S5 | Frozen pretrained features do not improve this reading interface.** Identical learned block heads consume frozen student or MIXAR features, with normal and blank-question prompts. All scores use the fixed 256 questions. External pretraining and rendering shift prevent a causal attribution to capacity or pretraining alone.

features are quantized to FP16 for caching and readout. All train, development and blank-question features are prepared without reading answer annotations. Recomputed quantized features agree exactly.

Each fresh head has four 514-to-384 projections and sixteen biases, totaling 789,520 learned parameters. Both train 4,000 updates, batch eight, with seeds 20265910/20265911, AdamW rate $3 \times 10^{-4}$, betas (0.9,0.95), decay 0.05, norm clip one, warmup 200, cosine to 10%, and EMA 0.999. Frozen encoders and native free writers are unchanged. An interrupted/resumed GPU preflight matches weights, EMA and RNG states exactly. Features take 184.781 s and 1.906 GiB; student and MIXAR readouts take 106.593 and 109.257 s, each at 0.170 GiB. These resource numbers exclude external pretraining.

Both completed heads precede inspection of the same reused 256 development IDs and 393 pairs. Student exact/F1 scores are 9.375/25.340%; MIXAR scores 2.344/11.588%. MIXAR-minus-student F1 is $-13.752$ points $[-18.247,-9.094]$, using 2,000 article draws, seed 20266019. Normal-minus-blank F1 is 2.246 [0.415,4.378] for MIXAR, versus 18.343 [13.949,22.519] for the student. Both questions are exact in zero versus twelve pairs. Three of seven gates pass; the frozen-foundation probe fails. The 117.085-s, 0.094-GiB audit uses cached readouts and excludes online encoder inference. Figure S5 reports all four conditions. Capacity, pretraining, prior task training and rendering differ; this is a scoped foundation probe, not a general ranking of encoders or native-font MIXAR competence.

## Supplementary Note 19: question-first causal reading

V60 tests whether passage states should be conditioned on the question before evidence selection. The context-first host layout places passage $X$ before question $Q$; the intervention reverses these two blocks, yielding causal passage states $h_i = f(Q, X_{\leq i})$. Framing and total visible length are unchanged. Both models start from identical V58 block EMA weights and retain the 19,597,363-parameter architecture, block law, image-only supervision, mode decoder and continuation writer. All visible prompt positions remain candidates. There is no answer offset or answer-aware cropping.

Each arm makes 4,000 updates with the same V58 batch-four schedule, optimizer rates, warmup, cosine, clipping, EMA and deterministic BF16/math-SDPA regime. Training/data seeds are 20266010/20266011. The new prompt-order helper is tested for exact preservation of source characters and correct blanking positions. An uninterrupted versus two-plus-two resumed GPU preflight matches all model/EMA tensors and RNG states exactly. Both production arms finish before their development scores open. Context-first and question-first training take 214.699 and 225.207 s, each at 0.966 GiB peak allocated CUDA memory. Cost remains incremental to the same previously trained foundation.

The same 256 primary IDs and 393 paragraph pairs are explicitly reused. Question pixels are blanked at each arm's actual question positions, retaining the paragraph, length and framing. Exact answers are 27 context-first and 21 question-first, versus three and two for the corresponding blank controls. F1 is 26.908% and 23.765%, versus 7.572% and 9.098% blanked. With 2,000 article-grouped bootstrap draws, seed 20266119, question-first minus context-first F1 is $-3.143$ points $[-6.544,0.446]$; exact match is $-2.344$ $[-5.490,0.810]$. Neither difference resolves zero. The question-first normal-minus-blank F1 gain is 14.667 [10.873,18.809]. Both answers are exact in 5/393 question-first pairs, versus 16/393 context-first pairs.

All 256 supplied motor images remain exact for both arms. The complete audit takes 192.723 s at 0.184 GiB allocated CUDA memory. Four of eight registered gates pass: boundary, clean split/resources, question-

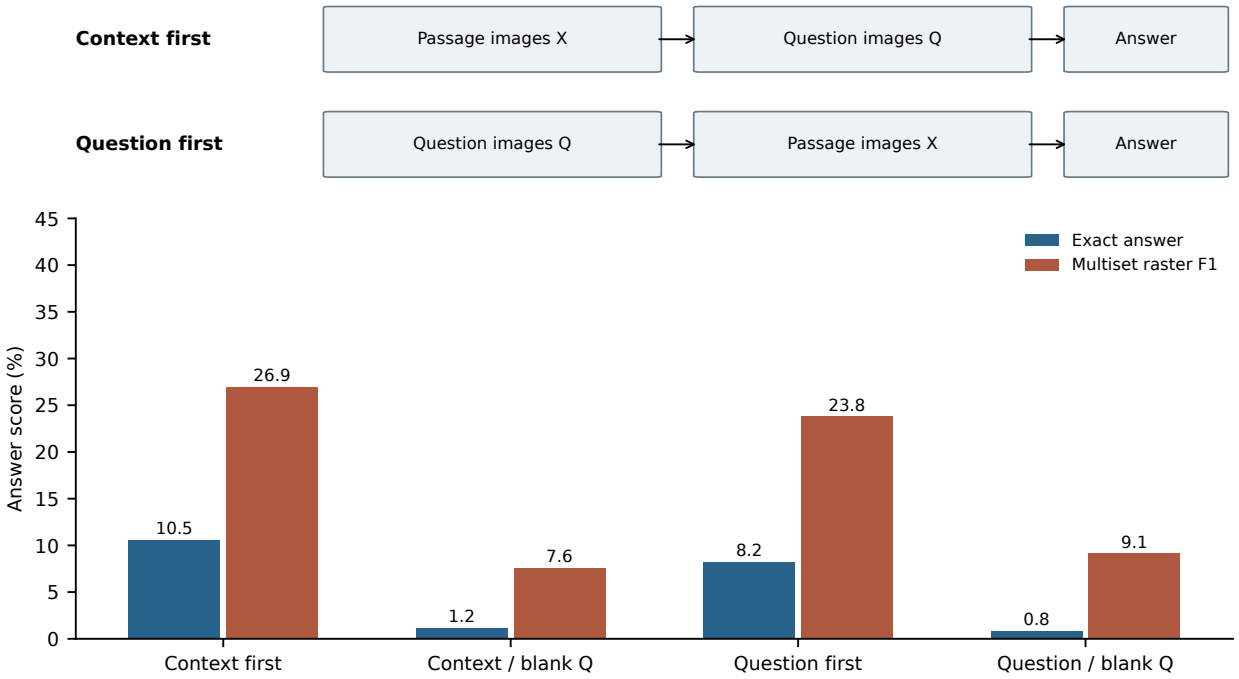



**Fig. S6 | Changing input order is insufficient.** Matched models receive the same source writing with the question before or after the passage. Question blanking uses each layout's actual positions. The question-first F1 difference is unresolved and its registered reading criteria fail.

dependent F1 and motor. The gain, absolute accuracy, absolute F1 and paired-question gates fail, so the decision is `reject_question_first_intervention`. This is one matched training seed and a task-trained, context-first foundation; neither a universal ordering result nor a final generalization test is established.

## Supplementary Note 20: full native pixel-language adaptation

The complete released MIXAR-477M[9] image-language model is adapted using the pinned release identified by its full revision, weight SHA-256 and architecture-code hashes in the accompanying evidence. This is supervised adaptation of external pretraining, not local foundation training from scratch. All 477,145,984 contextual and image-output parameters train. The source data are the same 3,211 training and 1,321 development questions used for the preceding reading experiments, with no article or context-hash overlap. Development is explicitly reused. Complete prompts contain at most 480 cells and first-reference training answers at most 16 cells.

Noto CJK glyphs at 26 pixels occupy 32-pixel binary cells, a rendering shift from the author's native bundled-font interface. Odd prompt/answer lengths receive one blank cell for alignment to 32-by-64 native blocks. Two all-white blocks follow each target answer. Native input uses white $+1$ and ink $-1$; the target Bernoulli probability describes white pixels. Spatial prompt bounds are known, but strings, codepoint IDs and source-answer offsets never enter the actor. The causal answer-only objective is

$$\mathscr{L} = -\frac{1}{|A|P} \sum_{t \in A} \sum_{i=1}^{P} \left[ y_{ti} \log \sigma(a_{t-1,i}) + (1 - y_{ti}) \log(1 - \sigma(a_{t-1,i})) \right], \tag{5}$$

where $P = 2048$, $A$ contains answer and termination blocks, and state $t-1$ depends only on preceding physical images. Prompt labels contribute no loss.

The registered run uses 1,000 updates, microbatch one accumulated eight times, uniform replacement sampling, AdamW $2 \times 10^{-5}$, betas $(0.9, 0.95)$, weight decay 0.01, warmup 50, cosine decay to 10%, norm clip one and EMA 0.99. Seeds are 20266110 for training and 20266111 for data. FP32 weights, optimizer and EMA coexist with BF16 forward/backward, deterministic algorithms, math attention and nonreentrant activation checkpointing. The final EMA is used without development checkpoint selection. A real four-update interrupted preflight reproduces model, EMA, optimizer and all RNG states exactly, with finite gradients throughout. Altering future answer pixels changes earlier logits by exactly zero in the recorded causality check.

Training through the final update takes 1,501.138 s, with 9.912 GiB peak allocated memory; loading takes 2.525 s. Cache preparation, final serialization and external pretraining are separate. All work uses physical GPU 0. Greedy decoding thresholds each white logit at zero, feeds back the generated image and stops at the first exactly white block,

otherwise capping at ten new blocks. Only terminal blocks and entirely blank trailing cells are removed. No malformed image is repaired or snapped to a character.

The locked audit compares adapted and released-base actors with normal and blank-question inputs on the same 256 primary IDs, and tests all 393 shared-paragraph question pairs for the adapted actor. All 1,321 development examples contribute likelihood. Pixel NLL decreases from 0.684156 to 0.154814 nats, yet exact answers are zero and exact-raster multiset F1 is 0.149%. Foreground F1 is 12.264%, compared with 8.473% blanked and zero for the released base under this shifted interface. Article-grouped bootstrap with 2,000 draws, seed 20266219, gives the adapted-minus-blank foreground difference as 3.791 points [2.521,5.056]. The exact-raster F1 difference is 0.149 [0,0.372], unresolved. Both-correct pair accuracy is 0/393.

No primary output is empty or capped for the adapted actor; the released base returns an empty answer on every normal primary prompt. The adapted mean length is 3.047 cells. Five of ten gates pass: image-only runtime, clean split/finite resources, full backward/resume/causality, independent replay, and capped-output rate. Exact answers, raster F1, adaptation gain, question-dependent raster F1 and paired correctness fail. Audit time is 145.690 s excluding loading, at 1.832 GiB allocated memory. The registered decision is `reject_native_free_reading`.

A learned-weight deployment bundle includes configuration and local custom architecture code. A fresh offline process with empty model and code caches reproduces actual answer tensors, PNG bytes and stopping events. It loads no original release weights and calls no external model, tokenizer or OCR. Exact replay is an implementation result, separate from language quality.

An auxiliary transcription plan was registered before development outputs were inspected. CPU Tesseract 5.3.4 receives anonymous answer-only images, enlarged fourfold with a white margin, without questions, references or model names. It recognizes only 40/256 retained reference images exactly, a 15.625% calibration rate. No generated answer is transcribed exactly, but the poor reference calibration precludes treating this as a reliable semantic score. The primary raw-image failure and strict gates remain unchanged. One seed, the shifted font interface, limited training data and absent general-language evaluation constrain the interpretation.

## Supplementary Note 21: conditional whole-image density

The matched experiment tests a latent expert shared across a complete native image block. Let $h_t$ be the visual transformer's causal state after the visible prefix and let $y \in \{0,1\}^{2048}$ denote the next $32 \times 64$ bitmap, with one denoting white in this native interface. The normalized law is

$$p_\theta(y \mid h_t) = \sum_{k=1}^{K} \pi_{\theta k}(h_t) \prod_{i=1}^{2048} q_{\theta ki}(h_t)^{y_i} [1 - q_{\theta ki}(h_t)]^{1-y_i}, \qquad q_{\theta ki} = \sigma(a_{\theta ki}), \tag{6}$$

where the gate is a softmax. Each expert is a neural function of the current visual state, without a fixed character identity or stored target bitmap. Marginalizing the expert retains a proper probability law. For distinct pixels, conditional covariance is

$$\mathrm{Cov}(Y_i, Y_j \mid h_t) = \sum_k \pi_k q_{ki} q_{kj} - \Big(\sum_k \pi_k q_{ki}\Big)\Big(\sum_k \pi_k q_{kj}\Big). \tag{7}$$

It can be nonzero even though pixels are independent within one expert. This elementary property permits coherent alternatives; it does not prove that training discovers valid writing or useful answers.

The objective is exact negative log joint likelihood, computed with `logsumexp`, divided by 2,048 physical pixels. It averages only answer and two blank termination blocks within each example, then averages over accumulated examples. Teacher forcing uses the same one-block causal shift as Note 20. Prompt pixels enter the context without a prompt loss. There is no transcript, answer-span offset or Unicode target inside the actor.

The single-head control has 477,145,984 parameters; eight experts have 489,998,208. Both initialize from the same externally pretrained MIXAR revision and use the unchanged Noto-26 cell interface, 3,211 training questions and 1,321 article-disjoint development questions. The eight-head projection repeats the released head with zero-mean perturbations of scale 0.01, seed 20266210, and a zero-initialized gate. Every backbone and head parameter receives gradients. The comparison matches examples, optimization and update budget, rather than parameter count or output computation.

Each arm uses 1,000 updates, microbatch one, accumulation eight, AdamW at $2 \times 10^{-5}$ with betas $(0.9, 0.95)$, decay 0.01, 50-update warmup, cosine decay to 10%, norm clip one and EMA 0.99. Weights, optimizer and EMA are FP32; forward/backward use BF16 with deterministic algorithms, eager/math attention and nonreentrant activation checkpointing. Training and data seeds are 20266220 and 20266221. Complete sampled-example orders must match exactly. Only the final EMA is audited. Both arms run sequentially on physical GPU 0, under a 20 GiB allocated-memory and four-hour ceiling per phase.

Generation thresholds each expert into one complete candidate bitmap, scores all candidates under the full mixture and emits the highest-scoring candidate. Searching component modes is approximate MAP; it is not a guarantee of the global mixture mode. Actual emitted pixels become subsequent context. The first exactly white native block ends

generation, otherwise ten blocks cap the continuation. Only the detected terminal block and wholly blank trailing cells are removed. Malformed nonblank strokes are preserved. The deployed bundle has no tokenizer, OCR, glyph inventory or external language-model call.

The primary audit retains the fixed 256 questions, blank-question controls, all 393 shared-paragraph pairs and all-development likelihood. Paired intervals resample articles 2,000 times, seed 20266229. An intervention pass requires positive lower intervals for both K8-minus-K1 strict-raster F1 and K8 normal-minus-blank F1, fewer than 20% capped outputs and all implementation checks. Each arm's separate bounded reading gate requires at least 10% exact answers, 25% raster F1, 2% both-correct pairs, a positive question effect, fewer than 20% caps and implementation checks. Expert-use counts and gate entropy are descriptive; they cannot select a new checkpoint or change these gates. Reused development, one seed, the shifted font interface and external pretraining limit the conclusions.

Both arms complete their registered budget and all replay checks. K=1 takes 1,887.5 s at 9.904 GiB peak allocated memory; K=8 takes 1,972.8 s at 10.174 GiB. Their complete training-example orders match exactly. The locked audit takes 527.4 s excluding model loading, at 1.904 GiB allocated memory. Four-update uninterrupted versus two-plus-resumed-two preflights match model, EMA, optimizer and every RNG state exactly for each arm. Future-pixel changes have zero effect on earlier predictions. Strict-FP32 cache errors remain below $2.1 \times 10^{-5}$; larger TF32 discrepancies are retained separately rather than hidden by changing the production recipe.

All-development NLL is 0.1542 for K=1 and 0.1248 for K=8. Primary strict-raster F1 is 0.234% versus 0.815%, a 0.580-point gain [0.089, 1.094]. Foreground F1 is 12.205% versus 15.827%, a 3.623-point gain [1.815, 5.554]. Both arms have zero exact primary answers and zero both-correct pairs. K=8's normal-minus-blank raster difference is 0.295 points [−0.006, 0.623]; its foreground difference is −0.044 points [−0.629, 0.509]. Neither resolves question dependence. Seven of eight intervention gates pass; both arms pass six of ten bounded reading gates. The registered decision is `reject_density_intervention`; neither arm qualifies as free reading.

K=8 selects experts 0, 1, 2, 4 and 7 on respectively 223, 4, 106, 605 and 3 primary generation steps, including termination. Mean gate entropy is 1.812 nats, compared with $\log 8 = 2.079$ for a uniform gate. The primary outputs all terminate before the cap, while one blank-question output reaches the cap. These descriptive observations do not identify semantic expert roles or justify selecting different parameters. The result supports improved bitmap fit at this budget, while the required question-dependent language remains unestablished.

### Auxiliary reader and post-hoc transcript sensitivity

A separately preregistered, answer-only Qwen3-VL 8B reader transcribes 122/128 training-reference images exactly (95.313%), returns every one of 16 blank and 16 random-ink controls empty, and produces valid JSON throughout calibration. The fixed instruction and anonymous images disclose neither questions nor reference answers to the reader. Three responses among 1,015 unique actor images are invalid, including two that reach the registered 256-output-token limit. The original diagnostic therefore reports `invalid_actor_transcription`; it has no complete fixed-case score.

An explicitly post-hoc sensitivity analysis reuses the first responses without retry, repair or new model calls. It retains all 256 cases in each condition and assigns each invalid bounded score possible range $[0,1]$. The resulting full-case means are below; ranges are identification intervals, not confidence intervals. Exact text means normalized transcript agreement with any retained reference. Character score is one minus normalized edit error, clipped to $[0,1]$.

| Condition | Valid / unknown | Exact text (%) | Character score (%) |
|---|---|---|---|
| K=1 | 254 / 2 | 0.781–1.563 | 1.388–2.169 |
| K=1, blank Q | 255 / 1 | 0.000–0.391 | 0.186–0.577 |
| K=8 | 256 / 0 | 0.000 | 2.806 |
| K=8, blank Q | 256 / 0 | 0.000 | 2.945 |

K=8's normal-minus-blank character difference is −0.139 percentage points, with article-bootstrap sensitivity envelope [−0.962, 0.638]. The envelope uses 2,000 article resamples, seed 20266379, taking the lower endpoint's 2.5th percentile and upper endpoint's 97.5th percentile. It covers the unknown-score assignments within this resampling scheme, not systematic reader errors, training-seed variation or reused-development selection. It establishes no new model qualification and changes neither failed decision. The external reader makes 1,172 calls over 1,092.1 s of owned runtime including cleanup; maximum resident GPU memory sampled after requests is 7,278 MiB. These costs are separate from actor inference and training. The evaluator server and runner exit, its port closes and GPU 0 memory is released.

## Supplementary Note 22: matched question–answer assignment

For two observed question images $x_0, x_1$ about the same paragraph and two accepted answer-image streams $y_0, y_1$, define $e_{ij}$ as mean answer and termination-pixel NLL for $y_j$ conditioned on $x_i$. Each actor scores all four streams using

the complete image generator. Let

$$D = e_{01} + e_{10} - e_{00} - e_{11}, \tag{8}$$

$$\mathscr{L}_{\text{positive}} = (e_{00} + e_{11})/2, \tag{9}$$

$$\mathscr{L}_{\text{assign}} = \tau \log(1 + \exp[-D/\tau]), \qquad \tau = 0.01, \tag{10}$$

$$\mathscr{L} = \mathscr{L}_{\text{positive}} + \lambda \mathscr{L}_{\text{assign}}. \tag{11}$$

Aligned uses $\lambda = 1$ and control $\lambda = 0$. Additive question-only and answer-only energy offsets cancel from $D$. Different answer lengths receive their own normalization, so $D$ is not a joint log-odds or mutual-information estimate. The algebra does not identify linguistic meaning.

There are 3,793 training-only pairs from 939 paragraphs and 3,016 questions. Accepted answer text sets and canonical reference images are distinct within each pair; two pairs with shared references are excluded. Sampling chooses paragraphs uniformly, then a pair within the paragraph. Both actors start from the same learned V62 eight-expert EMA, retain 489,998,208 parameters and use the unchanged Noto-26 interface and approximate component-mode decoder. No reference bank, OCR or teacher is deployed. All 2,000 update journals, containing four pair draws each, match byte for byte.

Microbatch is one pair with accumulation four. Each pair forms two positive and two swapped likelihood streams. AdamW uses learning rate $2 \times 10^{-5}$, betas $(0.9, 0.95)$, decay 0.01, 100-update warmup, cosine decay to 10%, clip one and EMA 0.99. Model/optimizer/EMA weights are FP32; forward/backward use BF16, deterministic eager/math attention and nonreentrant checkpointing. Model and data seeds are 20266320 and 20266321. Only final EMA is audited. Both arms execute sequentially on physical GPU 0 under eight-hour and 20-GiB allocated-memory ceilings per phase.

**Supplementary Table 4: completed question–answer assignment.** Image F1 values are percentages.

| Arm | Exact / 256 | Raster F1 | Foreground F1 | Both / 393 |
|---|---|---|---|---|
| Control | 2 | 3.229 | 20.227 | 0 |
| Aligned | 1 | 2.417 | 17.219 | 0 |

Control and aligned all-development NLLs are 0.113901 and 0.114186 nats per pixel. Aligned minus control raster F1 is $-0.812$ points $[-1.504, -0.202]$; foreground F1 differs by $-3.008$ points $[-3.978, -2.014]$. Normal-minus-blank raster effects are 1.765 points [0.504,3.251] for control and 1.677 [0.492,3.033] for aligned. All intervals resample source articles 2,000 times, seed 20266329, over the same 256 primary cases from 198 articles.

Each actor passes seven of ten reading gates. Both fail the unchanged 10% exact-answer, 25% raster-F1 and 2% both-correct-pair requirements. Question dependence, capped outputs and implementation checks pass. The intervention requires an aligned reading pass and a positive lower interval for aligned-minus-control raster F1; both requirements fail. The decision is `reject_assignment_intervention`. No decoder, checkpoint selection, threshold or primary case changes after scoring.

The separate accepted-reference diagnostic assigns the pair correctly in 324/393 cases for control and 330/393 for aligned. Both individual energy ranks are correct in 82/393 and 109/393. These are descriptive comparisons without registered difference intervals, and do not evaluate free writing. Training takes 7,355.156 and 6,737.336 s at 10.301 GiB peak allocated memory. The audit takes 582.104 s excluding loading, at 1.904 GiB. Timing excludes loading, final serialization and external foundation pretraining.

The preflight passes exact uninterrupted/resumed optimizer, EMA and RNG comparison, finite complete-backbone and assignment-only gradients, causal masking and physical stream assembly. Largest-pair allocation is 9.197 GiB. Same-shape future-pixel effects are zero; strict-FP32 cached-prefix error is $6.485 \times 10^{-5}$ against a $2 \times 10^{-4}$ bound. Both standalone bundles reproduce tensors, PNG bytes and stopping events with networking disabled, empty caches and no repository source path. Reused development, one seed, external pretraining and strict-raster limitations remain explicit. Sealed final partitions are unopened.

## Supplementary Note 23: inherited continuous writing actor

A separately executed successor retains the existing authenticated Qwen3-VL vision/contextual archive[17], omits its vocabulary matrices and attaches a continuous physical-image writer (Fig. S7). It has 7,548,043,521 parameters: 7,522,463,984 inherited/frozen and 25,579,537 trainable. The trainable subset includes rank-eight query/value/output adapters in 36 contextual layers, two continuous writing queries, a 19,374,096-parameter six-layer flow writer and a binary stop classifier. All 27 vision layers and three DeepStack feature sets remain present. The actor accepts observed grayscale image tensors and completed $32 \times 64$ blocks; it requires no character segmentation or alphabet.

Teacher forcing places each query pair before its answer image. The observed prompt may be bidirectionally image-encoded, while each completed answer block is an independent encoder image. A causal contextual mask

prevents current and future answer pixels from entering a prediction. Cached generation rereads only completed generated blocks. Explicit spatial/temporal rotary coordinates preserve image layout; sequential mask coordinates carry no character identity.

The writer models all 2,048 pixels jointly. With Gaussian $U$, uniform $t$ and signed target image $Y$, its flow objective is

$$Y_t = (1-t)U + tY, \qquad \mathscr{L}_{\text{flow}} = \mathbb{E}\|v_\theta(Y_t,t,h) - (Y-U)\|_2^2. \tag{12}$$

Explicit midpoint integration gives a continuous raster; zero thresholding produces binary output. A class-balanced binary stop objective is separate from white-space generation. Its sigmoid score has no independent calibration. No token IDs, OCR, font renderer, glyph bank or external model call enters actor inference. Encoded text remains permissible in offline dataset creation.

All 19 complete-model execution checks pass on GPU 0 at 17.181 GiB allocated and 18.053 GiB reserved peak memory. Geometry inspection covers 12,976 complete training examples from CMRC and Dolly instructions [12,18]. The largest contextual case has 918 positions, a $480 \times 1024$ prompt and 109 complete target blocks; the largest prompt is $512 \times 1024$. Both receive full contextual backward with Adam moments allocated, while 16 uniformly sampled target blocks receive writer loss. No prompt or answer history is truncated. Every trainable gradient is finite, and nonzero gradients reach first/last contextual adapters and the writing head. Restoring one saved optimizer update reproduces the next writer/adapters, optimizer, sampling and global RNG states exactly. Same-shape future-image and encoder-isolation interventions have exactly zero effect.

Two retained attempts fail cached/full-prefix comparison at relative maximum errors 0.18524 and 0.09815. The accepted implementation keeps contextual weight storage in BF16 but evaluates contextual projections, attention and residuals in FP32. Layer checkpointing avoids retaining a full FP32 body. Vision weights/computation are FP32; writer contractions may use BF16 autocast. The complete contextual call disables ambient autocast, verified at actual attention entry. Cached/full-prefix error becomes $3.910 \times 10^{-4}$ absolute and $1.210 \times 10^{-5}$ relative, below the unchanged 0.02 relative bound. These casts do not recover precision lost in the source quantization.

A separate offline helper temporarily reads source token rows and an output head to test the contextual initialization against the source engine. Across eight short text-only prompts, it matches 8/8 initial and 9/9 returned token choices, with mean KL 0.000264 nat after grouping reference top-ten tokens and the outside mass. This limited check is not full engine equivalence or image-language understanding. The helper vocabulary is never attached to the actor, and its owned reference service exits before native model loading. Its 103.832-s runtime and 18.701-GiB allocation include the temporary head and are separate from the image-actor preflight.

The preflight contains two optimizer updates and replay; its output is noise. It provides no writing or language quality score. A separate writer-only curriculum then trains 19,374,096 parameters for 10,000 updates on balanced Chinese/English blocks from 12,976 training examples. With the final EMA fixed, 128 development blocks yield foreground F1 of 34.000% under correct visual conditions and 21.420% with cyclically shuffled conditions, a gain of 12.581 percentage points. Neither route produces an exact block. Both registered component requirements, F1 at least 90% and gain at least 20 points, fail. Training peaks at 15.801 GiB; the resumed evaluation's separately reset peak is 15.514 GiB. The first evaluation attempt stops before output because its directory is missing. Creating that directory and resuming unchanged code at update 10,000 leaves final weights and training journals byte-identical, with no further learning. All 256 generated-image scores are recomputed from raw pixels. These are source-image reconstruction measurements on reused development data, not prompt-to-answer language. Independent autonomous answers, historical-form behavior and capability-matched efficiency remain required. External pretraining is inherited work, not training performed from scratch on the local GPU.

## Supplementary Note 24: matched writer-fitting diagnosis

The failure in Note 23 motivates a factorial experiment on 64 fixed training blocks. Each block is a physical $32 \times 64$ crop, which can contain partial letters or characters. Conditions are either the inherited reader's two 4096-dimensional features or a lossless embedding of the actual target pixels. The latter is an answer-supplying oracle control. Each condition is crossed with Gaussian velocity learning or direct one-pass image MSE, using the same writer architecture, initialization and 2,000-update batch exposure. All four early optimizer/EMA/RNG replays pass.

| Condition / objective | F1 (%) | Exact / 64 | Shuffled F1 (%) |
|---|---|---|---|
| Reader / flow | 34.567 | 0 | 26.716 |
| Reader / direct | 99.995 | 63 | 27.466 |
| Pixels / flow | 33.648 | 0 | 26.581 |
| Pixels / direct | 100.000 | 64 | 27.464 |

All 512 main generated-image scores and all 8,000 batch draws are independently verified against the original training pixels. The fitting result rejects an immediate flow-writing explanation at this configuration; it does not prove

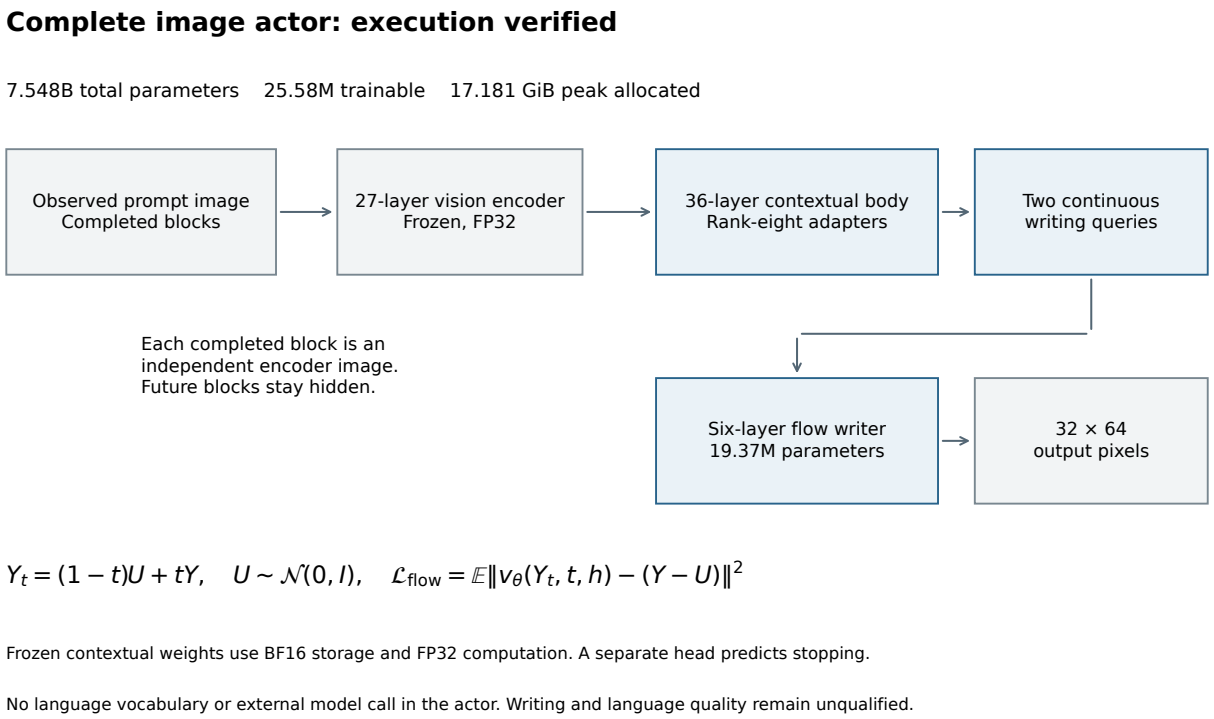



**Fig. S7 | An inherited image actor executes within one GPU, with language quality unqualified.** Observed prompt images and independently encoded completed blocks enter the retained visual/contextual body. Two continuous queries condition a flow over physical writing pixels. Blue modules contain trainable components; the contextual body also retains frozen inherited weights. A separate classifier predicts stopping. Parameter counts and memory describe actual complete-model execution. The diagram contains no generated-language example or quality claim.

generalization from inherited features. Source-feature extraction peaks at 15.251 GiB. Subsequent writer fits peak near 0.66 GiB after the unused semantic body is released, so the latter is not complete language-model memory.

An auxiliary native-codec comparator initially uses the wrong image polarity, giving F1 81.364%. That original report is retained and invalidated. The codec was trained with white-positive input. A separately recorded correction using its actual convention reconstructs all 64 training images exactly. Both comparator outputs and the correction are independently verified. The fixed raw rows in main-text Fig. 9 remain unmodified.

## Supplementary Note 25: broad direct writing and a fixed output codec

The broader direct writer retains the full inherited actor in memory but trains only its 19,374,096 writer parameters. It initializes from Note 24's reader/direct EMA, then makes 10,000 additional updates on balanced Chinese/English physical-block draws. Independent replay confirms 160,000 draws and 12,972 distinct consumed examples out of a 12,976-example pool. All 256 final correct/shuffled output scores are recomputed from raw images. Aligned foreground F1 is 56.563%, shuffled 18.520%, and exact counts are 3/128 and 0/128. Each route contains two blank blocks. The 38.043-point conditioning gain exceeds 20 points, but the unchanged 90% fidelity gate fails. The run is complete and rejected as a writing component.

Training peaks at 15.804 GiB. The recorded 1,555.127-s interval includes training, checkpointing and evaluation after restoration, excluding initial loading; it is not full wall-clock or from-scratch language training. The fixed fitting-stage initialization and different objective prevent treating this as a matched broad-corpus flow comparison.

The next audit uses the existing V34 continuous glyph codec with its original 6,000-update EMA, 7,423,361 frozen parameters and no additional optimization. Each target block is split into left/right $32 \times 32$ tiles. Its native white-positive encoder produces two continuous 768-vectors; the fixed decoder reassembles signed physical pixels. There is no quantization, character index, font renderer or stored-form lookup in this decoder. On the identical 128 reused-development blocks, aligned F1 is 99.998%, with 126 exact and no blank blocks. Shuffled codes give 21.413% F1, zero exact and no blank blocks. The 78.584-point gain and 90% fidelity requirements both pass.

The original source images, native codes and all 256 native decoder outputs are independently verified. The CPU audit takes 1.684 s, excluding earlier codec training and inherited foundation pretraining. This establishes a faithful supplied-code output representation. The target encoder sees the answer image, so the audit contains no autonomous semantic prediction.

## Supplementary Note 26: separate writing coordinates in a complete actor

The complete candidate retains all 27 vision and 36 contextual layers and replaces the failed reader-feature inverse with the fixed output codec (Fig. S8). A shared $4096 \to 768$ projection predicts each of two writing vectors. Rank-eight query/value/output adapters, two structural queries, the stop classifier and output projection contain 9,351,937 trainable parameters in a 7,539,239,282-parameter actor. The remaining 7,529,887,345 parameters, including the entire codec, are frozen. No language vocabulary is materialized in this actor. Offline image rendering may use source strings; actor generation receives observed pixels alone.

All 24 actual complete-model checks pass at 16.953 GiB allocated and 17.381 GiB reserved peak memory. Adam-state backward executes both the 918-position, 109-answer-block witness and the $512 \times 1024$ prompt witness. Finite, nonzero gradients reach the projection, queries, stop head and first/last contextual adapter outputs. Frozen parameters receive no gradients. A future image intervention has exactly zero effect on earlier contextual states, predicted writing codes and stop logits. Independent reader-image isolation is also exact. Cached/full contextual relative maximum error is $1.292 \times 10^{-5}$; predicted-code error is $1.903 \times 10^{-5}$, both below the unchanged 0.02 requirement. Restored parameters, optimizer, sampling and global RNG states replay exactly across a real optimizer update.

Setup takes 68.949 s and the complete preflight 78.022 s. Its two generated blocks are finite and deterministic with actual image feedback, but only two distinct optimizer updates precede them. Their strokes are fragmented and no language score is evaluated. A CPU source-bound receipt checks the committed implementation and GPU-report consistency; it is not a second GPU execution.

A separate registered CPU diagnostic tests local code perturbations on the same 64 training images as Note 24. Four seeded tangent directions per image give 256 evaluations at each scale. Codes are re-normalized after perturbation; the scale is not their resulting latent RMS error. The original codes and all 1,536 perturbed native outputs are retained and independently replayed.

| Scale | Mean latent MSE | F1 (%) | Exact / 256 |
|---|---|---|---|
| 0 | $1.23 \times 10^{-9}$ | 100.0000 | 256 |
| 0.01 | $1.00 \times 10^{-4}$ | 100.0000 | 256 |
| 0.03 | $8.99 \times 10^{-4}$ | 100.0000 | 256 |
| 0.1 | $9.93 \times 10^{-3}$ | 99.9986 | 253 |
| 0.3 | $8.43 \times 10^{-2}$ | 99.7517 | 73 |
| 1.0 | $5.86 \times 10^{-1}$ | 88.3693 | 0 |

The diagnostic takes 4.577 s on four CPU threads, excluding Python startup. It measures sampled local stability, not worst-case error, intrinsic dimension or semantic sufficiency. In particular, high foreground F1 can conceal a low exact-block rate. Learned prediction errors and free-running history need their own evaluation. For small continuous error $\delta$, the familiar local expansion

$$\|D_w(z+\delta) - D_w(z)\|^2 = \delta^{\mathsf{T}} J_{D_w}(z)^{\mathsf{T}} J_{D_w}(z)\delta + o(\|\delta\|^2) \tag{13}$$

explains why equal latent error can cause unequal pixel damage; it supplies no global guarantee, especially after thresholding.

The implemented conditional-learning stage uses the complete CMRC/Dolly training corpus with detached codec labels, decoded-image loss and balanced structural stopping. Its completed autonomous audit is reported in Supplementary Note 27; no evolving training loss is treated as a language result. Useful language, historical forms, final held-out evaluation and standalone deployment remain unqualified. Inherited semantic pretraining is external.

## Supplementary Note 27: complete instruction learning and autonomous rejection

The separate-space actor starts from the authenticated inherited archive and the original fixed codec, without the failed earlier writer or preflight adaptation. All 27 visual and 36 contextual layers remain present. Only 9,351,937 adapter, query, projection and stop parameters train; the language vocabulary module remains absent. Fused AdamW uses peak learning rate $10^{-4}$, betas $(0.9, 0.95)$, weight decay 0.01, gradient clipping at one, 500-update warmup and cosine decay to one tenth of peak. There is no EMA or development-based checkpoint selection.

Every one of the 12,976 admitted training examples occurs once in the first seeded permutation; 7,024 unique draws follow from a fresh second permutation. The complete run records 20,000 draws: 4,926 Chinese and 15,074 English. Independent CPU checks reproduce all data-order entries, target block counts, sampled block indices, final cursor and data-generator state. The final adaptation equals the update-10,000 checkpoint's trainable tensors. All saved trainable and optimizer values are finite. The registered early production replay independently checks one update's parameters, optimizer and RNG states exactly; the later journal audit does not recompute all 10,000 gradients.

Training-stage time is 16,773.459 seconds, including checkpoint and registered replay work, with separate initialization time 82.717 seconds. Peak allocated memory is 16.965 GiB on physical GPU 0. These costs exclude data

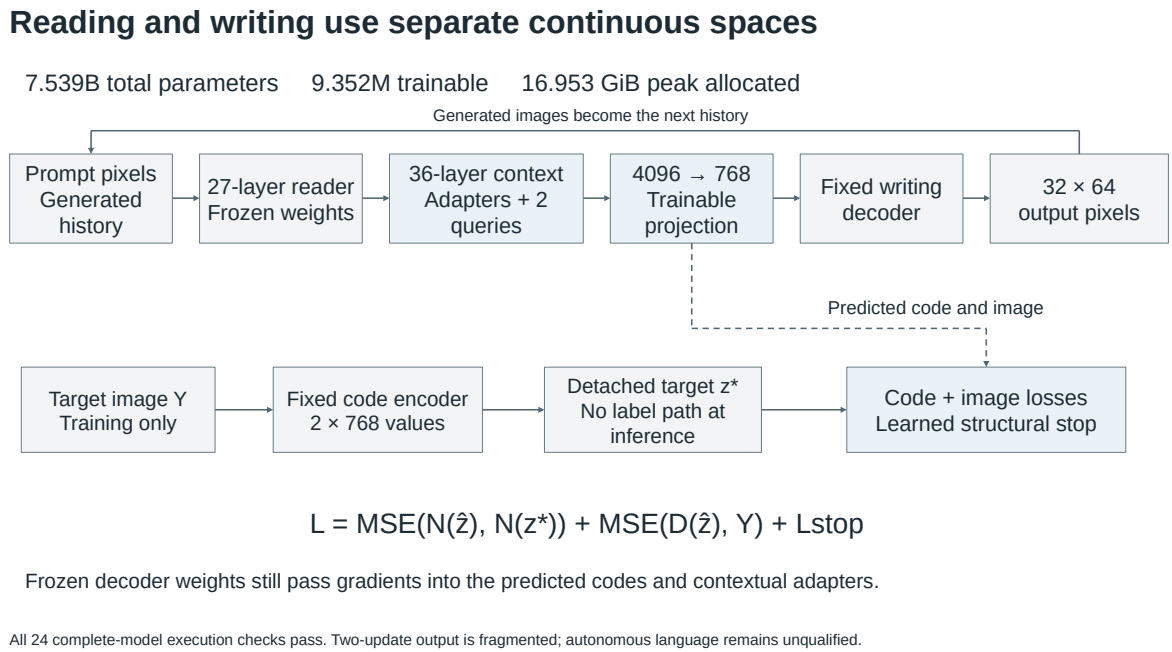



**Fig. S8 | A fixed writing decoder separates physical rendering from conditional code prediction.** Blue modules contain trainable components. Prompt and completed generated images enter the retained reader/contextual body; a small projection produces continuous writing coordinates. The output encoder supplies detached labels only during training. Gradients through the fixed decoder train the predicted codes and contextual adapters. The displayed parameter and resource counts come from actual complete-model execution, without a trained-language claim.

preparation, prior codec learning and external foundation pretraining. They establish local adaptation feasibility rather than foundation training from scratch or a matched efficiency advantage.

The source-frozen development audit contains 128 Chinese questions from 64 distinct two-question articles and 64 English instructions, eight per category. All 384 normal/control streams use a common cap of 128 physical blocks and learned stopping. Generation receives owning prompt pixels alone, without target length, reference codes, source strings or case metadata. Before generation, an input-layout correction removes reflow from 31 Chinese and one English blank control. The corrected controls reuse the normal prompt's exact fold ranges; all normal pixels, cases, references and gates remain unchanged. Independent NumPy reconstruction checks all 384 layouts.

Exact reference agreement requires correct stopping and identical full physical dimensions, including ordinary white padding. F1 counts all predicted/reference ink, with intersections only in the overlapping width; extra or missing ink contributes error. Alternative recorded references are accepted only when fully rendered by the fixed layout. Resampling uses 64 source articles, 2,000 draws and seed 20266831.

| Chinese criterion | Observed | Required | Decision |
|---|---|---|---|
| Exact answers | 0/128 | $\geq 50\%$ | Fail |
| Foreground F1 | 32.671% | $\geq 80\%$ | Fail |
| Both-correct pairs | 0/64 | $\geq 35\%$ | Fail |
| Question F1 gain | 13.163 pp | $\geq 10$ pp | Pass |
| Gain interval lower bound | 10.933 pp | $> 0$ | Pass |
| Learned stopping | 128/128 | $\geq 90\%$ | Pass |

All 192 original first references and 384 raw outputs are independently checked, including input geometry, PNG/tensor agreement, stopping and grouped aggregation. Chinese question-blank F1 is 19.508%; the question effect has interval [10.933, 15.730] pp. English normal F1 is 9.763%, versus 3.963% blank, with no exact recorded reference in either route and normal stopping on 54/64 cases. These English figures are reference agreement, not exhaustive semantic correctness. Generation sums to 783.228 seconds, with 65.599 seconds setup and 851.764 seconds total audit time; inference peaks at 15.685 GiB allocated memory. All data are reused development, the sealed final split remains unopened, and no failed output is repaired or replaced.

## Supplementary Note 28: exact-history prediction remains poor

The frozen final actor is measured on 32 source-bound cases, without optimization: eight Chinese and eight English training examples sampled from sorted admitted files with Python seed 20266840; the first eight Chinese audit cases; and the first lexicographic audit case in each English category. Selection does not use output scores. CPU preparation preserves every prompt/reference pixel, discards only the training cache's two explicit termination blocks, and fixes the sample identities before model calls.

Supplied-code reconstruction encodes and decodes each reference image through the fixed writing codec. Exact-history prediction instead uses the actor's actual cached contextual path, appending a reference block only after its own next-block prediction has been recorded. The driver continues to the reference length even after a predicted stop. This supplies history and length and is an assisted diagnostic. The 16 training autonomous streams use the original uniform 128-block cap; the 16 development streams are reused exactly from the locked audit.

| Group | Supplied codes | Exact history | Autonomous | Code MSE |
|---|---|---|---|---|
| Training, Chinese | 99.947 | 35.554 | 30.005 | 1.0635 |
| Training, English | 100.000 | 21.805 | 8.145 | 1.2776 |
| Development, Chinese | 99.987 | 29.131 | 25.632 | 1.2329 |
| Development, English | 100.000 | 24.444 | 10.542 | 1.2260 |

F1 columns are case averages in percent, with eight cases per row. Exact history includes reference-width assistance, so its advantage over autonomous F1 is not a pure history effect. No first predicted block, or block predicted with exact history, is exact. Initial stop probabilities and first physical blocks match between paths on all 32 identical starting prompts. This localizes a failure preceding generated-history drift, including on training examples, while leaving feature information, readout, loss and optimization as possible causes.

For reference signs $y_i \in \{-1,+1\}$ and supplied code $z^*$, define the decoded margin $m = \min_i y_i D_w(z^*)_i$. A positive margin and $\|D_w(\widehat{z}) - D_w(z^*)\|_\infty < m$ suffice for matching binary pixels. No recorded prediction satisfies this sufficient condition. This numerical comparison on retained scores supplies no global latent-space certificate. Code MSE normalizes both two-vector writing states as in the training comparison.

The diagnostic takes 208.112 seconds including actor initialization and peaks at 15.816 GiB allocated memory. Offline CPU verification reloads all source pixels and recomputes physical scores, margins and aggregation, including exact reuse of development streams. It does not independently replay the native neural forward pass. No optimizer update, OCR, source-model service call, vocabulary inventory or sealed example is used. The original autonomous language decision remains rejection.

## Supplementary Note 29: visual reading through numeric controls

The complete imported vision/contextual body retains its original authenticated Qwen3-VL 8B initialization and zero-output contextual adapters. Its language embedding module is absent. A one-time compiler converts the source chat/task frame into fixed numeric vectors; their deployed payload contains floating vectors only. These controls surround physical visual features and retain the image-specific rotary coordinates and DeepStack features. No source response or development answer enters an optimizer.

The first eight Chinese questions from the existing audit are checked at original scale and with a declared factor-32 observer. The latter uses minimum area 1,048,576 and maximum area 4,194,304, with Pillow RGB bilinear interpolation followed by grayscale conversion. Matched controls erase only question ink while preserving the original wrap coordinates. A fifth route uses matched source text. There are 40 helper generations in total.

| Route | Exact / 8 | Character accuracy (%) | Stopped / 8 |
|---|---|---|---|
| Original image | 1 | 59.721 | 8 |
| Original, question erased | 0 | 6.057 | 6 |
| Native-budget image | 4 | 69.354 | 8 |
| Native-budget, question erased | 0 | 3.561 | 7 |
| Matched source text | 3 | 77.110 | 8 |

Supplementary Fig. S10 separates the measured counts from the symbolic helper boundary. The source vocabulary projection and tokenizer are local variables in the separate diagnostic helper, never registered on the image actor. The helper decodes at most 64 source positions per answer with greedy source termination. Independent CPU verification checks all 32 source/transformed images and replays all 742 retained hidden-state projections. All argmax decisions agree, with no near-tie disagreement; maximum selected-logit error is $1.526 \times 10^{-5}$. This does not replay the entire visual/contextual forward pass. It also does not establish equality to native C++ resized pixels. Generation takes 232.533 seconds including 126.192 seconds of setup, with 18.983 GiB peak allocated memory including the temporary

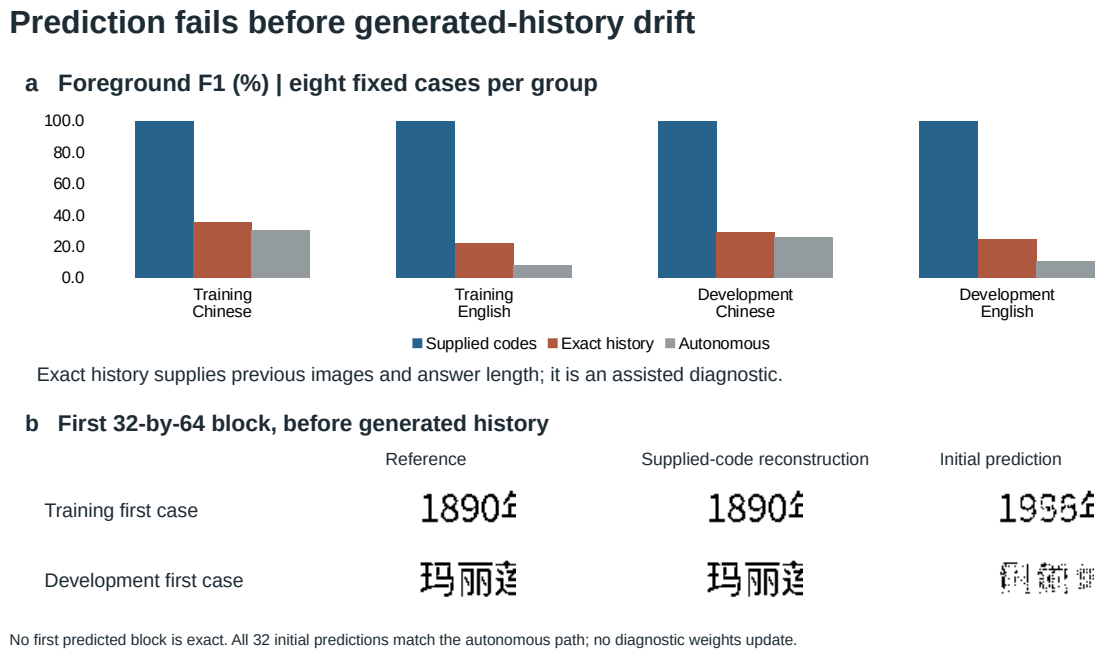



**Fig. S9 | Conditional writing fails with exact reference history. a**, Case-averaged foreground F1 for eight fixed cases in each split/source group. Supplied codes reconstruct the images faithfully, while the actor predicts poor writing even on training cases with correct history. The exact-history route also supplies length and forces continuation. **b**, The first 32-by-64 block of the first fixed Chinese training and development cases, displayed explicitly as block excerpts. Supplied-code reconstruction preserves the source marks; the initial prediction is already incorrect before generated history. All 32 initial predicted blocks match the autonomous path and none is exact. No image is repaired.

diagnostic head; CPU verification takes 38.511 seconds. No optimizer update or image answer is produced. The source-bound receipt is `references/evidence/native_visual_transfer_v71_result.json`.

Literal exact matching penalizes some valid aliases and formatting changes. Reference inclusion is not a semantic correctness measure. The eight reused cases do not support population estimates, and changes in observer scale and task framing are not an isolated causal intervention. The result motivates testing a complete image writer rather than qualifying one.

## Supplementary Note 30: causal joint flow in writing coordinates

Supplementary Fig. S11 shows training and deployment. The completed successor fails the language audit. For physical prompt $X$, let

$$M = \mathrm{sg}(F_{\mathrm{frozen}}(E_r(R(X));C)), \tag{14}$$

$$h_t = T_\theta(M, b, E_w(Y_{<t})), \qquad z_t = \mathrm{sg}(E_w(Y_t)). \tag{15}$$

Here $R$ is the declared observer, $C$ the numeric task frame, $b$ a learned structural start vector, and $M$ all 52 final task-frame contextual states. Memory is retained in BF16 in both caching and inference. The causal writer has width 512, eight layers, eight heads and feed-forward width 2,048. Layer-normalized memory and prior image codes project into its writing sequence with fixed sinusoidal positions. Query $t$ cannot attend to target block $t$ or any subsequent block. There is no character embedding table.

Each $z_t$ comprises two continuous 768-vectors. With independent Gaussian $e$ and uniform time $u$, train the joint velocity law

$$z_u = (1-u)e + uz_t, \tag{16}$$

$$\mathscr{L}_{\mathrm{flow}} = \mathbb{E}\|v_\theta(z_u, u, h_t) - (z_t - e)\|_2^2. \tag{17}$$

The implemented MSE averages coordinates and up to 16 uniformly sampled blocks within each complete example, then averages examples. A balanced continue/terminal logistic loss has equal weight. The velocity head has width 768 and three residual MLP blocks. At inference, 32 explicit midpoint steps produce a joint block code. Non-affine LayerNorm restores the codec's zero-mean, unit-variance coordinates; unit Euclidean normalization would use the

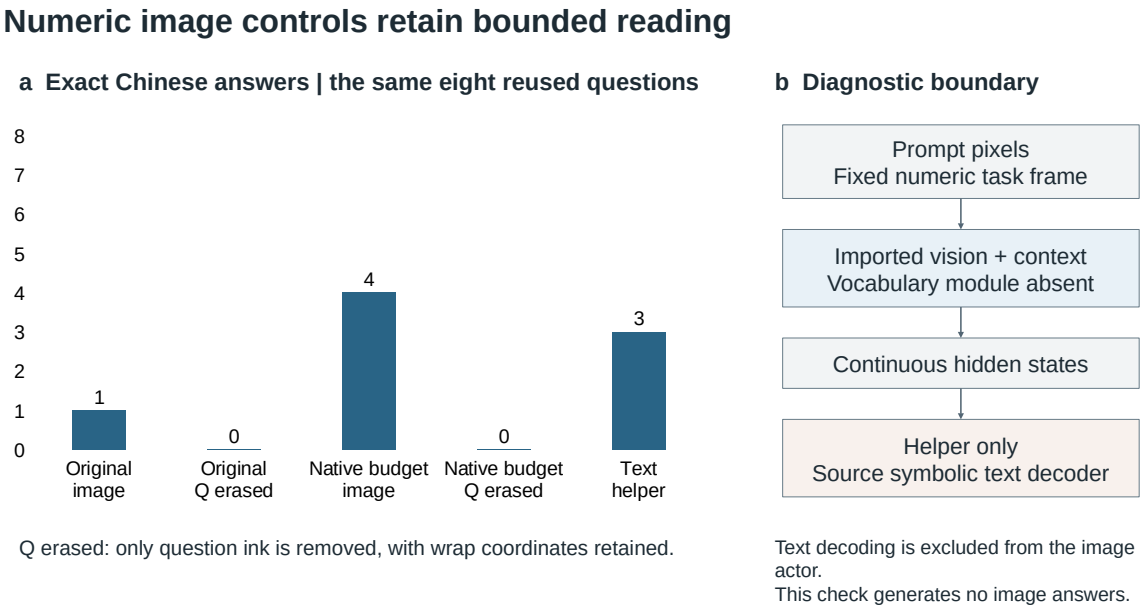



**Fig. S10 | Numeric image controls retain bounded reading in the inherited body.** **a**, Literal exact answers among the same eight reused Chinese questions for original images, matched question erasure, declared native-budget images, their erased controls and a matched source-text helper. All counts use the same denominator; no population accuracy or causal scale advantage is claimed. **b**, The vocabulary-free image interface produces continuous states. Only the separate diagnostic helper applies symbolic source decoding. All 32 source/transformed images and 742 retained projection decisions verify; the complete neural forward pass is not independently replayed. No image answer is generated in this diagnostic.

wrong scale. Physical decoding and feedback are

$$\widehat{Y}_t = \text{threshold}(D_w(N(\widehat{z}_t))), \qquad \widetilde{z}_t = E_w(\widehat{Y}_t). \tag{18}$$

Only $\widetilde{z}_t$, encoded from actual emitted pixels, enters history. The learned stop head uses threshold 0.5 and a uniform 128-block capacity. No reference length, code, OCR, tokenizer, persistent glyph inventory or teacher service enters autonomous inference. External source pretraining and the earlier fixed codec remain part of the provenance.

All 12,976 admitted complete CMRC/Dolly training images are retained. The cache removes only two explicit white termination blocks; ordinary padding and every real answer block remain. Prompt-only memory is computed before answer labels are loaded, and every source/cache payload is hashed. The first 128 examples of a fixed full-corpus permutation supply an execution preflight. Across 33 optimizer updates, future-target exclusion and exact optimizer/RNG/data-cursor resume pass. A separately forced continuation branch verifies re-encoding of actual emitted pixels and receives no quality credit. The 38,651,649-parameter writer peaks at 1.739 GiB in this execution check; this excludes the frozen reader and is not full-model inference memory.

The registered production schedule is 40,000 updates, batch 16, AdamW with learning rate $10^{-4}$, 500-update warmup and cosine decay to one tenth, betas $(0.9, 0.95)$, weight decay 0.01 and gradient clipping at one. New parameters and optimizer state are FP32; decoder contractions use BF16; there is no EMA. Fresh full-corpus permutations support exact resume. This exposure and objective differ from the failed earlier actor, so this comparison cannot isolate one component's contribution. The full original 192-case audit and matched blank controls follow training automatically. Sealed cases remain unopened because the development requirements fail. Training, inference, scoring and independent verification use `scripts/run_semantic_ilm_v72.py`, `scripts/infer_semantic_ilm_v72.py`, `scripts/audit_semantic_ilm_v72.py` and `scripts/verify_semantic_ilm_v72.py`.

All 12,976 examples are cached in 15,428.722 seconds. Training completes 40,000 updates, 640,000 draws and 49 complete corpus passes in 2,510.256 active seconds, peaking at 1.155 GiB allocated memory without the inherited reader. The 384 normal/control autonomous answers take 853.492 seconds, including 79.648 seconds setup, at 18.294 GiB with the reader. Earlier codec learning and external foundation pretraining are excluded.

Chinese original-image exact answers are 0/128, both-correct pairs 0/64, and foreground F1 29.386% versus 21.873% when questions are erased. The question gain is 7.513 points with 95% article-bootstrap interval [5.727,9.631]. Exact answers, absolute F1, both-correct pairs and minimum question gain fail; the positive effect interval and learned stopping pass. All normal Chinese outputs stop through the learned decision. English literal reference F1 is 14.394%

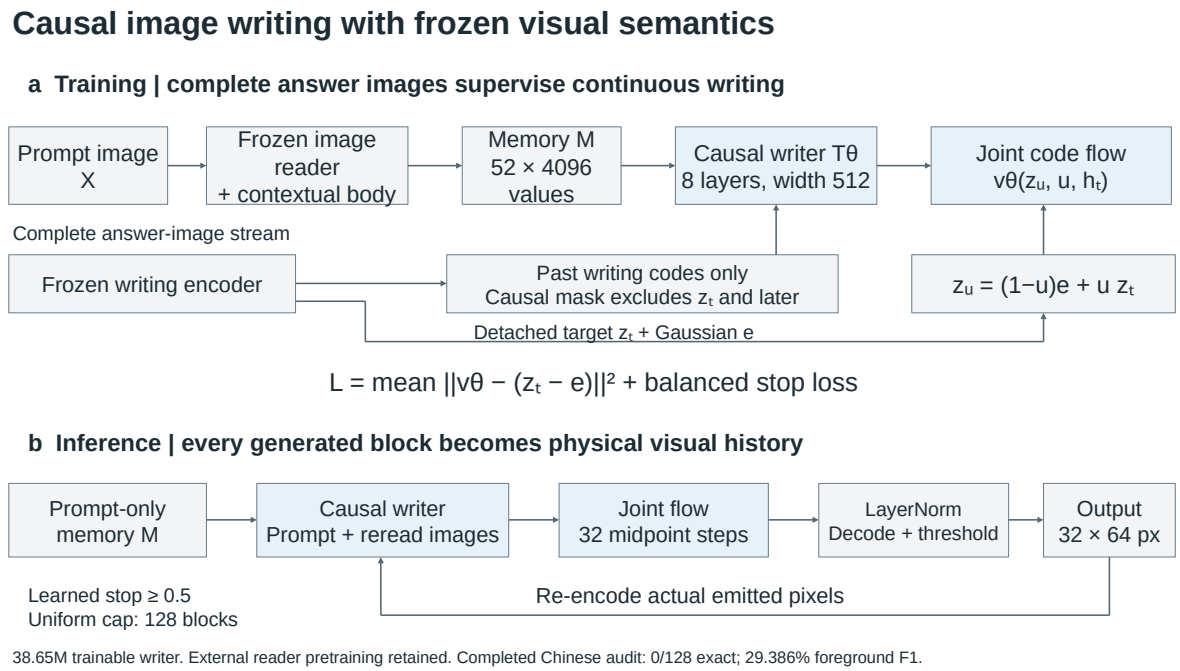



**Fig. S11 | Causal joint flow couples visual semantics to physical writing. a**, Prompt-only frozen contextual states condition a causal writing transformer. Detached image codes supply past history and a joint flow-matching target. The causal mask excludes the current and later target blocks; balanced structural-stop loss supervises termination. **b**, Autonomous inference integrates 32 midpoint flow steps, applies non-affine LayerNorm, decodes and thresholds physical output. Every emitted block is encoded again before entering history. Learned stopping uses threshold 0.5 and a uniform 128-block capacity. The writer has 38.65M trainable parameters; the larger frozen externally pretrained reader remains part of the complete actor. This diagram describes the completed velocity candidate, which fails its autonomous language audit; it contains no successful generated answer example.

versus 7.838%, with 0/64 exact reference images; these references do not exhaust semantically valid English answers. Independent CPU scoring authenticates all 384 raw outputs and 12,976 cache payloads, without replaying full neural generation. File-integrity success does not change the language rejection.

## Supplementary Note 31: a denoising obstruction and clean writing

Let $D = 1536$ and write the original velocity projection as $v = Ah + b$, with $A \in \mathbb{R}^{D\times 768}$. For $S = \mathrm{span}(A, b)$ and the orthogonal projector $Q$ onto $S^{\perp}$, $Qv = 0$. Consequently an exact trajectory obeys $Qz(u) = Qz(0)$ before its final LayerNorm. For an independent standard Gaussian $e$ and clean target $y$, the coordinate-averaged expected squared velocity error has the lower bound

$$\mathbb{E}\|v-(y-e)\|_2^2/D \;\geq\; \big(\mathbb{E}\|Qy\|_2^2 + D - \dim S\big)/D. \tag{19}$$

This concerns a fixed affine velocity output span, not all narrow denoisers or all image-language architectures. It does not prove that removing the obstruction is sufficient for language learning.

The trained BF16-rounded projection has affine rank 769, leaving 767 directions. On the first 128 sorted authenticated cache examples, containing 3,619 blocks and weighted equally per example, the noise-only floor is 0.499349 and target orthogonal energy 0.150753, giving 0.650102 in real arithmetic. A QR-based CPU trajectory check uses the actual trained weights, the first cached prompt, empty writing history and all 32 midpoint steps. Orthogonal-noise RMS changes from 0.699684821 to 0.699684823 in FP32, with drift RMS $1.10\times 10^{-7}$. CPU BF16 gives 0.699627645 to 0.699614454, with drift 0.0002073. Elementwise rounding breaks exact subspace invariance slightly; these are CPU numerical checks, not GPU answer-generation replays.

A fixed known-image control then trains narrow and full-dimensional velocity heads for 4,000 updates, batch 64, using one original CMRC writing block. On 64 fresh noise draws the full head gives 63 exact images, but worst F1 is 97.053%, below the unchanged 99% requirement. A registered precision/ integration diagnosis selects BF16 with 64 steps on those original draws. On one fresh 128-draw confirmation it gives 116 exact images and worst F1 87.607%, again failing. The confirmation is not used to select another setting. These failed outcomes remain part of the evidence.

The clean model instead predicts $d_\theta(z_u, u, h) \approx y$ directly, following clean-data denoising work [19]. Its final projection

executes in FP32, rather than casting a BF16 result after rounding. With $K = 32$, sampling begins at $z_0 = e$ and uses

$$z_{i+1} = \frac{K-i-1}{K-i} z_i + \frac{d_\theta(z_i, i/K, h)}{K-i}, \qquad 0 \le i < K. \tag{20}$$

This integrates the linear contraction exactly if the clean estimate is constant on the interval. The final update has zero old-state coefficient and evaluates no division at $u = 1$. A noise-dependent erroneous clean estimate can still produce an erroneous endpoint; the formula itself is not a fidelity guarantee. Decoding, thresholding and re-encoding actual emitted pixels follow Note 30.

The registered one-image clean control trains a fresh 36,403,200-parameter head for 4,000 updates, batch 64, using the selected original 32-by-64 CMRC block. The sampler receives noise, time and a zero continuous condition; the clean code appears only as a supervised training label. All 256 fresh draws reproduce the original binary image exactly: mean and worst F1 are 100%. Independent CPU scoring and fixed-decoder replay agree for every retained output. The phase takes 33.502 seconds after setup and peaks at 1.361 GiB, excluding earlier codec learning; it loads no inherited reader. The complete causal clean writer has 64,520,193 parameters, but the one-image control trains only its generation head. This result is known-image generation, not conditional reading, generalization or unseen-script writing.

Code and fixed protocols are retained in `scripts/diagnose_flow_rank_v72.py`, `scripts/probe_glyph_denoising_v73.py`, `scripts/probe_flow_precision_v73.py` and `scripts/probe_clean_endpoint_v74.py`.

## Supplementary Note 32: paired learning capacity

A fixed training-only control selects 16 pairs from the admitted CMRC corpus. Each pair has identical visible context but distinct questions, whitespace- normalized reference answer texts and physical target digests. Groups are sorted by visible-context SHA-256 and candidates by source name; the first eligible pair in each of the first 16 groups supplies 32 seen examples. Text and identifiers enter offline selection only, not model inputs.

The complete 64,520,193-parameter clean writer starts afresh and trains for 4,000 updates at batch 16: 64,000 draws and 2,000 complete selected-set passes. AdamW uses learning rate $10^{-4}$, 200-update warmup, cosine decay to one tenth, betas $(0.9, 0.95)$, weight decay 0.01 and gradient clipping at one. Model, sampler and noise seeds are 20267410, 20267411 and 20267412. For the clean predictor, time is $u = \mathrm{sigmoid}(-0.8 + 0.8g)$, $g \sim \mathcal{N}(0,1)$, with the clipped velocity-weighted clean objective

$$\mathcal{L}_{\mathrm{clean}} = \mathbb{E} \frac{\|d_\theta((1-u)e + uy, u, h) - y\|_2^2 / D}{\max(1-u, 0.05)^2}. \tag{21}$$

Coordinate loss is averaged equally per example over at most 16 sampled blocks; balanced continue/terminal logistic loss is added. The numerical body, FP32 projection, physical feedback and sampler follow Note 31. Future-target exclusion, exact optimizer/RNG/data-cursor restoration and a forced physical-feedback execution branch pass before production. All model and optimizer states and random streams are reset after that check.

A fresh inference process receives actual prompt pixels for all 32 questions, using seed 20267420 plus pair index, common within each pair. No answer length, target code, character identity, OCR or teacher is supplied. All actual reader memories exactly match their training-cache counterparts. Generation finishes 29/32 original answer images exactly, with mean F1 99.9967% and worst F1 99.9518%. Against frozen supplied-code reference images 31/32 are exact and 15/16 pairs have both answers exact. These exact counts use different image references and must not be conflated. Every answer stops by its learned decision. Mean owning-target F1 minus the partner target's F1 is 77.6155 points. Cross scoring does not add model calls.

The six fixed requirements are mean original F1 at least 99%, every original F1 at least 98%, supplied-code exact fraction at least 90%, both-exact supplied-code pair fraction at least 87.5%, learned stopping at least 95%, and owning-minus-partner F1 at least 10 points. All pass. Training takes 174.653 active seconds at 1.163 GiB without the inherited reader. Actual-image generation and evidence capture take 108.558 seconds, including 65.923 seconds setup, at 17.742 GiB with the reader. Earlier codec learning and external foundation pretraining are excluded.

Independent CPU verification checks source input pixels, every raw output, frozen-decoder references and physical scores. It does not replay full neural generation or every feedback encoding. The model has repeatedly seen every question; fitting them establishes learning capacity, not general reading, language knowledge or unseen-form generation. The subsequent full-corpus training and original full-answer audit are reported in Note 33. The protocol, runner and standalone image interface are `references/clean_learning_v74_protocol.md`, `scripts/run_clean_learning_v74.py` and `scripts/infer_clean_ilm_v74.py`.

## Supplementary Note 33: full clean writing and exact-history failure

The full-corpus successor initializes all 64,520,193 writer parameters afresh, independently of the fitted 32-question control. It reuses the authenticated prompt-only feature cache for 3,193 CMRC and 9,783 Dolly examples, the frozen

image/contextual reader and the fixed writing codec. Training completes 40,000 updates, 640,000 draws and 49 complete corpus passes. Batch size is 16; at most 16 blocks per example contribute equally weighted clean-code error. The logit-normal time law and clipped weighting follow Note 32. AdamW uses peak $10^{-4}$, warmup 500, cosine decay to one tenth, betas $(.9,.95)$, decay .01 and clipping one. Seeds for model, sampler and noise are 20267500, 20267501 and 20267502. The body uses BF16 and the final writing projection executes in FP32. Execution-only preflight verifies future-target exclusion, exact optimizer/RNG resume and physical feedback; its weights are discarded.

The final audit retains all 384 autonomous normal/question-erased outputs from the original 128 Chinese and 64 English requests, with seed 20267503 plus case index, shared by each normal/control pair. Only actual prompt pixels enter the actor. It receives no target length, codes, character IDs, OCR, language vocabulary head, glyph inventory or teacher service. The unchanged 32-step clean sampler re-encodes each emitted binary image for history, and the learned stop threshold is .5 with a 128-block safety cap.

Chinese original-image F1 is 37.631%, versus 23.609% under question erasure; exact answers are 2/128 versus 0/128. Both answers are exact in 0/64 shared- passage pairs. The original minimum exact, absolute-F1 and both-correct-pair requirements fail. Question gain is 14.023 points, with source-article bootstrap interval [10.861, 17.487]; all normal Chinese outputs stop by the learned rule. The original gain, positive-interval and stopping requirements pass. English literal-reference agreement is 0/64 exact and 16.098% F1; the references do not enumerate every semantically valid English answer. These are reused development examples, not sealed confirmation.

A separate author-specified copy diagnostic fixes 12 new requests: four Chinese, four English and four six-digit strings, arranged in six pairs. Their exact prompt rasters are absent from the training inventory. All 12 actual-image calls stop, but none matches its requested output exactly. Foreground F1 is 26.989%, 15.925% and 24.543% in the three groups, respectively. Owning-minus-partner reference F1 is only .123 points overall. This small diagnostic is descriptive, not a new population benchmark.

The same 32 training cases as Note 32 then diagnose the completed full-data model. First, perform actual-image free generation with seed 20267420 plus pair index. Only after each free call, use the true answer history and known length to construct an assisted stream. A separate fixed-decoder control receives the actual target codes. Free and assisted first blocks are equal in all 32 cases, and all actual reader memories exactly match the cache.

| Training-case route | Exact streams | Ink F1 (%) |
|---|---|---|
| Free generation | 0/32 | 45.771 |
| Correct history and length | 0/32 | 58.045 |
| Supplied writing codes | 28/32 | 99.9953 |

Assisted and supplied-code counts measure physical stream agreement at provided length, not learned stopping or autonomous answers. Free first-block F1 is 59.929%. These training failures precede corrupted long histories. The earlier 32-case fit differs in exposure and initialization; the result does not identify one unique cause of the full-data failure.

A retained first-block probe uses $z_t = (1-t)e + ty$ with the true target $y$, noise seed 20267597 and the owning prompt condition. At $t = 31/32$, the direct constructed input decodes to all 32 original blocks exactly. Denoiser prediction reduces exactness to 6/32, at 99.1349% F1. These are assisted near-clean probes, not free sampler states; their small stroke errors cannot by themselves explain the much larger free-generation deficit. At $t = 1$, the clean predictor still changes supplied codes. From complete sampler metadata and authenticated answer lengths, training sampled 7,007,799 blocks. Under its declared time law,

$$\Pr(t \geq 31/32) = 1 - \Phi\left(\frac{\log 31 + 0.8}{0.8}\right) = 6.0333 \times 10^{-8}. \tag{22}$$

The corresponding expected exposure is .423 blocks. This is a theoretical expectation; the actual original tail draws have not been replayed.

Full training takes 2,589.452 active seconds at 1.544 GiB without the inherited reader. The 384-output audit takes 679.857 seconds, including 72.053 seconds setup, at 18.390 GiB including that reader. Original feature caching, earlier codec learning and external foundation pretraining are excluded. Independent CPU verification authenticates all source/output files, reconstructs source references and recomputes physical metrics and original gates. The training diagnosis re-decodes all 32 supplied-code streams and recomputes retained latent errors. It does not replay the full neural actor. All failed outputs remain unchanged; sealed data stays unopened. The full-data, copy and diagnosis receipts are `references/evidence/clean_corpus_v75_result.json`, `references/evidence/copy_transfer_v75_result.json` and `references/evidence/corpus_fit_v75_result.json`.

## Supplementary Note 34: broader time coverage does not repair answers

A matched intervention starts both arms from the completed clean full-data checkpoint. Only the existing 36,403,200-parameter flow head trains; the causal body, condition projection, stop head and codec remain fixed. Each arm uses

all 12,976 training records for 4,000 updates at batch 16, with a fresh AdamW optimizer, peak $10^{-4}$, warmup 100, cosine decay to one tenth, betas $(.9, .95)$, decay .01 and gradient clipping one. Sample/noise seeds are 20267601 and 20267602. The control retains $t = \text{sigmoid}(-.8 + .8g)$; the intervention uses $t = \Phi(g)$, clamped to $[10^{-6}, 1 - 10^{-6}]$. The same Gaussian variates, noise, example indices and selected blocks are used. Nine retained common-draw digests agree exactly across arms. Both sample 700,759 writing blocks; the control realizes zero times at or above $31/32$, versus 22,067 for the uniform law. Both time exposure and the integrated objective weighting change; this is not an isolated sampler test.

Each final head is assessed on the same 32 training questions using their verified prompt-derived memory. These are cached-feature calls, with no new inherited-reader execution. Free inference receives no answer length or target codes and retains the original sampler, physical feedback, learned stopping, cap and paired seeds. The separate near-clean diagnostic receives noisy target codes and earns no autonomous credit.

| Measure | Logit-normal | Uniform time |
|---|---|---|
| Free exact answers | 0/32 | 0/32 |
| Free ink F1 (%) | 48.726 | 45.028 |
| Learned stopping | 32/32 | 32/32 |
| Near-clean exact blocks | 5/32 | 15/32 |
| Near-clean ink F1 (%) | 99.0658 | 99.8602 |

Uniform-time refinement therefore improves target-assisted high-signal reconstruction but lowers free-answer F1 by 3.698 points. It fails every prespecified training-fit repair criterion: near-clean F1 at least 99.9%, free F1 at least 80%, at least 16/32 exact free answers and free F1 gain of at least five points over control. Neither head advances as a qualified model. This failure rejects the bounded intervention, without identifying all causes or ruling out different objectives, capacity or training budgets.

The two training stages take 157.554 and 153.230 active seconds, respectively, excluding previous writer training, feature preparation, codec learning and external foundation pretraining. A discarded real-example execution check verifies exact optimizer/RNG/cursor resume and frozen-body preservation. Independent CPU verification reconstructs source references, authenticates all 64 retained free streams, re-decodes near-clean predictions and checks the frozen-body and common-draw hashes. It does not replay the whole neural sampler. This experiment uses training examples and makes no unseen-language claim. The protocol and result are `references/writing_time_coverage_v76_protocol.md` and `references/evidence/writing_times_v76_result.json`.

## Supplementary Note 35: content before the image writer

The failed full-corpus image writer is compared with a separate offline source decoder on exactly the same 12 copy requests and 32 fixed training questions. Every source prompt and canonical reference re-renders to the original physical strips exactly. The image helper receives the same folded prompt pixels and the unchanged native-budget observer and numeric task controls. The normal-image 52-state BF16 suffix exactly matches the original ILM reader memory for all 44 cases. The first helper projection uses precisely its last stored BF16 vector, without additional FP32 prefill precision.

Each case has a normal-image and source-text helper route. The 12 copy requests also have fully blanked prompt-image controls at the same geometry, giving 100 fixed calls. Text controls use the owning source prompt and do not append an answer. The source helper retains the frozen imported image/context body with zero adapter outputs and no registered vocabulary embedding module. Only this separate evaluator uses the authenticated source tokenizer, token row lookup and temporary output projection. It performs greedy decoding with a 64-source-token cap and authenticated source end IDs. Seed 20267710, four CPU threads, deterministic cuDNN/math attention and no TF32 are fixed. The temporary head is stored in BF16 and contracted in FP32. No optimizer updates, new independent image-actor calls or sealed-data access occur.

The image helper returns 8/12 exact copy answers, versus 6/12 for source-text controls and 0/12 with all prompt ink blanked. All 12 normal copy responses contain the owning reference while excluding the paired reference. Four are not exact because additional material, including quotation marks, is retained rather than removed by scoring. On the 32 training questions, the image helper returns 19/32 exact canonical answers, versus 24/32 for source-text controls. Owning-reference inclusion is 22/32 and 25/32, respectively. All 100 responses stop by source end IDs. Exactness requires termination and canonical-answer equality after outer whitespace trim; inclusion is a separate string measure, not semantic accuracy. The original image writer remains at 0/12 and 0/32.

On 22/32 training-image calls the first decoded source piece is a nonempty prefix of the owning reference and not of the paired reference. This first decision uses a vector already present in the compact ILM memory. It identifies some available answer information, without proving complete answer sufficiency. Subsequent source decoding retains the complete layerwise visual/context KV cache, whereas the image writer sees only 52 final-layer states. Correct full helper answers therefore cannot be attributed solely to the compact memory. The cases are reused diagnostic/training

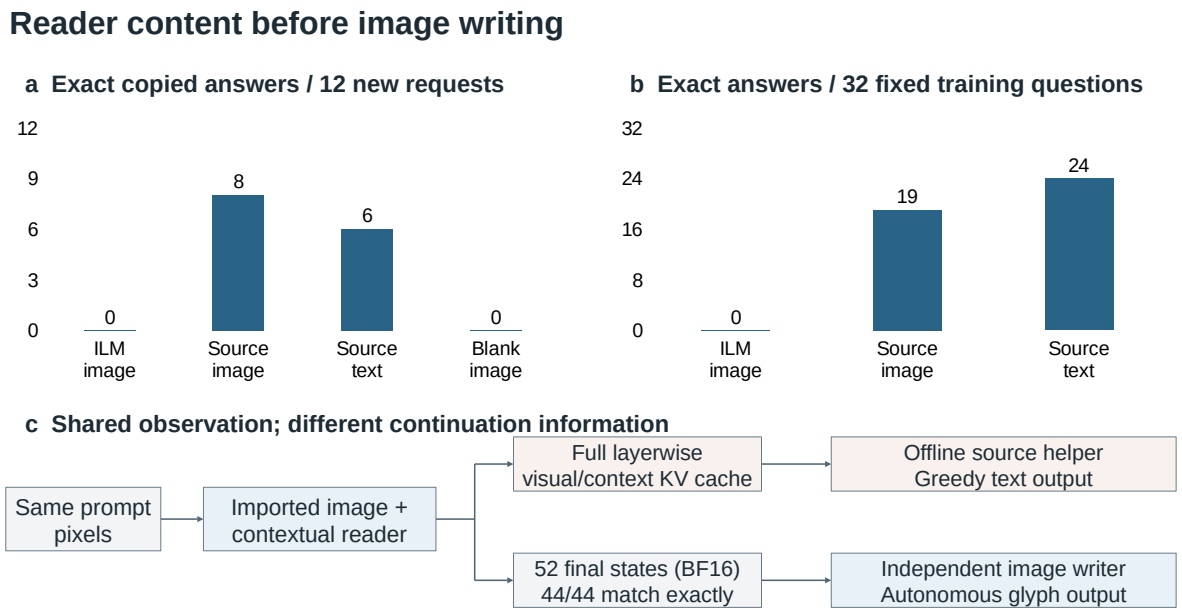



**Fig. S12 | Source-reader content and the image-writing boundary. a**, Literal exact counts among the same 12 copy requests: original ILM image output, source image helper, source text helper and fully blanked image helper. **b**, The same comparison on 32 fixed training questions, without a blank route. The original image scores are unchanged physical-raster counts. The source decoder uses outer-whitespace-trimmed canonical text equality, preserving extra quotations and explanations. **c**, All 44 normal-image suffix memories match the original BF16 bytes, but continuation information differs: the source text helper retains full layerwise KV history, whereas the image writer receives 52 final-layer states. No successful ILM image generation or compact-memory sufficiency is inferred. Both count charts and the information diagram are native editable PowerPoint objects.

examples and may have been exposed to source pretraining; no new population language or learning claim follows.

Independent CPU checks reconstruct all 56 physical/transformed image inputs, recompile controls and authenticate every retained helper response. All 725 source projection decisions replay as exact argmax from retained hidden states, within the fixed selected-logit and regret tolerances. The complete vision/context forward pass is not independently replayed. The helper stage takes 282.212 seconds, including 134.073 seconds setup, at 19.099 GiB allocated memory on GPU 0. External pretraining, earlier feature preparation, codec learning and ILM training are excluded. This is a capability diagnostic with different decoding information, not an efficiency comparison. The fixed protocol, raw-answer receipt and entry point are `references/reader_content_v77_protocol.md`, `references/evidence/reader_content_v77_result.json` and `scripts/probe_reader_content_v77.py`.

## Supplementary Note 36: context visibility and native image continuation

The source-reader intervention repeats the 44 normal-image cases from Note 35 after identical prefill. All existing cache positions and native rotary coordinates are retained, but subsequent queries can attend only to the last 52 task-frame keys and newly generated states. The first decision is supplied by the same stored BF16 state. Thus the intervention changes key visibility, without physically deleting cache entries or moving the remaining keys.

| Exact stopped helper answers | Full prompt KV | Suffix-visible KV |
|---|---|---|
| Copy requests ($n = 12$) | 8 | 0 |
| Fixed training questions ($n = 32$) | 19 | 2 |

All 44 normal-image memory suffixes and first decisions match their full-cache counterparts. Replacing the excluded keys and values with fixed poison values leaves the masked next state identical. Allowing attention to those poisoned keys changes the state, providing a positive control for actual mask exclusion. The unpoisoned full-cache state matches its retained source reference exactly. All 1,239 retained source projection choices replay as exact CPU argmax. This verifies the recorded projection decisions and mask fixture, not an independent replay of the entire visual/contextual model. The suffix occupies 15,335,424 bytes across 36 layers. All eight full-correct copy responses and 17 of 19 full-correct training responses are lost. The experiment rejects this frozen suffix interface; it does not prove that a

trained compressor must fail.

The successor retains the complete native prompt KV, frozen inherited body and qualified image codec. Let $E$ encode a physical answer block $Y_j$ into two 768-dimensional continuous tile vectors. A learned image encoder $A_\phi$ places those two observations into the frozen native contextual body $F_\psi$. With prompt keys and values $K(X)$, the state after $j$ completed blocks is

$$h_j = F_\psi\big(K(X), A_\phi(E(Y_1)), \ldots, A_\phi(E(Y_j))\big)_{2j}.$$

The first block uses the original initial reader state $h_0$, without another cache query. Later blocks use the complete retained context. The causal mask excludes future answer observations and right-padded prompt keys while each example keeps its original rotary offset. The learned clean predictor outputs all 1,536 coordinates; a separate learned stop determines variable length. Decoded, thresholded physical pixels are re-encoded for subsequent inference. No tokenizer, character inventory, language output projection, OCR or teacher service enters this deployed image actor. External source pretraining remains.

For $z = E(Y_j)$, the new clean prediction objective draws $z_t = (1-t)\varepsilon + tz$, $t \sim \mathscr{U}(0,1)$, and minimizes the mean coordinate error divided by $\max(1-t, 0.05)^2$. Its 32-step sampler uses

$$z_{k+1} = \frac{r-1}{r} z_k + \frac{1}{r} C_\theta(z_k, k/32, h_j), \qquad r = 32-k.$$

The last step removes the explicit prior-noise coefficient, not model error. Neither this objective nor codec fidelity proves conditional language learning. In particular, a correct first source piece need not imply that $h_0$ suffices for a full multi-character image block. First-block and full-stream quality therefore remain distinct requirements.

Actual execution checks traverse all 36 native layers and yield zero forward error against the retained reference, finite nonzero input gradients and no change in frozen parameters or prefix KV. Checkpointed training gradients, causal exclusion, padded-prefix equivalence and incremental continuation pass; the maximum incremental-state difference is $8.40 \times 10^{-5}$ against a $10^{-4}$ tolerance. Two untrained generated blocks are reread as actual images. The complete model contains 7,582,980,210 parameters, including 46,899,713 new trainable image-interface parameters. The execution peak is 16.638 GiB.

A supplemental verifier authenticates all retained execution artifacts, reproduces the original two GPU feedback hashes exactly, and compares retained GPU codec vectors with independent CPU encodings. Maximum absolute difference is $2.98 \times 10^{-6}$ against a declared $10^{-4}$ tolerance. Numerical CPU/GPU agreement is distinguished from bitwise same-device identity. Checks of retained native tensors do not independently replay the full 36-layer forward/gradient computation. Original execution and verification errors remain in the source evidence record. No optimizer updates occur in these checks, and no trained image-language result follows from them.

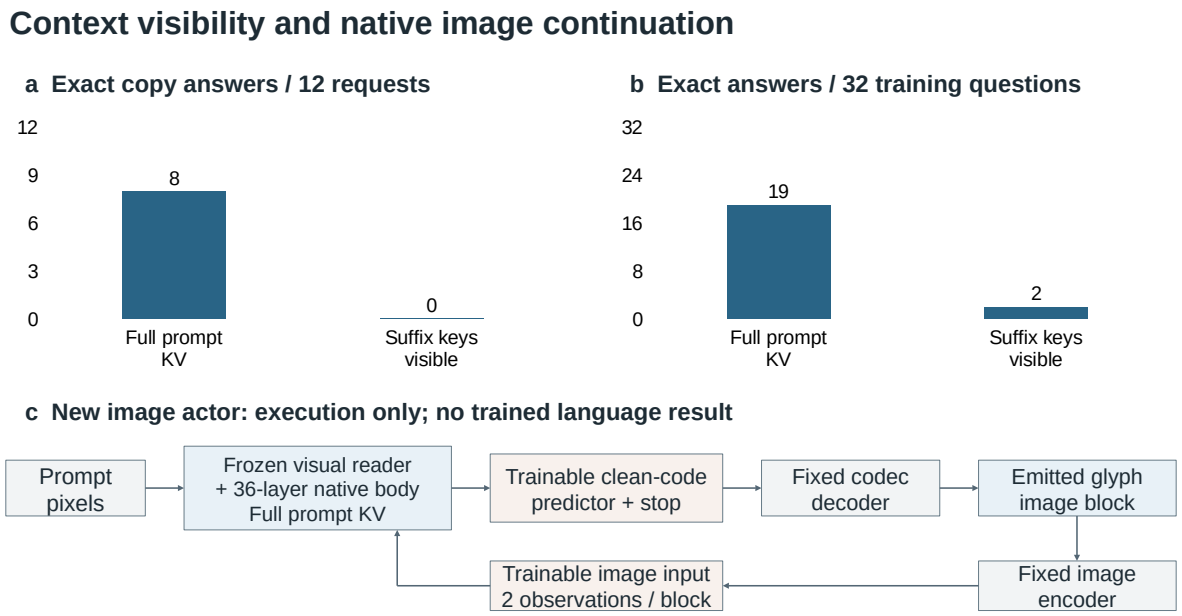



**Fig. S13 | Context visibility and an untrained native image actor. a,b**, Exact canonical stopped source-helper answers with full prompt KV or only suffix keys visible, on the same 12 copy requests and 32 training questions. Original positions, first decisions and BF16 prefill memories are unchanged. These are offline source-decoder diagnostics, not image actor language scores. **c**, The successor retains the externally pretrained native body and full context, learns continuous image input, writing and stopping, and rereads emitted pixels. It has 7.583B complete parameters and 46.900M trainable interface parameters. The first output block still uses the initial reader state. The 16.638 GiB measurement qualifies actual execution, with zero optimizer updates, not language quality. Both charts and the diagram are native editable PowerPoint objects.

## Supplementary Note 37: completed native training and prediction failure

The native-context actor in Note 36 subsequently completes the declared 40,000-update Chinese optimization. All 3,193 admitted CMRC training records are used, with 160,000 example draws and 10,016 actual prompt prefills. The complete trained actor contains 7,583,221,875 parameters, including 46,899,713 trainable image-input, writing and stopping parameters. The inherited visual and contextual weights remain frozen and externally pretrained. This total describes the complete trained actor; the earlier execution-fixture count in Note 36 remains a distinct historical measurement. Active training time is 45,426.51 s, with 18.298 GiB peak allocated memory on GPU0.

The original complete image/code cache, codec and 32-step clean-code sampler are unchanged. Training uses batch four, AdamW $(0.9, 0.95)$, weight decay 0.01 and gradient norm clip one. Learning rate warms linearly to $10^{-4}$ over 500 updates and decays by cosine to $10^{-5}$. A shuffled source sampler selects 32 prompt images per CPU RAM pool. Each is prefilled once, checked against its original prompt-only memory, and reused for 128 updates with a separately shuffled batch sampler. The final pool contributes 64 updates. This bounded-memory reuse is not independent sampling from the whole corpus. Only true preceding image codes enter the causal training history; the current and future answer blocks are excluded from their own conditions. Inference re-encodes the actual emitted pixels. Internal white answer regions are retained, and only the two structural white termination blocks are removed.

The original fixed development audit generates 384 streams: normal and question-erased versions of 128 Chinese and 64 English questions. The control uses the same random numbers and the normal prompt's layout. English examples are excluded from this optimization and are reported as untrained transfer.

| Question images | Exact images | Foreground F1 (%) | Stopped (%) |
|---|---|---|---|
| Chinese normal ($n = 128$) | 0 | 36.504 | 100 |
| Chinese question erased ($n = 128$) | 0 | 25.404 | 100 |
| English normal ($n = 64$) | 0 | 5.861 | 100 |
| English question erased ($n = 64$) | 0 | 5.965 | 100 |

The Chinese both-correct count is 0/64 question pairs. Mean question gain is 0.111009, with 95% article-bootstrap interval $[0.085778, 0.140177]$, using 64 articles, 2,000 resamples and seed 20266831. The six unchanged gates require exact-image rate at least 0.5, foreground F1 at least 0.8, both-correct rate at least 0.35, question gain at least 0.1, a positive lower gain bound and stopping rate at least 0.9. The first three fail; the remaining three pass. The model therefore fails qualification. Stopping does not establish the correct answer length, and question sensitivity does not establish a correct answer. This development set has been reused; no sealed evaluation examples are opened by this audit. Exact raster agreement and aligned ink overlap are not independently measured semantic accuracy.

The fixed-cohort diagnosis reuses all 32 training questions in the same sixteen distinct-answer, same-passage pairs as Note 33. Actual prompt-prefill memories and initial states match the authenticated training cache. Each free call receives only its prompt prefix. Targets become available to the correct-history and supplied-code controls after that free call finishes.

| Path on the 32 training questions | Exact images | Foreground F1 (%) |
|---|---|---|
| Autonomous writing | 0 | 46.235 |
| Correct history and reference length supplied | 0 | 50.241 |
| Target codes supplied to the fixed decoder | 28 | 99.9953 |

Free first-block F1 is 71.544%. The first blocks in free and correct-history paths are identical in all 32 cases, before their histories differ. All free streams stop; all 32 reference-history stop traces agree with the supplied continuation and terminal positions. Correct-history images receive no autonomous-answer credit. Architecture and training histories differ from the earlier writer, so matching this cohort is not a single-factor causal comparison. The poor correct-history predictions locate a failure in conditional code prediction under the tested representation and training procedure, without selecting a unique architectural or optimization cause.

The first-block conditioning probe keeps the target and one noise draw fixed while swapping the owning question state for its same-passage partner. At $t = 0$, target-code MSE is 0.78324 with the owning state and 1.38188 with the partner; at $t = 1$, it is 0.03024 and 0.08427. Constructed target-assisted inputs are not free sampler trajectories. Conditional sensitivity therefore coexists with inaccurate sampled writing.

Independent CPU checks reconstruct all 384 development prompt layouts and rescore the emitted pixels. Aggregation shares the producer's statistical implementation. The 32-case verification independently reconstructs prompts, replays all supplied-code decodes, decodes retained correct-history coordinates and recomputes image agreement. It does not independently replay the complete inherited contextual forward pass. The original diagnostic launch error is retained and contributes no prediction. Source hashes, every per-case score and all 32 unaltered comparison sets remain in `references/evidence/native_chinese_v80_result.json` and `references/evidence/native_corpus_fit_v80_result.json`.

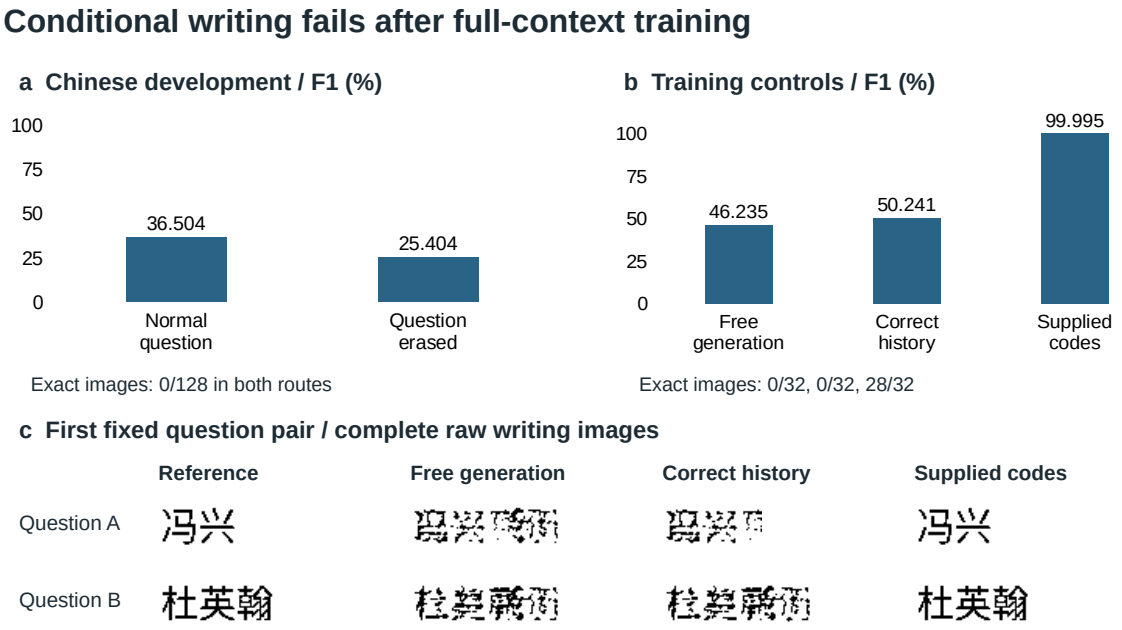



**Fig. S14** | **Full-context training leaves a conditional writing failure. a**, Foreground F1 for 128 Chinese development questions with normal or erased question ink. Both paths have zero exact reference images. **b**, The same 32 training questions under free generation, correct-history prediction with supplied answer length, and supplied-code reconstruction. The exact counts are 0/32, 0/32 and 28/32. Correct-history images are controls, not autonomous answers. **c**, The first question pair in the fixed cohort, showing complete raw references, free answers, correct-history outputs and supplied-code reconstructions. All 32 cases are retained in the source receipt; no output repair is applied. Charts, labels and image objects are editable in the matching PowerPoint. The measurements establish a failed image writer, not semantic accuracy or general language capability.

## Supplementary Note 38: matched readout repair remains unsuccessful

The conditional-prediction failure in Note 37 motivates a matched readout test on the same 32 training questions in sixteen distinct-answer, same-passage pairs. Both arms restore the completed native image interfaces. The control continues the frozen native transition. The adapted arm adds one continuous initial query and trains rank-8, alpha-16 adapters on the query, value and output projections in all 36 inherited contextual layers. This is a composite intervention. Adapter effects are enabled inside writing-layer calls and their checkpoint recomputation; prompt prefill uses the unchanged base weights. Each prompt KV cache remains fixed through optimization.

Both arms receive 2,000 updates, batch four, the same shuffled case order and noise draws: 8,000 draws or 250 exposures per question. Fresh AdamW uses $(0.9, 0.95)$, weight decay 0.01 and gradient clip one. Learning rate warms to $10^{-4}$ over 100 updates and decays by cosine to $10^{-5}$. The objective, uniform noise-time law, 32-step sampler, 0.5 learned stop and 128-block cap are unchanged. Every emitted image is re-encoded for actual-pixel feedback. Autonomous generation receives no target code or reference length.

| Writing path | Control F1 (%) | Adapted F1 (%) | Exact images |
|---|---|---|---|
| Autonomous generation | 71.177 | 72.748 | 0/32; 0/32 |
| Correct history and length | 77.372 | 79.629 | 0/32; 0/32 |
| Supplied target codes | 99.9953 | 99.9953 | 28/32; 28/32 |

All autonomous streams stop. First-block F1 is 99.330% in the control and 98.174% after adaptation. Later writing still deteriorates under correct history, so generated-history errors are not the sole explanation. The provisional repair criteria require at least 24/32 exact free images, free F1 at least 0.98 and improvements over control in both quantities. Only the F1 improvement criterion passes. These are training-fit measurements, not an unseen-language qualification or an independently measured semantic score. The result does not uniquely select a representation, optimization or loss mechanism as the remaining cause.

The adapted actor has 7,583,225,971 complete parameters, with 53,096,961 trainable. The control has 7,583,221,875 and 46,899,713 respectively. External foundation pretraining and prior codec learning are outside these local costs. The added query accounts for the 4,096 additional complete parameters; the adapters reuse already allocated tensors. No lightweight-from-scratch training or efficiency advantage is established.

The original execution completes both optimizations but stops final evaluation at adapted case 00: its first autonomous and full-history pixels are identical, while an exact floating stop-probability check rejects a $3.908 \times 10^{-14}$ difference. That failed execution is preserved. A separately specified evaluation restores the same final checkpoints with zero optimization and uses the same incremental start for free and correct-history controls. All 32 prior control autonomous streams and the one retained adapted stream replay exactly. The common incremental first pixels and stop are identical in all cases. The retained full-history diagnostic gives F1 77.358% and 79.632%; its pixels differ from incremental control in 50 and 42 positions over 107 blocks per arm. Existing normwise hidden-state bounds pass, while the original absolute $10^{-4}$ comparison fails on two control cases and one adapted case. These discrepancies are reported explicitly and do not account for the large quality shortfall.

CPU verification independently scores all 64 final autonomous answers, decodes 128 retained control-coordinate streams and 64 supplied-code streams, and checks owning prompts, targets, raw pixels, PNGs and source identities. The complete contextual body and sampler are not independently replayed. Evaluation, including model setup and prompt prefill, takes 221.75 s and peaks at 17.675 GiB allocated memory on GPU0; these are not training costs. The source receipt and all 320 unaltered PNGs are retained under `references/evidence/adaptive_readout_v81_result/`.

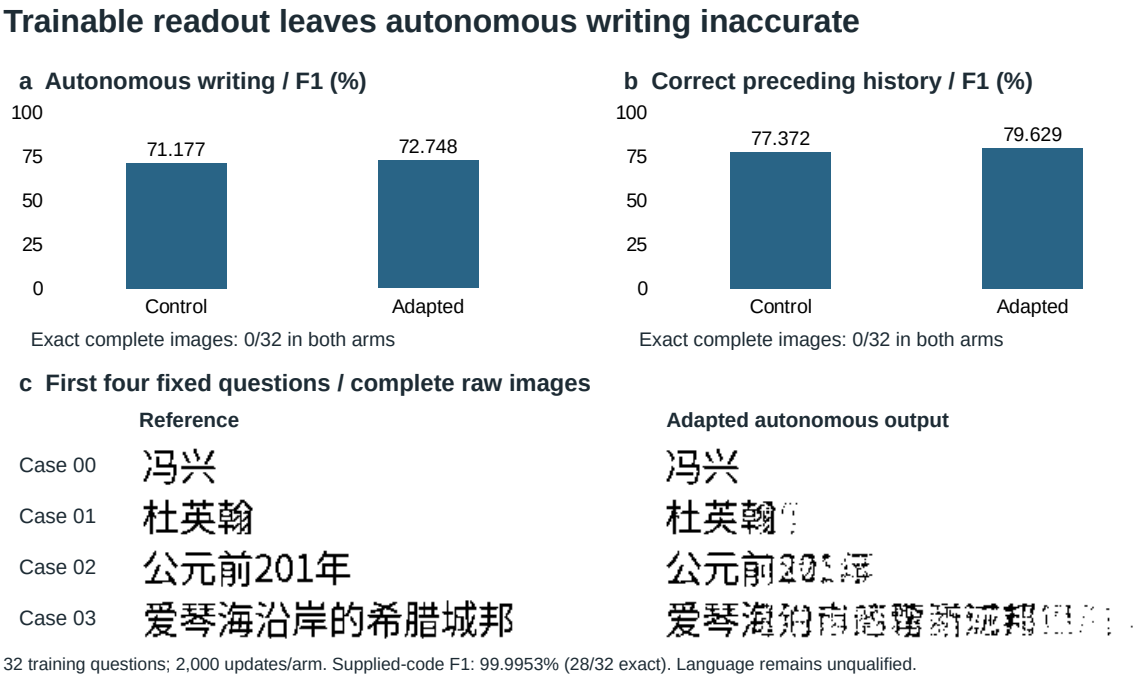



**Fig. S15** | **Readout adaptation fails the autonomous writing requirement. a**, Foreground F1 on the same 32 training questions after matched continued optimization. **b**, The correct-history and known-length control on those questions. Neither path has any exact complete images in either arm. Supplied-code reconstruction remains at 99.9953% F1 and 28/32 exact images. **c**, Complete raw references and adapted autonomous outputs for the first four fixed cohort questions, covering two question pairs without selection by outcome. The original PNG bytes are embedded with unchanged aspect ratios; all 32 cases are retained in the receipt. Native charts, labels and image objects remain editable in the matching PowerPoint. Image agreement does not establish semantic accuracy or successful general language generation.

## Supplementary Note 39: decoder-sensitive loss worsens writing

The accurate supplied-code control in Note 38 motivates testing whether the regression geometry underweights physically important writing errors. Let $z \in \mathbb{R}^{1536}$ be the two-tile target code, $x_t = (1-t)\varepsilon + tz$, and $c_\theta(x_t, t, h)$ the clean-code prediction under prompt/history state $h$. The frozen map $R$ applies the sampler's per-tile normalization and decodes 2,048 image logits. The treatment replaces code MSE by

$$\ell_\alpha(c, z; x_t) = \frac{\|c-z\|^2}{1536} + \frac{\alpha}{2048}\|J_R(x_t)(c-z)\|^2.$$

The Jacobian is evaluated at the observed, detached noisy code, not at the target or predicted endpoint. Its product with the residual is computed without constructing the full matrix. The time weighting $\max(1-t, 0.05)^{-2}$, structural stopping loss and autonomous sampler are unchanged. For fixed $x_t, t, h$, this positive-definite quadratic loss retains the conditional-mean optimum. That function-space identity does not guarantee finite-model convergence, physical image fidelity or language learning.

Control ($\alpha = 0$) and treatment ($\alpha = 0.001$) restore the same adapted Note 38 checkpoint and receive 2,000 updates each, batch four, on the same 32 seen training questions: 8,000 draws per arm. Case sampling and noise RNG are matched. The initial padded execution narrowly fails its original state comparison. A separate execution computes each owning history at its true length in both arms, preserving all budgets, losses, seeds, criteria and causal positions. This passes the complete preflight, including exact model/optimizer/sampler replay, finite gradients, frozen-prefix checks and generated-pixel feedback. No quality threshold is widened.

| Writing path | Control F1 (%) | Metric F1 (%) | Exact (each) |
|---|---|---|---|
| Autonomous generation | 85.984 | 70.009 | 0/32 |
| Correct history and length | 92.430 | 75.116 | 0/32 |
| Supplied-code reconstruction | 99.9953 | 99.9953 | 28/32 |

Both free streams stop in every case, but learned stopping is not evidence of correct length. Six control and eight treatment answers are shorter than their references. The treatment fails every original repair criterion: at least 24/32 exact free images, F1 at least 0.98, and improvement over control on both measures. Neither arm reaches the absolute training-fit requirement. Scaling this repair is rejected.

A post-hoc diagnosis authenticates all 388 exported PNGs and independently reproduces the six full-output summaries from pixels. First-block F1 is 99.467% for control and 97.018% for treatment. Remaining-stream F1 is 76.237% and 47.493%; correct history raises those tails to 87.724% and 56.682%. First-block averages cover all 32 questions; tails cover the 28 references longer than one 64-column block. Missing ink and excess generated ink remain errors. This localizes inaccurate conditional writing before generated-history drift alone explains the outcome; it does not identify a unique cause within representation transfer, optimization, objective or sampling. Foreground agreement is not semantic accuracy.

Each complete actor retains 7,583,225,971 parameters with external pretraining; 53,096,961 are trained locally. Peak allocated memory is 18.893 GiB per arm. Recorded control/metric invocation times are 3,894.676/4,053.182 s, excluding shared setup, prefill and initial outputs. An interrupted metric attempt had no durable checkpoint and was discarded before restarting from the declared warm state. Its additional compute is excluded from those timings. All 19 logged rows through update 180 reproduce after restart, and all protected control/initial files remain unchanged; full model/optimizer equality at that interruption point was not measured.

CPU verification scores all 64 final autonomous answers, decodes retained assisted/supplied coordinates, matches the case/noise RNG and replays all 64 initial outputs. It does not independently replay the complete final contextual body and sampler. A subsequent fresh checkpoint invocation reproduces one original case's pixels, stops, feedback and reader memory exactly; file/socket tracing observes no training-data or ancestor-checkpoint access and no internet-family connection. That one-case restoration is execution evidence, not language qualification. All 384 original PNGs and per-case results remain in `references/evidence/decoder_metric_v82_result/`.

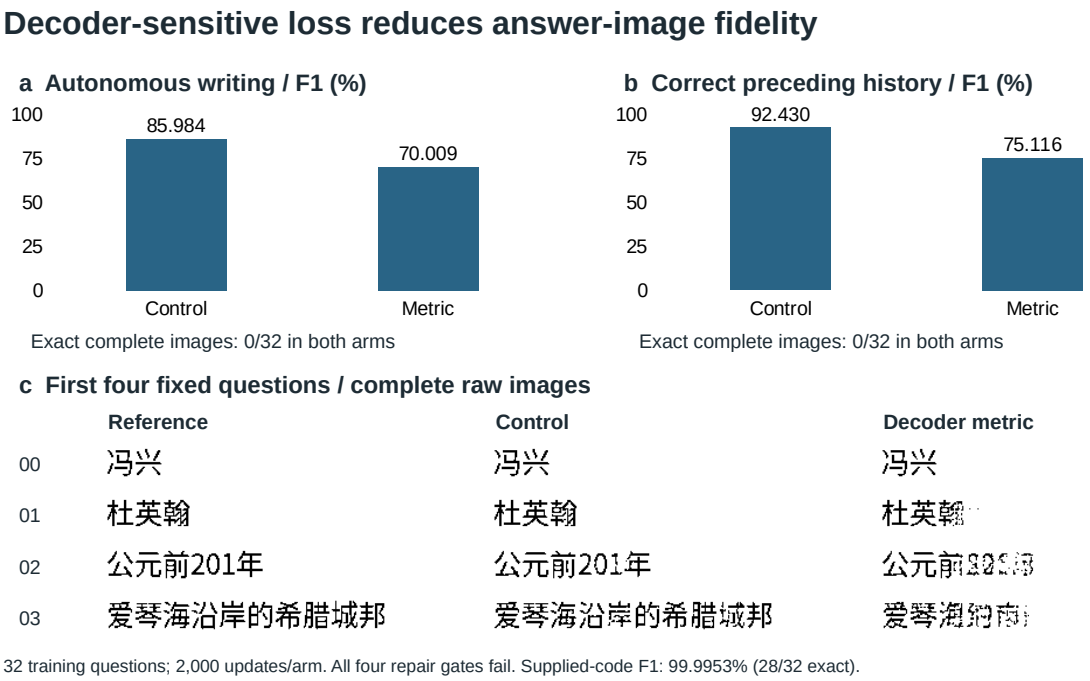



**Fig. S16** | **Decoder-sensitive reweighting worsens autonomous writing. a**, Free foreground F1 on the same 32 seen training questions after matched optimization. **b**, Prediction with correct preceding answer history and known length. Both arms retain 0/32 exact complete images in both paths. Supplied-code reconstruction remains at 99.9953% and 28/32 exact. **c**, Complete unmodified reference, control and metric PNGs for the first four fixed questions. Original image bytes and aspect ratios are preserved in editable native PowerPoint objects; native chart caches/workbooks retain full-precision measurements. All four original repair gates fail. These image scores do not qualify language understanding or an efficiency advantage.

## Supplementary Note 40: native paired-image copying remains unreliable

A separate experiment retains the pretrained continuous image-generation path rather than attaching a new writing decoder to a contextual language body. The source is the undistilled FLUX.2-klein-base-4B, revision `a3b4f4849157f664bdbc776fd7453c2783562f4d`. Its transformer has 3,875,544,576 parameters. Rank-16, alpha-16 residuals on 103 native linear layers add 24,358,912 trained parameters; the deployed transformer therefore contains 3,899,903,488 parameters, with an 84,046,115-parameter VAE. External source pretraining is retained and excluded from local costs.

The native VAE posterior mode, patch normalization and packing encode source and target images. Let $z_Y$ encode the target and $z_X$ the input. For independent $\varepsilon \sim \mathcal{N}(0,I)$ and $\sigma \sim \mathcal{U}[0,1]$, training minimizes

$$z_\sigma = (1-\sigma)z_Y + \sigma\varepsilon, \qquad \mathcal{L} = \mathbb{E}\,\|v_\theta(z_\sigma,\sigma;z_X,c) - (\varepsilon - z_Y)\|_2^2 .$$

The implementation averages squared error over native latent coordinates. This is conventional conditional flow matching; no new convergence or language-efficiency theorem follows from it. All samples use the same generic numeric condition $c$, compiled offline from layers 9, 18 and 27 of a shared externally pretrained Qwen3-4B encoder. This hybrid conditioner is not claimed identical to the stock FLUX encoder. No changing caption or answer embedding enters inference; the variable question content enters through image pixels. The deployment path contains no tokenizer, language embedding table, OCR or font-based replacement of generated writing.

Readable preparation contains 128 training and 32 validation copies plus 256 training and 32 validation CMRC question-answer pairs. Article partitions for QA are disjoint within preparation. Only the first 32 training copies are used in these two optimization runs; no natural QA is optimized. Source pages are 1024-square and target pages 512 by 256, with 32-pixel writing and explicit clipping checks. Neither an answer-aware crop nor target-dependent layout selection is used. Chinese validation targets include 13 characters absent from these 32 local training targets; external pretraining remains available.

Both arms start fresh from the pinned source. Training uses batch one, accumulation one, AdamW at $10^{-4}$, weight decay 0.01, seed 20268321, FP32 adapter masters and optimizer state, BF16 native computation and checkpointed activations. All 512 sampled IDs and sigma draws match across arms. A precision error discovered during the initial preflight is corrected before optimization: adapter residuals are cast to the native linear output dtype. The corrected zero-adapter prediction matches the native source exactly. Two-update preflights execute real optimization and establish finite nonzero gradients and parameter changes; these are execution checks, not quality tests.

The padding treatment removes only rows below fixed height 256. Every removed pixel is exactly white in all 160 copy inputs; inverse white padding reconstructs every source RGB pixel exactly. No source ink is moved and target PNG bytes remain identical. Native reference positions decrease from 4,096 to 1,024; target positions remain 512. However, changing canvas extent changes VAE features: all 64 target tensors are bit-exact across caches, while maximum input-feature difference in the retained extent is 1.6796875. This is a preprocessing comparison, not an identical-latent attention-mask intervention.

| Measurement | Square | Strip |
|---|---|---|
| Training invocation time (s) | 795.624 | 287.182 |
| Peak allocated GPU memory (GiB) | 9.501 | 8.437 |
| Original eight-case screen, OCR exact | 2/8 | 3/8 |
| Complete training cohort, OCR exact | Not run | 6/32 |
| Complete validation cohort, OCR exact | Not run | 10/32 |

The fixed screen uses the first four training and first four validation cases, with matched sampling seeds, 50 native steps, guidance one and a 512-by-256 answer canvas. The unadapted source gives 0/8 exact OCR strings in both input formats. Full strip observation preserves the actual eight samples and adds 56 calls under a separately fixed seed rule, yielding 64 logical outputs, not 64 new calls. Fresh generator processes receive only input images, a constant condition and learned state. Targets and recognition are separate. The recognizer uses Tesseract 5.3.4, `chi_sim+eng`, PSM 6, NFC and fixed whitespace normalization, retaining case and punctuation. Reference calibration is 27/32 training and 28/32 validation, so OCR counts are exploratory and uncertain readings require the raw pixels.

Raw inspection independently rejects the unchanged requirement of 30/32 legible exact requested strings alone per split. Missing Chinese characters, doubled English letters, duplicated lines and extra marks remain, including on training examples. Correct copies coexist with these errors. All 64 input, target and output pages and eight unadapted controls remain publicly retained under `references/evidence/paired_padding_result/`. CPU export authenticates all 72 generated PNGs against retained postprocessed decoder tensors; this is not an independent replay of the neural computation. All original failures remain in the evidence. The approximately 2.77-fold reduction in one training invocation excludes source acquisition, preparation, setup failures and separate controls. There is no independent training-seed replication, capability-matched text baseline, natural QA result or demonstrated efficiency at language quality.

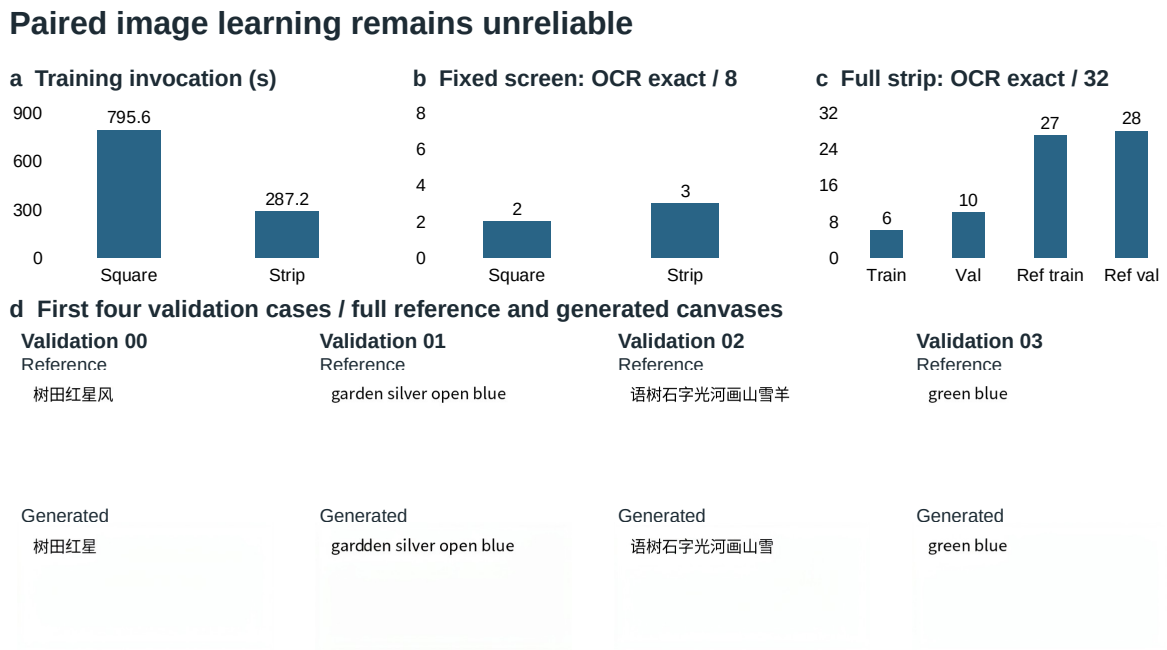



**Fig. S17** | **Faster paired-image training still fails reliable writing. a**, Wall time for one 512-update training invocation per input format. **b**, OCR exact match on the same eight fixed copying cases. **c**, Complete strip-model observation on 32 training and 32 validation cases, with clean-reference OCR calibration shown separately. The square model has no complete-cohort result. **d**, Complete unmodified reference and generated canvases for the first four validation cases in the original order. All 64 outputs remain in the evidence archive. Native editable charts and embedded workbooks retain the measurements; original PNG bytes and aspect ratios are preserved. Omissions visibly reject reliability. Copying and execution cost do not establish language understanding.

## Supplementary Note 41: visual copying improves without question understanding

The native Show-o2-1.5B-HQ source[20] retains a language body, semantic image frontend, continuous image velocity head and Wan codec. The complete checkpoint contains 2,830,777,056 parameters, of which 1,543,304,192 belong to the language body; the codec adds 126,892,531. The 1.5B source label is not the complete deployed footprint. Source revision, model and codec hashes are pinned in `references/native_unified_assets.lock.json`. Constructor-only component hydration avoids redundant base-weight downloads, then strictly loads the complete checkpoint and checks its tied language embeddings. The source neural functions are unchanged. Existing external pretraining is retained and excluded from the local resource measurements.

On the fixed twelve-case source diagnostic, image prompts give 0/12 canonical exact text answers. Eight copying-prompt transcriptions also fail. Original text gives 5/12 canonical exact answers, including four English copies and one natural question; all eight requested copy contents occur, with unwanted quotation marks on four Chinese responses. Readable source-image codec reconstructions preserve the marks. These observations separate useful text-language access from unreliable access through this visual composition, without selecting a unique cause. They do not replicate the source authors' published benchmark conditions. Separately supplying eight desired strings to the native image generator gives 0/8 strict OCR-exact outputs, against 10/12 on the diagnostic reference images. Some English outputs remain legible; Chinese outputs are blank or incorrect. These are assisted source-writing controls, not autonomous image answers.

The visual-access experiment adds gated rank-16, alpha-16 residuals to 158 native semantic-encoder and fusion linear layers. All source weights, including the language body/head and image generator, remain frozen. The 8,489,984 learned parameters affect only clean source-image prefix formation. The disabled gate retains the base generator path. A real two-update preflight verifies finite gradients through all up/down residuals, exact zero-residual prefix equivalence and an unchanged disabled-gate prefix after those updates. The complete learning run starts from fresh residuals rather than that preflight checkpoint.

Let $E(X)$ be the continuous source-image encoding, $R_A$ the adapted visual frontend, $H$ the native contextual body and $U$ its frozen language head. For target answer labels $y_1,\dots,y_m$, the auxiliary objective is

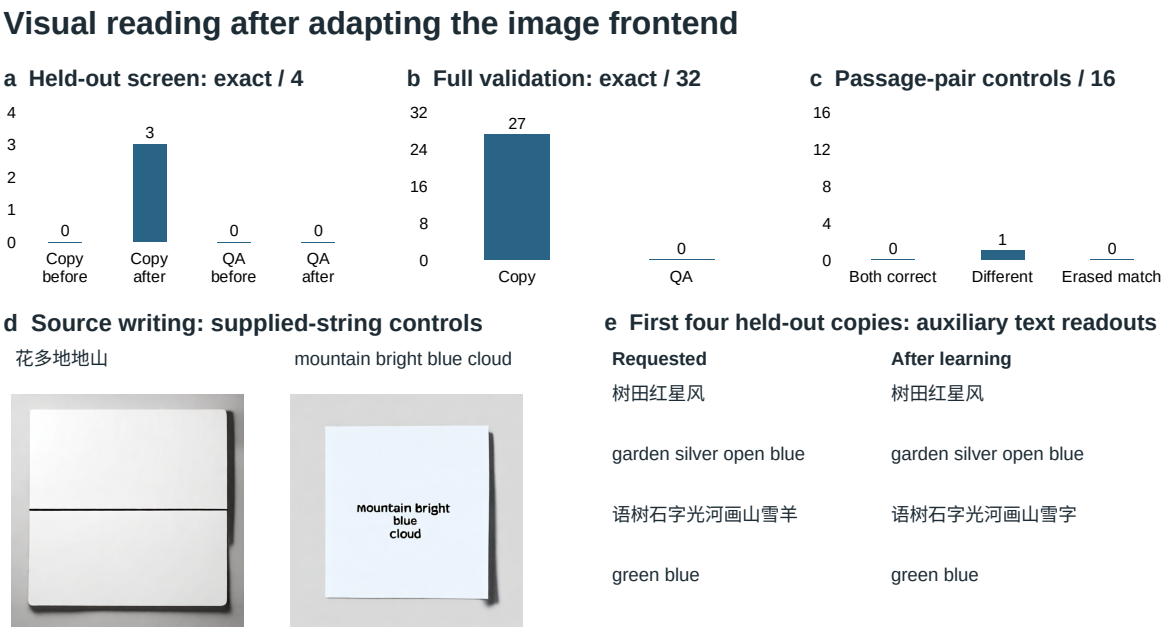
$$\mathscr{L}_{\text{access}} = -\frac{1}{m}\sum_{j=1}^{m} \log \operatorname{softmax}(UH[C, R_A(E(X)), y_{<j}])_{y_j}. \tag{23}$$

The generic control $C$ is constant; the changing question enters through pixels. Correct preceding answer labels are available during this loss, while autonomous evaluation receives neither answer histories nor references. This explicitly uses Unicode/token supervision and a temporary text readout. It is not vocabulary-free training or the required image-only output model.

All 128 prepared copying examples and 256 natural questions enter training; 32 copies and 32 questions are held out. QA articles are disjoint between these splits. Native FP32 VAE samples are cached once without resizing or answer-aware cropping; the twelve original diagnostic latent tensors match their retained observations exactly. Input copy strips retain all original ink, and QA pages retain complete passages at $1024 \times 1024$. FP32 adapter masters and AdamW states use BF16 native computation, batch one, alternating copy/QA lists, seed 20268921, learning rate $10^{-4}$ with 100-step warm-up and cosine decay to $10^{-5}$, betas $(0.9, 0.95)$, weight decay 0.01 and gradient clipping at one. All 4,096 optimizer records are contiguous and finite; all 384 declared training IDs occur and no validation ID is optimized. The single training invocation takes 2,758.904 s with 6.894 GiB allocated peak, excluding acquisition, source pretraining, cache construction and subsequent evaluations. The adapted native model contains 2,839,267,040 parameters plus the codec; trained parameters alone do not describe its size.

Native incremental KV and last-position logits reproduce all 512 decisions across the twenty original FP32 diagnostic calls before use in assessment. The matched BF16 before/after screen gives training-copy exact scores 0/4 and 4/4, held-out copy scores 0/4 and 3/4, and natural-question scores 0/4 and 0/4. The separate complete 64-case observation gives 27/32 exact stopped copies: 11/16 Chinese and 16/16 English. Its remaining five copies substitute, insert or omit characters. Natural QA gives 0/32: 27 outputs are the year 1999 and five are 1998, unrelated to the requested answers. All streams stop. Exact match uses NFC and declared whitespace canonicalization, preserving case, punctuation, quotation marks and every written character. These are direct native text readouts, not OCR scores of generated images. The overlapping screen cases are retained separately and never added to the 64-case denominator.

The question intervention uses all sixteen distinct-answer held-out passage pairs. Original font geometry and every source pixel are checked before erasing from the question paragraph to the bottom, including the constant answer marker. Every context pixel remains unchanged; both erased versions of a passage are identical. Each pair receives the same latent and readout seeds across its two normal pages and one shared erased page. There are 32 normal calls and sixteen independent erased calls. Normal QA again gives 0/32; no pair has both answers correct. Only 1/16 normal pairs gives different outputs. Erasure changes 3/32 normal outputs, leaving 29/32 unchanged, and none of the sixteen shared erased answers matches either reference. Thus generic stopped output cannot be interpreted as following the question. The separate 64-call and 48-call invocations take 26.916 and 33.030 s, respectively, excluding source loading;




4,096 updates; 384 training examples; only visual adapters learned. Text readouts are not image answers. General ILM remains unachieved.

**Fig. S18 | Visual copying does not establish question understanding. a**, Exact stopped auxiliary text readouts on four held-out copies and four held-out questions before and after adaptation. **b**, Complete held-out copying and QA observations, 32 cases per task. **c**, Sixteen passage-pair controls: both questions correct, normal outputs different, and the shared erased output matching either reference. Different outputs alone are not correct answers. **d**, Complete first two original supplied-string source-writing outputs; desired strings are explicitly provided, so these are assisted controls. **e**, First four held-out copying responses, displayed as editable text from the temporary Unicode head. They are not answer images generated by the learner. Raw case order, PNG bytes and aspect ratios are preserved; all observations remain available.

these are not quality-matched efficiency comparisons.

All raw readouts, complete input/reference pages, sixteen erased pages and the optimizer journal remain in `references/evidence/native_visual_access_result/`; the export binds 179 files. The learned checkpoint hash is `e2937d2fa986fac0e7ca1964607d5dca06194b698b6a25ad0644635f585427fb`. Source writing controls and input codec reconstructions remain separately retained. CMRC passage images and references preserve their CC-BY-SA-4.0 attribution [12]; copying strings are synthetic. These public exploratory cases are not a sealed test, and there is no independent training-seed replication. Short visual copying improves, while page-level question selection and independent image-answer generation remain unqualified.

## Supplementary Note 42: dense supervision and native image geometry

The completed curriculum extends the adapters in Note 41 by 2,560 updates, with the same 384 training images and all 64 held-out images excluded. The first 1,024 updates supervise complete-page transcription; the remaining 1,536 cycle transcription, QA and requested-only copying. Totals are 1,536 transcription updates (four passes), 512 QA updates (two passes) and 512 copying updates (four passes). Every optimizer record is present and finite, and exact per-ID coverage follows the saved schedule. The 8,489,984 learned visual/fusion parameters and frozen source components are unchanged. This invocation takes 1,946.753 s with 7.072 GiB peak allocated VRAM on one RTX 4090 D, excluding source pretraining and separate assessments.

Complete requested copying improves from 27/32 to 29/32 exact stopped answers (Chinese 13/16, English 16/16), still below the declared 30/32 requirement. Chinese natural QA remains 0/32; outputs are twenty instances of `1998` followed by the Chinese year marker and twelve of `1990` with that marker. No paired questions are both correct. Erasing the question leaves 30/32 normal responses unchanged. The independent eight-page transcription screen improves from 0/4 to 4/4 exact complete short copying pages, with aggregate Levenshtein character error falling from 0.603 to zero. The four full Chinese QA-page transcripts remain 0/4 exact; error increases from 0.999 to 1.734, with two reaching the 1,024-decision cap. Insertions can make this error exceed one. Unrelated repeated paragraphs do not constitute transcription of the source pages. No matched continuation arm isolates the curriculum from ordinary continued optimization.

**Prospective English transfer.** Sixteen SQuAD 1.1 articles[21] supply two fixed questions each, with all annotated alternatives and verified answer offsets retained. Whole passages and questions are rendered in 32-pixel type on 1024-square pages. Each model receives all 32 normal pages, 32 passage-erased pages and sixteen shared question-erased pages. No English passage enters the optimizer. All conditions for a passage share latent/readout seeds. Original source-page OCR is exact on 22/32 pages, with aggregate character error 0.001522; primary reference images are 31/32 OCR-exact. OCR is offline calibration only. Both completed adapter checkpoints answer 0/32 normal questions correctly. They produce unrelated Chinese years rather than the requested English facts. Their paired-question scores are both 0/16. These public development cases may occur in foundation pretraining; this is not an official benchmark score or an uncontaminated foundation-model test. The source release is CC-BY-SA-4.0.

**An effective source-interface correction.** The same frozen HQ source also answers 0/32 of these questions at the original 1024-square geometry, with only three normal outputs reaching EOS. The pinned official MMU configuration instead uses a whole-image bicubic resize and centre crop to 432 square. Since all eighty English inputs are square, resizing crops no content. The spatial sequence length changes from

$$P_s = (s/16)^2, \qquad P_{1024} = 4096, \quad P_{432} = 729.$$

The image path changes from $R_A(E(X))$ to $R_A(E(T_{432}X))$, where $T_{432}$ resizes the complete image. Source pixels, latent features and spatial positions therefore change jointly. No answer-aware crop or supplied answer conditions inference. All source/transformed hashes, original references, per-passage seeds, generic instruction, BF16 body, FP32 codec and top-one 96-decision readout are fixed.

Without new training or adapters, the source now gives 11/32 exact stopped normal answers, all 32 terminate, and 3/16 passage pairs have both answers correct. Fourteen normal pairs differ and question erasure changes 30/32 outputs. Passage erasure leaves two answers correct; nine of the eleven correct normal answers become incorrect. Correctness lost without the passage supports passage dependence on those cases, whereas the two retained answers cannot establish use of it. Different outputs alone do not establish accuracy. The original 4,096-update adapters and final curriculum checkpoint each give only 5/32 exact answers at 432, with 1/16 both-correct pairs. They are weaker than the unadapted source in this intervention, not a validated repair.

Resized normal-page OCR has 0.004207 character error, with 10/32 complete pages exact; sampled FP32 native-codec reconstructions have 0.006210 error and 0/32 complete pages exact. No recognition result filters cases or enters model conditioning. The source's eighty calls take 131.576 s at 1024 and 14.872 s at 432, excluding loading, with 11.129 and 6.738 GiB peak allocated VRAM. These are costs of the specific interface intervention, not LLM parity or efficiency at matched useful language quality.

**Scope and evidence.** The intervention identifies a substantial interface effect on this fixed English cohort; it does not isolate spatial interpolation, image scale or sequence length as the unique mechanism. It does not establish native-scale Chinese transfer or explain every earlier image-writer failure. All original failed outcomes remain unchanged. Source and adapter readouts use an auxiliary Unicode head, not generated answer images. Complete model and codec counts and inherited external pretraining remain as specified in Note 41.

Three exports bind 13, 121 and 128 files, respectively:

```
references/evidence/native_visual_alignment_result/
references/evidence/native_english_transfer_result/
references/evidence/native_reader_geometry_result/
```

They retain the complete optimizer schedule/journal, all transcription, QA and control responses, full English source/reference images, all eighty resized images and 32 codec reconstructions. Original source PNGs are reused by explicit hashes. The completed curriculum state is identified by checkpoint SHA256:

```
016429ec5454cdd681693d881df4c6b7d52fb43dcae7b9e15857d2d9cd32b483
```

These observations require calibration of the source image interface before another learning run; they do not qualify the independent image-input/image-output ILM goal.

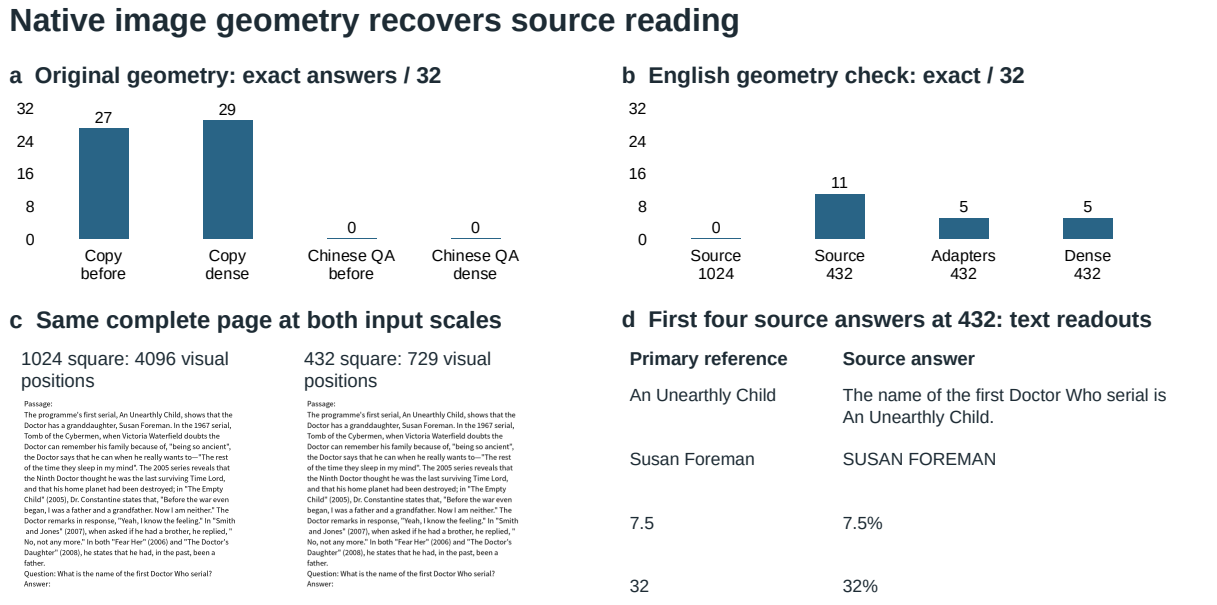



**Fig. S19 | Native image geometry recovers source reading, while learned adapters remain weaker. a**, Complete requested copying and Chinese QA at the original geometry before and after the dense curriculum. **b**, Exact stopped English answers on the same 32 questions for frozen-source 1024, frozen-source 432 and the two completed adapter states at 432. **c**, Complete first English page at both input scales, shown at matched display size; full original canvases and PNG bytes are embedded without cropping. This comparison documents the geometric intervention rather than two independent samples. **d**, The first four native-source text answers and their primary references, retained in fixed order. Exact assessment accepts any annotated alternative and requires EOS; extra wording and capitalization remain significant. These are auxiliary text readouts, not answer images written by the learner. All 320 model calls and failed answers remain available.

## Supplementary Note 43: state alignment and dependence on the canvas

The frozen source's native-resolution English result in Note 42 does not transfer to Chinese. On the fixed 32 Chinese questions, the source gives 0/32 exact stopped answers from either the original 1024-square image or its 432-square whole-image resize, versus 16/32 from the original visible text. The latter is a privileged source-language control. Independent full-page recognition of sampled codec reconstructions gives character errors 0.086875 at 1024 and 0.705221 at 432; these are readability observations, not proofs of latent information loss. Native-scale semantic windows retain resolution but give 0/32 Chinese and 2/32 English QA; pooling them to the native prefix gives 0/32 in both languages. Neither frozen replacement qualifies reading.

**A source-preserving learned correction.** For a complete image $X$, let $C(X)$ be the native 729-position visual feature sequence from its whole-image 432-square resize. Let $F_A(X)$ encode the original 1024-square image using nine overlapping native semantic windows, with learned visual/fusion adapters $A$, then pool the fused 64-by-64 field to 27 by 27. Both sequences have width 1,536. The corrected features are

$$Z(X) = C(X) + G\big(F_A(X) - C(X)\big), \qquad G_0 = 0.$$

The bias-free full-rank $G$ has 2,359,296 parameters; 158 gated rank-16 visual/fusion adapters add 8,489,984. All original weights, the coarse frontend, language body/head and image generator remain frozen. At zero correction the complete native source prefix is exact. Pooling is lossy; retaining all input pixels before pooling is not a lossless-context claim.

The same frozen language body supplies offline original-text targets before any correct answer history. For its final assistant-role state at layers $\mathscr{L} = \{4, 8, 12, 16, 20, 24, 28\}$, define

$$N(h) = \frac{h}{\sqrt{1536^{-1}\sum_d h_d^2 + 10^{-6}}}, \qquad \mathscr{L}_{\text{state}} = \frac{1}{7 \cdot 1536} \sum_{\ell \in \mathscr{L}} \|N(s_\ell) - N(t_\ell)\|_2^2.$$

Optimization minimizes answer cross entropy plus $\mathscr{L}_{\text{state}}$, with unit coefficients. Correct preceding answer history enters only the cross-entropy term. The native mask blocks every future answer position from the pre-answer states. The temporary FP32 causal check has normalized maximum error $7.6294 \times 10^{-6}$, below $2 \times 10^{-5}$.

All 640 training records appear four times: 128 copies, 256 Chinese questions and 256 English questions from 128 paragraphs in 32 articles excluding all sixteen English assessment articles. Preparation rejects only missing font coverage or complete-page overflow, without teacher-outcome selection. AdamW uses $(0.9, 0.95)$, decay 0.01, norm clipping at one, 64-update warm-up to $2 \times 10^{-5}$ and cosine decay to $2 \times 10^{-6}$. There is no validation checkpoint selection. All 2,560 observations and finite gradients are retained. The invocation takes 3,112.084 s at 6.370 GiB peak allocated VRAM, excluding loading and the 238.713 s offline cache. The complete adapted model has 2,841,626,336 parameters plus the 126,892,531-parameter codec and external pretraining; 10,849,280 trainable parameters do not define the full model size.

| Fixed primary assessment | Native source | Learned correction |
|---|---|---|
| English copying | 7/16 | 16/16 |
| Chinese copying | 0/16 | 0/16 |
| English QA | 11/32 | 16/32 |
| Both English questions correct | 3/16 | 4/16 |
| Chinese QA | 0/32 | 0/32 |
| Both Chinese questions correct | 0/16 | 0/16 |

English gains six correct answers and loses one on the same small, repeatedly inspected cohort. Fifteen of sixteen correct learned answers become incorrect without the passage; one remains correct without it. Chinese remains unreliable: eleven of sixteen pairs give identical responses to different questions. The first two references, *India–West Pacific region* and *India Pondicherry*, both receive the shorter Chinese answer *India*. All normal Chinese responses stop, but stopping and related fragments are not correctness. Four of the six declared reader requirements fail.

The original preflight replays four native-source readouts, but compares the corrected prefix/mask on only one training probe. The additional complete four-case corrected-path check is performed after optimization, before final scoring: four prefixes/masks and eight readouts match exactly. This timing deviation is retained. The controller reuses the completed optimizer and 160 after-readouts; neither is restarted. No transcript, teacher state, correct history or recognition output conditions autonomous assessment.

**Seen-prompt fit.** A separate prospective diagnosis selects the first eight training records per task and language before the primary after-readouts. All 96 calls (source image, learner image and original-text source) stop. The learner

copies 8/8 English but 0/8 Chinese strings; all eight Chinese failures change characters, including the first fixed case in Fig. S20. Seen English QA is 6/8 against original annotation alternatives, or 5/8 against its particular cross-entropy target; strict Chinese QA is 0/8. The original-text control has three exact Chinese copies, four further strings intact inside quotes and one translated string. Several QA responses differ in language, surrounding text or date formatting, while others give wrong quantities. Its strict 1/8 Chinese QA count is not a count of every semantically related answer. No criteria are changed after inspection. The observed copying errors establish a fit obstruction in this recipe, without identifying its unique architectural or optimization cause. These are seen data, not generalization.

**A separate attention-only intervention.** Starting instead from the dense checkpoint of Note 42, a second learner freezes the complete calibrated visual frontend and trains fresh rank-16 residuals only in the 112 native q/k/v/o projections. Copying retains its original strips and QA its complete square pages; no pooling or image resize is introduced. All 640 bilingual records appear four times. The 4,358,144 new parameters receive answer cross entropy, with AdamW, 64-update warm-up to $5 \times 10^{-5}$ and cosine decay to $5 \times 10^{-6}$. There is no text-state teacher or validation selection. The complete adapted model has 2,843,625,184 parameters plus the same codec and external pretraining. The 2,560-update invocation takes 1,667.861 s at 6.345 GiB peak allocated memory, excluding loading and its 262.431 s prefix cache. The real three-update preflight preserves disabled-source logits exactly, changes all 112 zero output matrices and gives finite nonzero gradients in all 224 residual tensors. Four disabled/zero source pairs replay exactly; the normalized FP32 causal check has zero error. Nevertheless, the complete 160-call assessment retains 29/32 copies (Chinese 13/16, English 16/16) but gives 0/32 QA in both languages, with no both-correct pairs. The first English title and person questions both receive the unrelated year 1961. All six reader requirements fail. These costs and outcomes belong to this separate attention learner, not to the coarse/fine correction above. Its complete export is `references/evidence/native_reader_attention_result/`.

**A controlled canvas failure.** The 13/16 Chinese copying result in Note 42 uses 1024-by-256 strips, whereas the correction study above uses complete 1024-square copying inputs. The cross-model difference therefore cannot be attributed to pooling alone. An independent 480-call control fixes the earlier dense checkpoint, its seeds, generic instruction and all ink. All 160 original-layout decision sequences replay exactly before altered layouts are evaluated. For every copying request, white bottom padding gives the original square exactly:

$$X_{\text{square}} = \text{pad}_{\text{white}}(X_{\text{strip}}).$$

The visual field changes from 1,024 to 4,096 positions. Exact copying falls from 29/32 to 7/32: Chinese 13/16 to 0/16 and English 16/16 to 7/16. No request character is resized, removed or moved. Ten padded copies fail to stop within the 96-decision cap. A separate tight crop retains all pixels unequal to white, adds a 16-pixel margin and rounds outward to multiples of 16. It uses no recognition, transcript, font metric, answer or semantic role. Replacing its white exterior reconstructs every source exactly, but copying is 0/32. Fewer visual positions, 161–304 for these copies, do not guarantee correct reading.

Natural QA remains 0/32 in both languages in all three arms. The legacy and square QA inputs are identical. Tight cropping changes their visual features, positions and context lengths despite preserving the full visible request. Pixel-level invertibility does not imply invariance of the learned mapping $R(E(X))$. This control demonstrates geometric brittleness, not the unique mechanism of failure. The three 160-call arms take 86.856, 120.337 and 64.060 s excluding loading, with peaks 11.161, 11.161 and 10.506 GiB. Faster incorrect inference is not efficiency at matched useful quality.

**Separating source features from context layout.** A prospectively fixed 128-call intervention uses the same frozen dense checkpoint and all 32 copying requests. Let $S$ be the 1,024 strip features and $Q$ the 4,096 square features; $Q_{\text{top}}$ denotes its first 1,024 positions, covering the same upper 256 pixels. All request ink lies inside that strip. Before any hybrid call, all 64 original strip and square decision sequences replay exactly. Every pair has bit-identical nonvisual headers and suffixes. The two hybrids use the native short or long mask, respectively, without changing model parameters, instructions or readout.

| Visual features | Prefix length | Chinese copies | English copies |
|---|---|---|---|
| $S$ | 1,063 | 13/16 | 16/16 |
| $Q$ | 4,135 | 0/16 | 7/16 |
| $Q_{\text{top}}$ | 1,063 | 0/16 | 6/16 |
| $[S; Q_{\text{tail}}]$ | 4,135 | 2/16 | 16/16 |

Short context alone does not restore copying from the square's features. Substituting strip features into the long context restores English copying, but Chinese remains substantially below the strip source. These hybrid prefixes need not

encode real images. Nominally blank-region features can contain global information; context changes also move the suffix and change positions and available keys. Thus the comparison narrows the representation failure without isolating a unique internal cause. All 128 raw responses are retained. The invocation takes 68.220 s after loading at 11.161 GiB peak allocated memory; the complete frozen model has 2,839,267,040 parameters plus the same codec and external pretraining. There are no optimizer updates, natural-QA measurements or image answers in this intervention.

The full exports preserve all raw responses, input rasters, original alternatives, seeds, executed-code hashes and reconstruction checks:

```
references/evidence/native_state_alignment_result/
references/evidence/native_state_fit_result/
references/evidence/native_canvas_control_result/
references/evidence/native_canvas_factorization_result/
```

The state study's public per-update journal is compact; its full per-parameter gradient journal remains local with SHA256. The canvas export binds all 480 source PNGs. CMRC 2018 and SQuAD 1.1 retain CC-BY-SA-4.0 attribution; the Show-o2 source/model are NUS Show Lab, Apache-2.0. These auxiliary Unicode head results do not demonstrate continuous image answers. The failed Chinese requirements do not admit scaling or a writing-stage qualification.

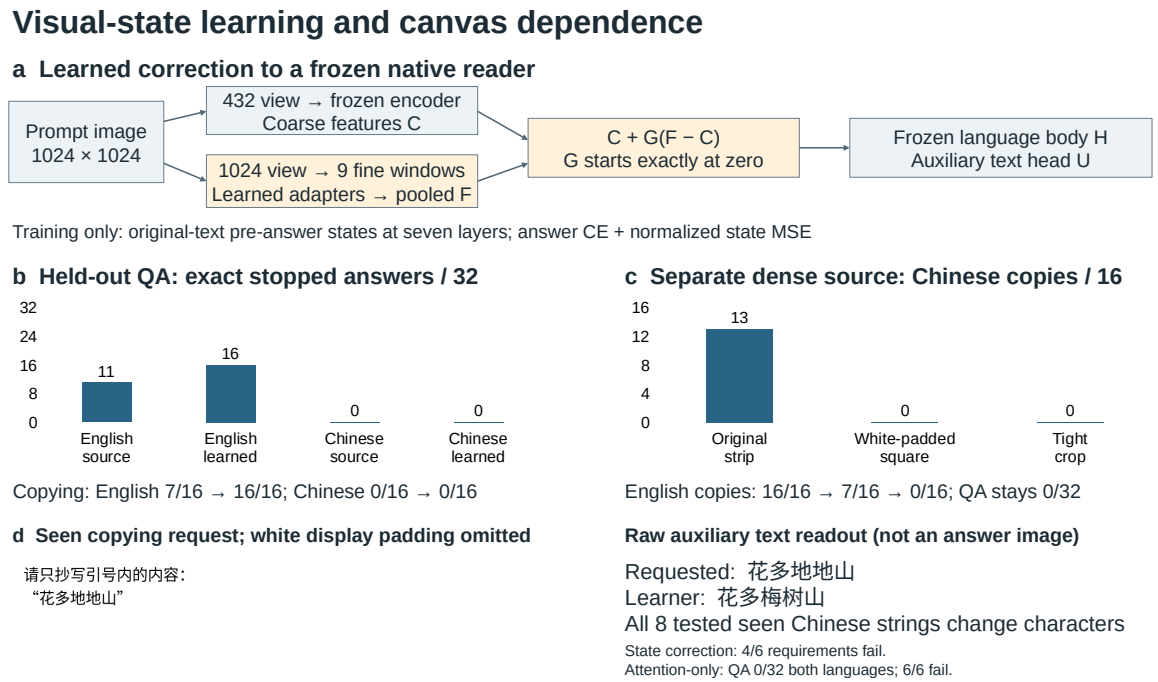



**Fig. S20 | Limited English transfer and a controlled failure of geometric invariance. a**, Source-preserving coarse/fine correction; orange layers learn, while the native language body and temporary text head remain frozen. Offline text-state targets and correct answer history are training supervision only. **b**, Exact stopped English/Chinese QA on the same 32 questions before and after the 2,560-update correction. Copying counts are shown separately. **c**, The separate earlier dense checkpoint under three whole-request layouts; all weights and sampling seeds are fixed, and every ink pixel is preserved. This is not a cross-model comparison. **d**, The first fixed seen Chinese copying request and raw auxiliary text response. Only white exterior is omitted for display; the learner receives the complete square. The visible request pixels are retained exactly. The separately labelled attention-only result in the footer belongs to a different learner, which retains copying but fails QA. The editable text is a Unicode-head readout, not an answer image generated by the learner. All failures remain in the complete exports.

## Supplementary Note 44: exact source-pixel copying and weak reading

The free-writing failures leave open whether accurate physical writing and answer selection can be separated. A new component predicts a contiguous region of the current writing image and copies its original columns. It is an extractive component, related to pointer-based copying [11] and the earlier image-block model in Note 17. It cannot write an answer absent from its input and does not replace the independent free-generation objective.

**An image-only conditional distribution.** Let $X \in \{0,255\}^{32\times W}$ be a complete prepared writing strip. Its anonymous units $u_i = [a_i, b_i)$ are maximal consecutive columns containing black ink. Units need not be characters: a radical or letter can occupy several units. The calibrated image/context body [17] encodes the whole strip after white-column folding and native image-budget resizing. Nine bilinear samples at each unit's original folded location supply a contextual descriptor $v_i \in \mathbb{R}^{4096}$. All final numeric task-frame states $q_k \in \mathbb{R}^{4096}$ are retained. The contextual computation is FP32 with frozen BF16 language weights and FP32 vision weights imported from the shared Q4 archive; descriptors are cached in BF16. The language vocabulary module and output projection are absent. Neither source text, tokenizer, recognition nor a teacher service supplies inference answers.

The learned head has width 256. With trainable layer normalizations and projections, define $v_i = P_v(\mathrm{LN}(v_i))$ and $\bar{q} = K^{-1}\sum_k P_q(\mathrm{LN}(q_k))$. A shared nonlinear network maps $[v_i; \bar{q}; v_i \odot \bar{q}; g_i]$ to start and end scores $(s_i, e_i)$; $g_i$ contains four physical position/width features. Two 256-wide GELU layers precede the two scores. For all visible spans $\mathscr{A}(X) = \{(i,\ell) : 1 \le \ell \le 128,\ i+\ell-1 \le n(X)\}$,

$$p_\theta(i,\ell \mid X) = \frac{\exp(s_i + e_{i+\ell-1} + b_\ell)}{\sum_{(j,m)\in\mathscr{A}(X)} \exp(s_j + e_{j+m-1} + b_m)}.$$

The motor $C_{i,\ell}(X)$ copies every original pixel between $a_i$ and $b_{i+\ell-1}$ and adds a two-pixel horizontal white frame. All 32 rows are unchanged. Identical complete images share probability mass:

$$P_\theta(Y \mid X) = \sum_{(i,\ell):C_{i,\ell}(X)\equiv Y} p_\theta(i,\ell \mid X).$$

Equality removes only all-white horizontal exterior. Training minimizes the negative log total mass of all reference-image alternatives and their matching source occurrences. No gold answer offset or candidate whitelist enters neural conditioning. Inference selects the image of maximum marginal mass, not necessarily the single most likely span. Its predicted end boundary terminates the copied block; finite candidate selection is not learned free-stream stopping or a density for arbitrary new images.

**Data and complete execution.** The fixed training set contains two different-reference questions about each of 128 CMRC paragraphs (256 requests), plus 64 random copying strings, half Chinese and half English. Development uses 32 disjoint-article paragraphs with two questions each and 32 new copying strings. Question- and passage-erased views add 128 inputs without changing layout. All 544 inputs are encoded; assessment produces 224 development answers and the first 32 training-QA answers. The source and targets use complete 24-pixel Noto Sans CJK JP writing inside 32-pixel strips. Pango logical positions and original annotations prepare source-rendered reference crops offline; optimization receives their pixels and marginalizes every matching occurrence. One training paragraph pair is excluded before model calls because its source ink exceeds the vertical canvas. All 160 selected paragraph pairs have different reference images. Training/development article titles do not overlap. This is reused public development, not a sealed foundation test. The interface supports prepared binary strips, not arbitrary scan segmentation.

Before new encodings, four retained V77 task-frame memories replay exactly. The full frozen source has 7,528,898,801 parameters; the head adds 2,378,370, giving 7,531,277,171 runtime parameters and inherited external pretraining. AdamW uses peak rate $3\times10^{-4}$, 100-update linear warm-up, cosine decay to $3\times10^{-5}$, $(0.9, 0.95)$ moments, decay 0.01 and clipping at one. FP32 optimization completes 3,200 updates at batch eight. Every one of 320 training IDs appears 80 times, for 25,600 draws; all losses and gradient norms are finite. Only the final checkpoint is assessed. No held-out outcome selects data, hyperparameters or weights.

Source setup and calibration take 139.548 s; encoding takes 637.448 s with 18.031 GiB peak allocated VRAM. Head optimization takes 37.689 s at 0.316 GiB after caching; this is not the memory or cost of the full model. The 256 cached-feature readouts take 17.289 s. Four additional actual PNG calls load the complete runtime in 78.726 s, then take 5.198 s at 17.948 GiB. Their visual descriptors, task memories, geometry, units and answer pixels reproduce cached inference exactly. They include both the first wrong question and next correct question, plus the first new Chinese and English copies. No source weights are downloaded or duplicated.

| Fixed image assessment | Exact | Unordered column F1 |
|---|---|---|
| New Chinese copies | 16/16 | 100% |
| New English copies | 16/16 | 100% |
| Seen Chinese QA, first 32 | 32/32 | 100% |
| Held-out Chinese QA | 9/64 | 36.114% |
| Question ink erased | 2/64 | 18.868% |
| Passage ink erased | 0/64 | 9.422% |

Only one of 32 held-out paragraph pairs has both questions correct. The normal-minus-question-erased exact gain is $7/64 = 10.938$ percentage points, below the declared 25-point floor. New copying passes its 30/32 requirement; natural answers (48/64), paired answers (20/32) and question dependence fail. Column F1 is an unordered diagnostic, not semantic accuracy. Exact image agreement is stricter than semantic equivalence; the visible wrong spans also show substantive errors. The first common-name question receives a taxonomic description rather than its two reference names (Fig. S21). Other failures extend a correct prefix or cut through a character. Ink-unit boundaries do not guarantee character or answer boundaries.

The export `references/evidence/contextual_pixel_spans_result/` binds 657 files, every raw answer/input/reference image, all optimizer records and the native calls. Independent PNG scoring reproduces all 256 outcomes and reconstructs each answer from its declared original source columns. All 544 feature files authenticate. All 128 erasure controls leave no partly erased ink runs. The final checkpoint SHA256 is `85369b1c1547f6fd02e5cc7ea601e202bb03ed8360e6e4964e94b5beec476a91`. Execution uses `scripts/run_contextual_pixel_spans.py`; actual image inference uses `scripts/infer_contextual_pixel_spans.py`; independent export uses `scripts/export_contextual_pixel_spans.py`.

This establishes accurate copying on its tests and a registered seen-data fit, while rejecting this frozen-context head as a qualified natural reader. It identifies a generalization gap in selection/boundaries, without selecting a unique cause or demonstrating that more frozen-head training repairs it. V58 is not a matched control: source, units, data, pretraining, budget and assessment differ. The full ILM still requires grounded language learning and independent free image generation. CMRC retains Cui et al.[12], CC-BY-SA-4.0 attribution; Qwen3-VL retains Alibaba, Apache-2.0 attribution.

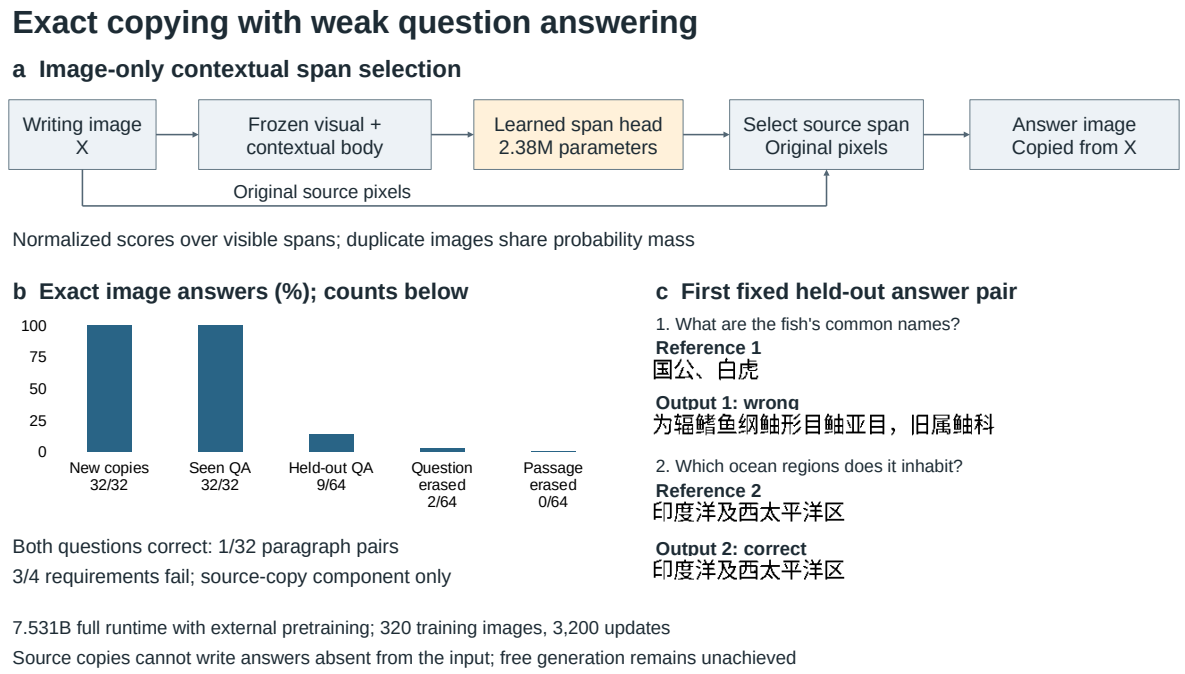



**Fig. S21** | **Exact source-pixel copying does not qualify general reading. a**, Complete input writing pixels enter the frozen contextual source and the original-pixel motor. Only the span head learns; source descriptors and numeric task-frame states condition its normalized dynamic image distribution. There is no language vocabulary module or text answer decoder. **b**, Exact image-answer percentages; categories show the actual numerators and denominators. New copies include 16 Chinese and 16 English strings; seen QA is the registered first 32 training questions. Held-out QA uses 64 questions from 32 paragraphs. Only one pair is both correct, and three of four requirements fail. **c**, The first two fixed held-out questions concern the same fish. English question labels describe the original Chinese requests; all reference and output images are unaltered source-rendered PNGs. The wrong first output is a taxonomic passage, while the second matches its ocean-region reference. The model copies visible source regions and cannot synthesize an absent answer. The complete 7.531B runtime includes external pretraining; the 2.38M learned head is not a small foundation trained from scratch.

## Supplementary Note 45. Task memory and contextual image-answer learning

The component in Note 44 separates physical source-pixel reproduction from selection of an appropriate answer. Two subsequent observations examine the conditioning path: a fixed-head task-memory substitution, followed by a matched comparison of frozen and adapted contextual features. The former uses the earlier head, whose normal development score is 9/64. It is not a third arm of the later comparison.

### Task-memory substitution with the earlier head

Let $H_\phi(V,M,G)$ denote the learned distribution over source spans, where $V$ contains contextual visual descriptors, $M$ contains task-frame memory, and $G$ contains physical geometry. For two questions $i, j$ about one passage, the intervention is

$$H_\phi(V_i,M_i,G_i) \longrightarrow H_\phi(V_i,M_j,G_i).$$

Weights, source pixels, unit boundaries and $V_i, G_i$ remain fixed. This constructs hybrid features rather than a native encoding of a new image. Moreover, $V_i$ can contain question information, so the substitution does not isolate one internal cause of the original errors.

| Cohort | Original correct | Swap: own | Swap: partner | Image changes |
|---|---|---|---|---|
| Seen questions | 32/32 | 5/32 | 26/32 | 27/32 |
| Development | 9/64 | 1/64 | 9/64 | 53/64 |

Paired reference-image sets are disjoint and both remain available in each owning source. Mean span-index total variation is 0.8423 on training and 0.8556 on development. The log odds of the owning versus partner reference move toward the partner in every one of 32 training and 64 development substitutions. Original owning references outrank their partners in 45/64 development cases, but only 9/64 selections beat all competing spans. Two-reference preference therefore cannot replace complete-answer scoring.

The diagnostic replays all 96 original selected images and boundaries and retains all 192 original/substituted images. A separate NumPy/PIL check reconstructs them from source columns and recomputes probabilities, paired counts and total variation. Cached CPU probabilities differ from the native GPU values by at most $4.293 \times 10^{-6}$; they are not bit-identical. The audit makes 192 head calls, no source-encoder calls and no optimizer updates. Its 39.644-second invocation includes 1.631 seconds setup and peaks at 0.551 GiB RSS, excluding the earlier 7.531B source encoding and the separate verification duration. It rejects exact insensitivity of this memory branch on these cases without repairing the reader.

### Matched contextual adaptation

The later comparison fixes the source-pixel motor, admitted data, draw schedule, initial head and final-checkpoint rule. Both arms optimize the image-level likelihood from Note 44, summing probability over all admissible source crops that equal a target under the declared exterior-white normalization. Control learns the 2,378,370 head parameters from authenticated frozen features. Adaptation also learns rank-eight query, value and output adapters in all 36 contextual layers:

$$W_\ell^{\text{eff}} = W_\ell + \frac{\alpha}{r} U_\ell D_\ell, \qquad r = 8, \quad \alpha = 16.$$

The up matrices initially vanish. The 6,193,152 additional parameters give 8,571,522 optimized parameters in adaptation. Both complete runtimes contain 7,531,277,171 parameters and external Qwen3-VL pretraining. Vision and base weights remain fixed. Contextual base storage is BF16, with FP32 contextual computation and learned parameters; the original BF16 feature boundary is preserved.

Each arm sees the same four shuffled passes over 3,193 admitted Chinese QA images and 64 training copies: 13,028 draws, batch four by sequential accumulation, and 3,257 updates. Eighteen QA records are excluded before model calls. Head/adapter peak learning rates are $10^{-4}/10^{-5}$, with 100 warm-up updates and cosine decay to one tenth of peak. AdamW uses $(0.9, 0.95)$, decay 0.01 and joint norm clipping at one. The intervention is this complete regimen, including the changed clipping parameter set. Development does not select a checkpoint.

| Fixed cohort | Control | Adapted |
|---|---|---|
| New Chinese/English copies | 32/32 | 32/32 |
| Seen Chinese questions | 28/32 | 27/32 |
| Unseen Chinese questions | 23/64 | 26/64 |
| Both questions correct per paragraph | 3/32 | 3/32 |
| Question erased | 1/64 | 0/64 |
| Passage erased | 0/64 | 0/64 |

Of the 64 matched normal questions, 20 are correct in both arms, six only after adaptation, three only in control, and 35 in neither. Both pass the 30/32 copy floor and 25-percentage-point question-erasure gain floor, but fail the 48/64 natural-answer and 20/32 paragraph-pair floors. Gains of three answers, zero pairs and zero copies miss the separate improvement rule of at least eight extra answers, four extra pairs and no more than two lost copies. These floors are decision rules, not significance tests. Normal unordered-column F1 is 61.622% and 67.444%; it does not measure semantic correctness or readability.

Question erasure preserves the correct ink but moves other ink in 11/64 inputs. Passage erasure removes every correct source crop, so its zero score is forced by this motor's support. Public development is repeatedly reused and is not a sealed test.

### Execution, costs and scope

Independent verification authenticates all schedules, four draws per training ID, finite final Adam moments at step 3,257, and both final checkpoints. Fifteen head tensors change in each arm. Control changes no contextual tensor; adaptation changes all 216 adapter tensors. No optimizer update is replayed. All 512 native arm/input calls retain original PNGs. Independent scoring reconstructs every output from declared source columns and reproduces the aggregate and paired decisions. Eight additional standalone calls reproduce pixels and events exactly without opening training manifests, a feature cache or the original head checkpoint.

Original setup takes 93.118 seconds. Feature preparation takes 3,387.907 seconds, with 2,937 new encodings and 320 authenticated reused payloads. Control/adapted optimization takes 126.665/30,856.570 seconds, at 15.265/18.075 GiB peak allocated and 15.504/22.082 GiB reserved memory. Control retains the full resident body during its cached-head optimization. For four standalone calls, setup plus inference is $77.695 + 5.022$ seconds for control and $76.854 + 5.056$ seconds for adaptation; both peak at 17.948 GiB allocated. These local costs exclude external pretraining.

Assessment is interrupted four times across five controller invocations. The prescribed calls accumulate as 218 control, then 38 control and 72 adapted, then 184 adapted. All 328 prior indexed outputs survive the final recovery unchanged. Complete assessment and additional reload duration are unmeasured. The retained 45.634-second control interval covers only 38 calls; 223.855 seconds covers only the last 184 adapted calls. A completed-control resume records almost no current-invocation work, not the cost of 256 calls. The public execution notes retain these timing boundaries. The termination sender remains unknown. All model stages use one RTX 4090 D.

The source-copy law assigns zero probability to answers absent from its admissible crops. Neither arm qualifies a general reader or independent free image-language generator, and this regimen does not qualify for scale-up. Historical writing and handwriting remain untested. CMRC2018 retains Cui et al.[12], CC-BY-SA-4.0 attribution; Qwen3-VL retains Alibaba, Apache-2.0 attribution.

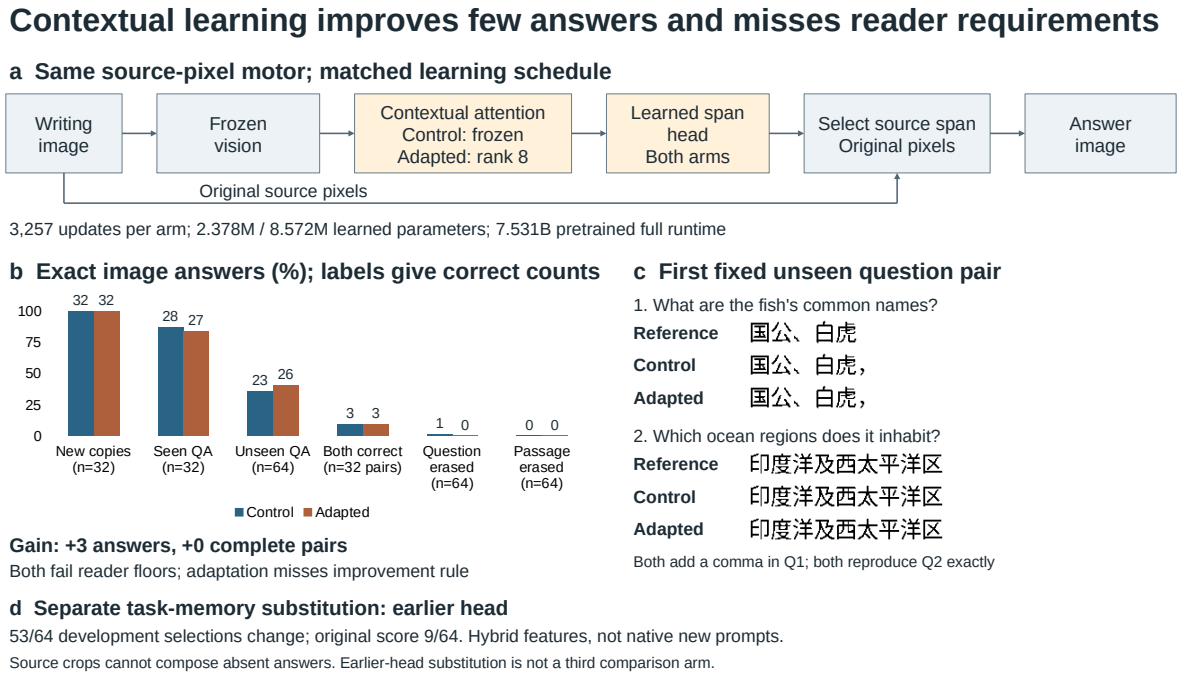



**Fig. S22 | Contextual adaptation misses the registered reader improvement. a**, Both arms retain the same externally pretrained source and original-pixel motor. Control learns the head; adaptation additionally learns contextual query, value and output adapters. Each receives 3,257 updates on the same schedule. **b**, Exact image-answer percentages with counts and denominators. New copies include 16 Chinese and 16 English requests; seen QA contains 32 training questions. Unseen QA contains 64 questions in 32 paragraph pairs. Both arms pass copy and question-erasure-gain floors but fail natural-answer and pair floors. The three-answer gain and unchanged three complete pairs miss the prospective improvement rule. Passage erasure removes the correct answer from source support and cannot isolate comprehension. **c**, The first fixed question pair, shown with complete unmodified reference and output PNGs. Both arms add a trailing comma to the common-name answer and reproduce the ocean-region reference. English labels describe the Chinese requests. **d**, A separate earlier-head intervention changes 53/64 development selections when paired-question task memory is substituted. Its original normal score is 9/64, distinct from the new 23/64 control. Hybrid features do not encode native new prompts or establish a repaired reader. The full runtime is 7.531B and pretrained; outputs are source crops.

## Supplementary Note 46. Matched image and text-prefix histories

The image writer and source-language helper use different physical history units. This audit joins their existing traces on all 32 shared training questions. It tests whether a comparison by writing position is supported by identical visible content. It performs no new contextual model forward, vocabulary projection, optimizer update or alignment treatment.

Let $Y$ be the complete canonical answer raster, padded to 64-column blocks, and let $P_j(Y)$ preserve its first $64j$ columns while setting the remaining canvas to white. Let $R_o(t_{:k})$ render a complete reference-text prefix with the full reference's original font, canvas and draw origin. The eligible prefixes at boundary $j$ are

$$\mathscr{K}_j = \{k : R_o(t_{:k}) = P_j(Y)\}.$$

A saved helper state can be compared only when its decoded preceding text equals an eligible complete prefix. The image writer consumes two learned image embeddings per block; the helper consumes native language pieces. Equal ordinal indices therefore do not identify equal histories. Independently centering each prefix would also move its vertical origin; the audit retains the complete reference's origin.

| Position | Image boundaries | Exact prefix raster | Saved matching state |
|---|---|---|---|
| Initial | 32 | 32 | 32 |
| Intermediate | 75 | 7 | 2 |
| Terminal | 32 | 32 | 19 |

At 68/75 intermediate boundaries there is no complete canonical text-prefix raster. Five further boundaries admit such a raster but have no matching retained helper state. All terminal rasters are representable; thirteen lack a corresponding saved state. Missing states and different helper answers remain in the denominator. All 216 eligible saved source-prefix positions decode to complete UTF-8, so split-byte prefixes do not explain the missing correspondences in this cohort.

All owning prompt hashes and BF16 reader-memory hashes agree. The 32 initial states agree after common BF16 rounding. These controls do not imply that later image-history states retain the helper's original language behavior. Only two intermediate correspondences can be evaluated from the retained traces:

| Question | Image blocks | Image positions | Helper positions | Distance |
|---|---|---|---|---|
| `TRAIN_1779_QUERY_1` | 3 | 6 | 6 | 1.497 |
| `TRAIN_2146_QUERY_1` | 1 | 2 | 6 | 1.407 |

Figure S23 retains both preceding prefix images. For states $h, g$ of width $d$, the reported distance is

$$\delta(h,g) = \sqrt{\frac{1}{d}\left\| \frac{h}{\sqrt{\mathrm{mean}(h^2)+10^{-6}}} - \frac{g}{\sqrt{\mathrm{mean}(g^2)+10^{-6}}} \right\|_2^2}.$$

The first correspondence has equal position counts; the second does not. Neither distance is calibrated as a language score or establishes a cause of the corrupted answers. Nineteen terminal matches have median distance 1.404, with the same interpretation limit.

The earlier state-alignment learner in Note 43 supervised Show-o2's pre-answer prompt states. It did not test alignment of these Qwen image-history transitions. The present result supplies neither a complete teacher-state dataset at the existing block boundaries nor an evaluated repair. A future treatment must define how teacher and actor histories correspond before comparing their states. Future-conditioned targets can serve as predictive supervision, but cannot be described as states of an identical preceding history or supplied to autonomous inference.

The audit reads saved states and tokenizer-string metadata only. It materializes no neural source weights. The string table is hashed; the complete shared source archive is not rehashed in this metadata-only operation. The invocation takes 13.270 seconds and about 1.000 GiB process RSS. An independent NumPy/PIL check reconstructs UTF-8 prefixes, rescans physical correspondences and recomputes BF16 rounding, distances and all totals. It verifies stored arrays rather than rerunning either contextual model or the font renderer. The 361 retained prefix/state files occupy 5.18 MB and are bound by the public receipt. The renderer is not a universal character-segmentation rule.

Execution uses `scripts/audit_history_prefix_alignment.py`, with its separate `-verify` stage. The protocol and producer precede the audit at `bfa8026`; the verified public result is `references/evidence/history_prefix_alignment_result.json`. The fixed task-memory diagnostic in Note 45 uses `scripts/audit_contextual_query_memory.py` and its independent verifier. The matched comparison uses `scripts/compare_contextual_pixel_learning.py`; standalone inference uses `scripts/infer_adapted_pixel_spans.py`; complete independent export uses `scripts/export_contextual_pixel_comparison.py`. Completed data, protocols, checkpoints and failed invocations remain preserved. These observations add no natural-language capability or free image generation.

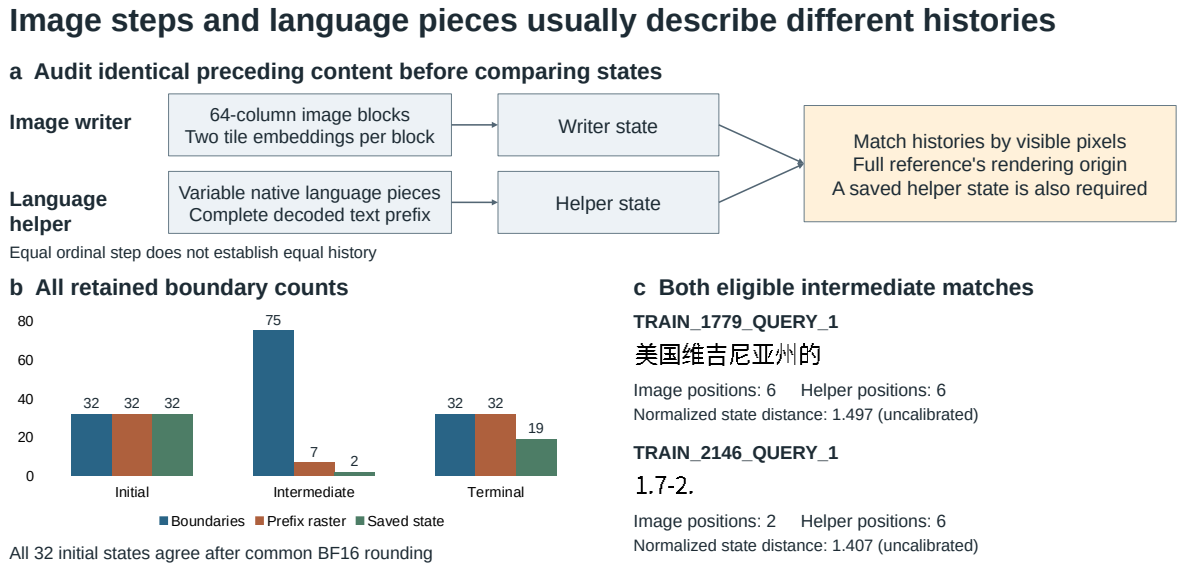



**Fig. S23 | Most retained image steps lack a matching complete text-prefix state. a**, The image writer consumes two continuous tile observations per 64-column block; the helper consumes a variable number of native language pieces. A state comparison requires identical preceding pixels under the full reference's rendering origin, rather than an equal ordinal step. **b**, Complete initial, intermediate and terminal counts. Seven of 75 intermediate boundaries have a canonical complete-prefix raster; only two have an eligible retained helper state. Sixty-eight have no such raster and five lack the corresponding state. All 32 initial states agree after common BF16 rounding; thirteen terminal boundaries have no retained matching state. **c**, Both eligible intermediate matches, with their complete canonical prefix rasters. The Chinese example has six image positions and six helper positions; the numeric example has two image positions and six helper positions. Their normalized distances are uncalibrated representation measurements. All 216 saved source-prefix positions in this cohort decode to complete UTF-8. The audit uses existing traces, makes no new neural calls and does not train an alignment or establish a reader/writer repair.

## Supplementary Note 47. Recognition supervision and autonomous writing loss

The paired-image writer in Note 40 uses the externally pretrained undistilled FLUX.2-klein-base-4B and its native VAE. A fixed generic condition is compiled offline; changing question content enters only through the image branch. The next experiment adds a frozen training-only reader, a pinned GOT-OCR conversion[22]. Its measured parameter count is 560,528,640; it is absent from native inference. Its vocabulary and teacher-forced labels are supervision tools, not the student's deployed output interface.

For normalized native target features $y$, image-reference features $x$ and independent $\varepsilon \sim \mathcal{N}(0,I)$, the assisted objective is

$$z_s = (1-s)y + s\varepsilon, \qquad s \sim U[0,1], \tag{24}$$

$$\widehat{y}_s = z_s - s v_\theta(z_s, x, s), \tag{25}$$

$$\begin{aligned}\mathscr{L}_{\text{assisted}} &= \operatorname{mean}\left[(v_\theta - (\varepsilon - y))^2\right] \\ &\quad + 0.05\,\mathbf{1}_{s\le 0.5}\,\mathrm{CE}\big(R(D_{\text{disp}}(\widehat{y}_s)), t + \mathrm{EOS}\big).\end{aligned} \tag{26}$$

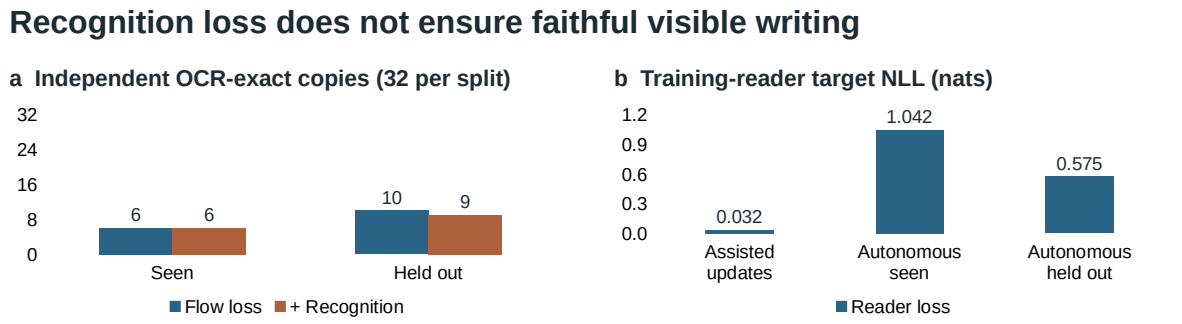
Here $D_{\text{disp}}$ uses native unpatchification, inverse batch normalization, frozen VAE decoding, RGB normalization and clipping. $R$ is the frozen reader, and $t$ is the training reference string. CE scores only answer positions, including EOS. Differentiable image preprocessing permits input gradients without updating reader weights.

Both arms start with fresh rank-16 adapters, use the same first 32 copying pairs for 512 updates, and retain batch one, seed 20268321, AdamW learning rate $10^{-4}$, decay .01 and clipping one. All 512 sampled IDs and noise times match the earlier flow-only control. A preflight reproduces its native flow term exactly before adding recognition. Complete independent CPU OCR scores are 6/32 seen and 9/32 held out, versus 6/32 and 10/32 for flow only (Fig. S24). Clean references score 27/32 and 28/32. Visible omissions and repetitions independently fail the 30/32 writing floor. One run per arm does not establish that all recognition supervision is ineffective. The treatment takes 374.748 s and 12.268 GiB peak allocated memory, versus 287.182 s and 8.437 GiB for control; acquisition and separate assessment are excluded.

Complete clean-reader calibration reads 62/64 targets exactly, Chinese 32/32 and English 30/32. Mean blank-minus-clean target NLL is 9.785251 nats. The first calibration fails preprocessing agreement. A recorded correction matches native Pillow's intermediate clipping with a continuous separable resize surrogate, preserving the original thresholds. Both calibration reports remain available. Frozen-reader pixel gradients and the student's recognition-to-velocity gradients are finite and nonzero.

The added recognition term is active on 255/512 updates and averages .031519 nats. A separate complete diagnostic scores actual autonomous decoder tensors without changing any writer weights. Mean target NLL is 1.041635 on seen outputs and .574959 on held-out outputs, against .033706 and .043822 on the clean references. Assisted updates have noisy correct answers and changing checkpoint states; free outputs use the final state and independent noise. These measurements do not isolate one causal factor.

The reader is not an independent judge. Its greedy transcripts are exact on 14/32 seen and 19/32 held-out outputs, exceeding the separate CPU counts without making the pages correct. In held-out case 026 it returns the expected string despite a visibly duplicated line; target NLL is .042427 versus .000192 for its clean reference. It similarly overlooks a repeated character in case 002. These blind spots motivate retaining direct physical image supervision rather than treating reader reward as writing accuracy. All 64 raw outputs, all transcripts and hashes of actual neural pixels are retained. No natural QA is optimized or assessed in this copying experiment.



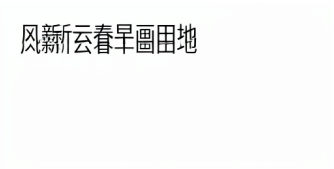

255 assisted updates and final-state free outputs are different regimes; this is not a causal decomposition.

**c Reader ignores a visible duplicated line**

**Reference**

风新云春早画田地

**Actual generated page**



Reader returns the expected string.
Gold NLL: 0.0424; clean reference: 0.000192.
Recognition does not count every visible repetition.

6/32 and 9/32 with the independent evaluator; 14/32 and 19/32 with the training reader.
The reader is not an independent judge. No reliable writer or semantic QA is established.

Same first 32 image pairs; 512 updates; source backbone and condition fixed. Complete 64-case outputs retained.

**Fig. S24 | Recognition supervision does not ensure correct visible writing. a**, The complete independent OCR observations after matched 512-update learning on the same 32 copying pairs. Both fail the writing requirement. **b**, Recognition loss on assisted training updates and on the final checkpoint's full autonomous output cohort. The conditions and checkpoint states differ, so the gap is not a controlled causal decomposition. Plot data are rounded to six decimals; displayed labels give three. **c**, Unmodified reference and actual generated page for held-out case 026. The training reader returns the expected string while the page duplicates its line. Its more permissive transcripts must not replace the independent evaluator or raw-image review.

## Supplementary Note 48. Direct supervision of autonomous answer images

We next optimize the final page produced by the native sampling procedure, following the differentiable-sampling principle of DRaFT[23]. This is an application of an established method, not a new backpropagation algorithm. Let $x$ be the native encoding of the input image and $c_0$ the same generic condition for every example. With independent initial noise, the deterministic native Euler trajectory is

$$z_{50} \sim \mathcal{N}(0, I), \tag{27}$$

$$z_{k-1} = z_k + (\sigma_{k-1} - \sigma_k) v_\theta(z_k, x, c_0, \sigma_k), \quad k = 50, \ldots, 1. \tag{28}$$

The implementation preserves the source scheduler's native BF16/FP32 arithmetic, image-position IDs, time schedule and output casts. No answer-reference image, target code, target/noise mixture or varying text condition enters this sampler. A mutable scheduler counter is kept outside checkpointed neural evaluations, avoiding repeated state advances during backward replay.

Let $d = D(z_0)$ be the native VAE output before display normalization, let $p = (d+1)/2$ be its unclipped RGB field, and let $q$ be the exact native clipped display pixels. For target page $y$, define $m_i = 1$ when its mean RGB is below .9. Both target classes are nonempty. The final-image objective is

$$E_i = \tfrac{1}{3} \sum_{c=1}^{3} |p_{ic} - y_{ic}|, \tag{29}$$

$$B(p, y) = \tfrac{1}{2} \left( \frac{\sum_i m_i E_i}{\sum_i m_i} + \frac{\sum_i (1 - m_i) E_i}{\sum_i (1 - m_i)} \right), \tag{30}$$

$$\mathscr{L}_{\text{final}} = \mathrm{CE}(R(q), t + \mathrm{EOS}) + 10 B(p, y). \tag{31}$$

Targets enter only these losses after generation. The unclipped physical branch retains a gradient when white output is saturated. The recognition and inference paths retain native clipping. The target page fixes a canonical writing style; this test does not establish style-invariant or historical script generation.

Before optimization, both complete 50-step trajectories and final images for the first Chinese and English seen examples reproduce native inference bit-exactly, including the retained starting checkpoint's outputs. The first preflight fails this check before any update: historical input features were encoded outside autocast, whereas native inference uses BF16 autocast. The maximum feature difference is .0625 for both checked inputs. Fresh native encoding restores equality without relaxing the criterion; this does not establish that the difference caused prior content errors.

Two real preflight updates verify finite nonzero gradients on predictions at steps 0, 24 and 49 and on the image-reference features. The first actual autograd/checkpointed forward also reproduces native pixels exactly. All 206 adapter tensors change while base, VAE and reader weights remain frozen and without parameter gradients. Peak allocation is 12.033 GiB. These updates are discarded before the pilot and do not establish writing quality.

The pilot starts again from the completed recognition-treatment checkpoint and runs 32 additional updates. CPU RNG seeds 20268402 and 20268403 determine example selection and independent noise seeds; every draw reproduces. Sampling with replacement visits 22 of the 32 previously seen examples. AdamW uses learning rate $10^{-5}$, decay .01 and global clipping one. Every update differentiates through all 50 steps. A fresh process then uses the original 64 input-only cases and seeds, 50 native steps, guidance one and a $512 \times 256$ output canvas. It loads no reader, target images, tokenizer or varying text condition.

| Independent OCR observation | Starting | After 32 updates |
|---|---|---|
| Seen, all strings | 6/32 | 15/32 |
| Held out, all strings | 9/32 | 13/32 |
| Seen Chinese / English | 1/16; 5/16 | 3/16; 12/16 |
| Held-out Chinese / English | 2/16; 7/16 | 3/16; 10/16 |

Fifteen outputs become OCR exact and two cease to be exact. OCR is not manual reading accuracy, but some corrections are visible: the first held-out English example restores the missing letter in `open`; seen case 015 removes an extra line. Failures also remain visible, including omitted characters in seen case 004, copied instructions in case 012 and held-out case 030, and a duplicated line in held-out case 026. Held-out case 028 adds a previously absent quotation mark. The complete original-order before/after galleries reject the 30/32 writing requirement in both splits. All final PNGs equal their quantized retained neural tensors; no output is repaired or selected from rerolls. This authenticates artifacts, not an independent neural replay.

The gain meets the prospective exploratory rule of at least three additional held-out OCR-exact strings with no aggregate seen regression (Fig. S25). It permits consideration of another controlled learning experiment, not writing admission or unchanged large scale-up. There is no matched continuation using 32 additional updates of the old

objective and no independent-seed replication. Extra optimization, corrected input precision, the recognition term and the physical term are not isolated. Improved font recognition can contribute to the measured gain.

Training takes 1,135.452 s at 12.057 GiB peak allocated memory. The separate 64-case native inference invocation takes 518.400 s at 8.031 GiB. These exclude acquisition, preceding training, preflight and separate CPU evaluation. The deployed transformer has 3,899,903,488 parameters, including 24,358,912 locally trained parameters, plus an 84,046,115-parameter VAE. The frozen reader has 560,528,640 training-only parameters. External pretraining is retained. No small from-scratch foundation or efficiency gain at matched language quality is established. Natural QA, etymology, general language understanding and broader writing forms remain unproved.

The complete source records are `references/evidence/recognition_guided_writing_result/result.json`, `references/evidence/autonomous_writing_loss_result/report.json`, `references/evidence/autonomous_writer_preflight_result/result.json` and `references/evidence/autonomous_writer_result/result.json`. Reproduction uses `scripts/train_recognition_guided_writer.py`, `scripts/probe_autonomous_writing_loss.py`, `scripts/train_autonomous_writer.py`, the unchanged native inference entry point `scripts/infer_paired_image_cohort.py`, and separate CPU evaluation `scripts/score_paired_writing.py`. Every actual output is authenticated by the corresponding export script and retained tensor hash.

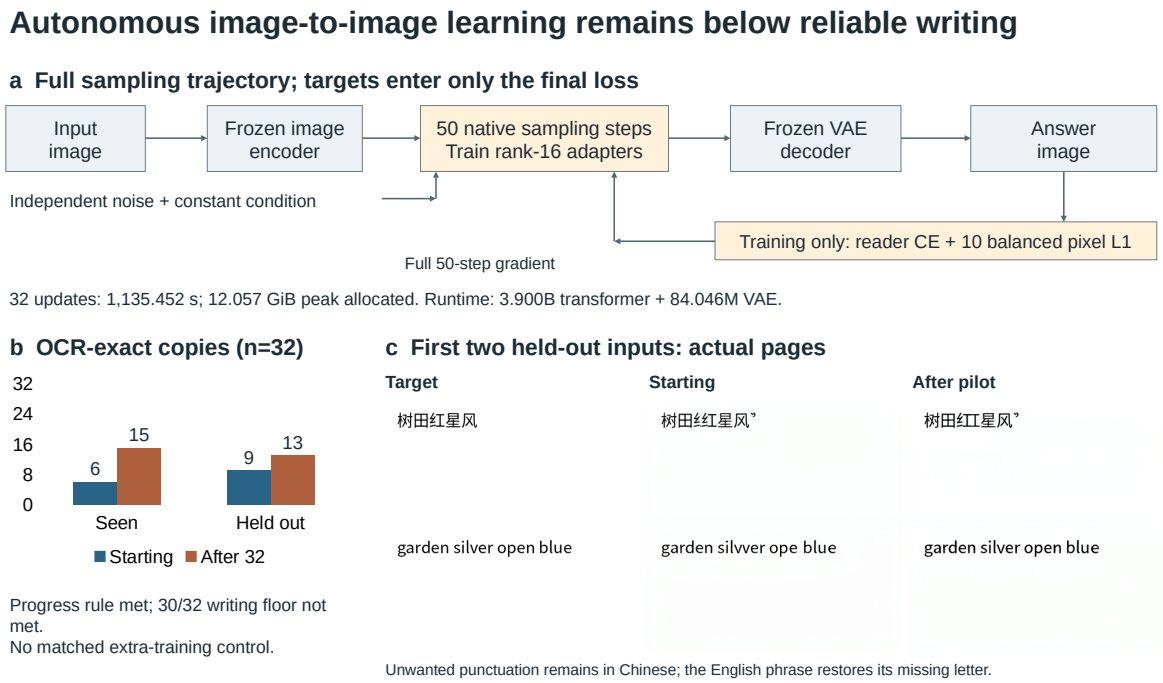



**Fig. S25** | **Direct final-image continuation improves some copies but remains below reliable writing. a**, The complete native noise-to-image sampling procedure. Input-image features and one fixed generic condition supply context; targets enter only the final training loss. Gradients traverse all 50 steps. **b**, Complete independent CPU OCR observations before and after the registered pilot. The exploratory progress rule passes, while the writing floor fails. This is one continuation without a matched extra-training control. **c**, The first two held-out cases, fixed before evaluation, with unmodified target, starting-state and final-state pages. Chinese still contains unwanted punctuation; English restores a missing letter. These are copying examples, not semantic QA. All 64 cases and their failures are retained.

## Supplementary Note 49. Additional-training control and balanced exposure

Statements about missing control in Note 48 and Fig. S25 describe the initial pilot assessment. The final-image pilot is next compared with additional optimization of the old assisted objective. Both continuations start from the same recognition checkpoint, use fresh AdamW with learning rate $10^{-5}$, decay .01 and gradient clipping one, and reproduce the same 32 example IDs and noise seeds. Native input encoding under BF16 autocast is held common. The control retains the old velocity-MSE objective and its recognition term, active only when $\sigma \leq .5$. It does not differentiate through final noise-to-image sampling. All 206 adapter tensors change; base, VAE and reader parameters stay frozen. The first Chinese and English starting trajectories reproduce native pixels.

The control gives 5/32 seen and 10/32 held-out OCR-exact strings, compared with 15/32 and 13/32 in the final-image pilot. Held-out Chinese remains 3/16 in both. On held-out examples, four become exact only in the pilot and one only in control; all five are English. The comparison meets its exploratory rule of at least three additional held-out exact strings without aggregate seen regression. That rule was specified after observing the pilot and before observing control. This is a retrospective comparison of objective bundles, not an independently replicated or compute-matched causal estimate. It does not isolate recognition, physical error or trajectory differentiation.

The control training invocation takes 56.962 s at 12.251 GiB peak allocation, including two initial native replays taking 16.192 s. Its optimization loop takes 33.790 s. Separate native inference takes 513.381 s at 8.031 GiB. An initial inference invocation was interrupted after 19 outputs when a legacy whole-repository identity check detected an unrelated source commit. All interrupted artifacts are preserved; the replacement invocation uses identical inputs and seeds and reproduces all 19 PNGs and retained tensors exactly. The complete 64-case cohort is assessed without quality selection.

We then continue the final-image pilot for 128 additional updates under the unchanged objective in Note 48. Unlike the pilot's sampling with replacement, the registered schedule makes four complete permutations of its 32 training inputs: four updates per input, 64 Chinese and 64 English draws. CPU seeds 20268502 and 20268503 fix order and noise seeds; model seed is 20268501. The pilot's weights and AdamW states are retained. Every optimizer state reaches step 160. No resume is used. All updates differentiate through all 50 native steps; independent noise, guidance one and the $512 \times 256$ output canvas are unchanged. No QA example is optimized.

| Independent OCR | Old loss +32 | Final image +32 | Then +128 |
|---|---|---|---|
| Seen, all | 5/32 | 15/32 | 13/32 |
| Held out, all | 10/32 | 13/32 | 17/32 |
| Seen Chinese | 0/16 | 3/16 | 4/16 |
| Held-out Chinese | 3/16 | 3/16 | 4/16 |
| Seen English | 5/16 | 12/16 | 9/16 |
| Held-out English | 7/16 | 10/16 | 13/16 |

The balanced continuation's prospective rule requires all three conditions: at least 6/16 held-out Chinese, 15/32 seen strings and 10/16 held-out English. The Chinese and seen-retention conditions fail; only English retention passes (Fig. S26). More exposure under this recipe is therefore not admitted by that rule. Seen case 002 becomes OCR exact, while 003, 025 and 027 cease to be exact. Held-out cases 007, 015, 022, 023 and 028 become exact, while 010 ceases to be exact. These are reused development inputs; font recognition can contribute to the observations. Clean-reference OCR remains 27/32 seen and 28/32 held out, rather than perfect accuracy.

Complete original-order target/pilot/final galleries are inspected. Seen case 004 still omits characters; 012 copies instructions and 025 repeats a letter in `small`. Held-out case 000 adds a quotation mark, 002 repeats a character, 026 duplicates its line and 030 copies instructions. More than two obvious failures in each split independently reject the unchanged 30/32 requirement for correct, legible requested strings alone. This is not a quantitative manual annotation of every glyph. No PNG is repaired or rerolled.

All 128 updates have finite nonzero gradients and change all 206 adapter tensors, with frozen parameters without gradients. Initial native replays and the first actual differentiable forward reproduce pilot pixels exactly. The first checked prediction-gradient norms at steps 0, 24 and 49 are .020289, .024817 and .145401. Export authenticates all 128 pre-update PNGs and all 64 final PNGs against retained neural tensors. This verifies artifact bindings; it is not a second independent neural replay. Training takes 4,500.117 s at 12.032 GiB peak allocation; separate native inference takes 512.959 s at 8.031 GiB. Acquisition, preceding training and separate CPU OCR are excluded. The inherited transformer, VAE, reader and trainable counts remain those in Note 48. No lightweight foundation training, semantic learning, historical writing or efficiency at matched language quality is established.

Complete assessments, protocols, raw outputs and source hashes are in `references/autonomous_writing_control_result.md` and `references/balanced_autonomous_writer_result.md`. Reproduction uses `scripts/train_autonomous_writing_control.py`, `scripts/train_balanced_autonomous_writer.py`, `scripts/infer_paired_image_cohort_v2.py` and separate CPU scoring/export. Native v2 checks inference sources and data dependencies while permitting unrelated Git commits. Each result retains the subsequent raw-review receipt.

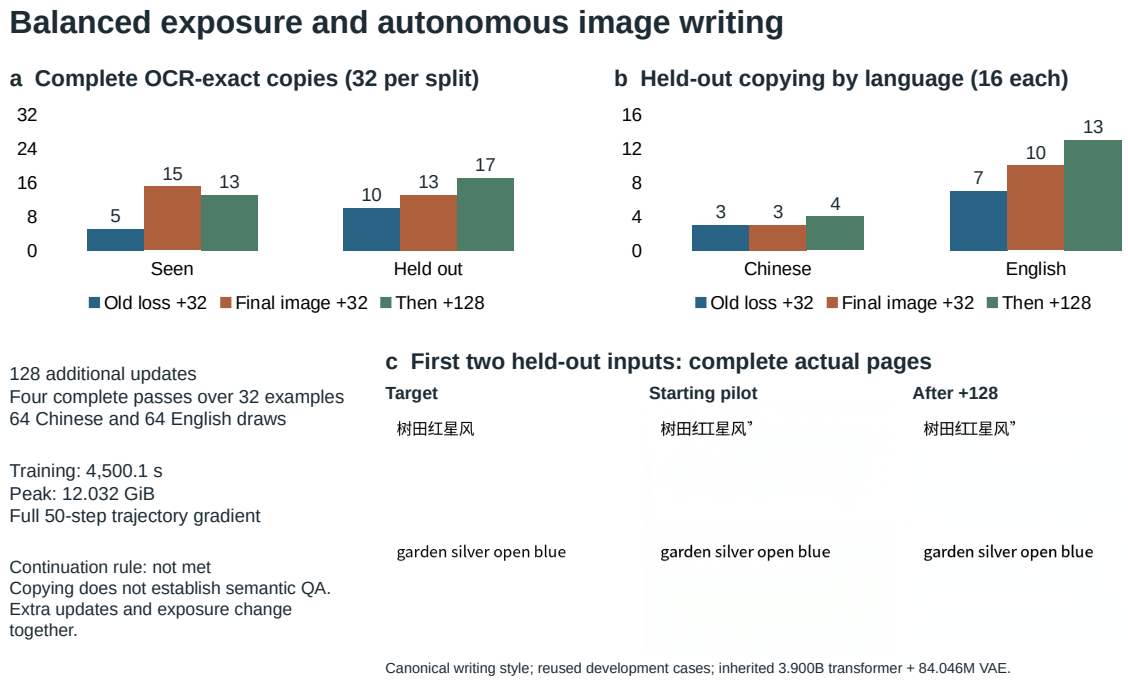



**Fig. S26 | Balanced exposure improves some held-out copies but fails its continuation requirements. a**, Complete independent OCR-exact counts on 32 seen and 32 held-out inputs: old-objective 32-update control, final-image 32-update pilot, and the pilot followed by 128 balanced updates. The first two match example draws and native input precision, but differ in objective and compute. **b**, Held-out counts by language, 16 each. Chinese reaches only 4/16; seen aggregate performance also falls. Two of the three prospective continuation requirements fail. **c**, The fixed first two held-out cases, with unchanged target, pilot and final pages. All 64 actual outputs, including failures outside these examples, are retained. These are copying observations with external pretraining and reused development inputs, without independent-seed replication or semantic QA.

### Reproduction

From the repository root, use the following entry points. GPU stages use physical GPU 0; preparation and independent pixel checks run on CPU.

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=.  python
scripts/train_causal_visual_block_diffusion_v53.py -device cuda:0

CUDA_VISIBLE_DEVICES=0 PYTHONPATH=.  python
scripts/eval_causal_visual_block_diffusion_v53.py -device cuda:0
```

Completed output directories are protected against accidental replacement. For the exact density, with the same GPU selection, use

```
scripts/train_exact_raster_density_v54.py
scripts/eval_exact_raster_density_v54.py
```

Independent image inference and export use

```
scripts/infer_exact_raster_density_v54.py.
```

For the matched continuation, run the following with `-arm frozen` and then `-arm joint`, completing both before the evaluator:

```
scripts/train_joint_exact_raster_v55.py
scripts/eval_joint_exact_raster_v55.py
```

The independent runtime uses `scripts/infer_joint_exact_raster_v55.py`. The form diagnostic and locked pixel decoder audit use

```
scripts/diagnose_raster_form_validity.py
scripts/eval_raster_beam_v56.py.
```

Independent pixel-beam inference uses `scripts/infer_raster_beam.py`. For grounded visual reading, complete both training arms before auditing:

```
scripts/train_visual_copy_language_v57.py -arm copy
scripts/train_visual_copy_language_v57.py -arm no_copy
scripts/eval_visual_copy_language_v57.py.
```

Native answers and self-contained export use

```
scripts/infer_visual_copy_language.py.
```

For complete image blocks, train both arms before the audit:

```
scripts/train_visual_answer_blocks_v58.py
scripts/eval_visual_answer_blocks_v58.py.
```

Native inference uses

```
scripts/infer_visual_answer_blocks.py.
```

Frozen-feature preparation, matched readout training and audit use the `cache_frozen_visual_features_v59`, `train_frozen_visual_readout_v59` and `eval_frozen_visual_readout_v59` scripts. The matched input-order experiment runs both orders before auditing:

```
scripts/train_question_first_reading_v60.py
-order context
-order question
scripts/eval_question_first_reading_v60.py.
```

Native pixel preparation, full adaptation and locked audit use the existing Python 3.12 MIXAR runtime, with `CUDA_VISIBLE_DEVICES=0`:

```
scripts/prepare_native_pixel_qa_v61.py
scripts/train_native_pixel_instruction_v61.py
scripts/verify_native_pixel_export_v61.py
scripts/eval_native_pixel_instruction_v61.py.
```

Independent image inference uses scripts/infer_native_pixel_language.py. The matched whole-image experiment uses the persistent sequential coordinator

scripts/run_correlated_native_v62.py.

It verifies both complete training reports, standalone offline replays and the locked audit in that order. Independent image inference uses scripts/infer_correlated_native_language.py; the exported bundle's run.py does not need the repository or the original foundation checkpoint. The optional, separately calibrated vision diagnostic is scripts/audit_correlated_native_transcripts_v62.py. That reader stays outside the actor and changes no primary gate. The complete assignment comparison uses scripts/run_question_assignment_v63.py; its independent raw-score verification is scripts/verify_question_assignment_result_v63.py. The separate successor preflight is scripts/check_grounded_image_actor.py, and its helper-only source check is scripts/check_inherited_context.py. The writing-only curriculum is specified in references/grounded_writer_learning_protocol.md; it is not an additional completed language result.

The matched writer diagnosis and broader direct stage use scripts/diagnose_grounded_writer_v65.py and scripts/train_direct_grounded_writer_v66.py; their independent verifiers retain the failed fidelity decisions. The separate output-codec audit is scripts/audit_codec_output_v67.py, with native replay in scripts/verify_codec_output_v67.py. Complete separate-space actor execution is checked by scripts/check_codec_visual_language_actor.py. The fixed instruction-learning protocol is references/codec_language_learning_v68_protocol.md; its training entry point is scripts/train_codec_language_v68.py. Prepared prompt images and blank controls use scripts/prepare_codec_language_audit_v68.py. The evaluator scripts/eval_codec_language_v68.py accepts only the final 10,000-update state and has a separate CPU data/metric preflight. Its independent raw-stream verifier is scripts/verify_codec_language_v68.py. These entry points produce the completed, failed autonomous result in Supplementary Note 27. The source-bound integrity and output-verification receipts are retained in references/evidence/.

The frozen-actor diagnostic is scripts/diagnose_codec_language_v68.py. In the compatible native PyTorch environment, use -prepare-only with CUDA hidden, then run the same command on physical GPU 0 with the deterministic cuBLAS configuration, and finally use -verify-only with CUDA hidden. Supply the same fresh -out directory for those phases; completed measurements are never overwritten. The source-frozen selection, assisted-history scope and measurement checks are defined in references/codec_language_v68_diagnosis_protocol.md.

The completed native-context run in Note 37 uses the compatible existing Python 3.12 environment and authenticated local image, code and weight caches. GPU phases require CUDA_VISIBLE_DEVICES=0 and CUBLAS_WORKSPACE_CONFIG=:4096:8. Its training and complete audit entry points are scripts/train_native_chinese_v80.py and scripts/audit_native_chinese_v80.py; the latter's -verify phase runs with CUDA hidden. The fixed training-cohort controls use scripts/probe_native_corpus_fit_v80.py, with separate -stage prepare, -stage evaluate and -stage verify invocations, and the same explicit source run and fresh output directory. Preparation and verification run with CUDA hidden; evaluation uses the GPU environment above. Exact arguments, source identities and retained failure handling are specified in references/native_chinese_v80_protocol.md, references/native_corpus_fit_v80_protocol.md and docs/native-image-continuation.md.

The matched readout experiment in Note 38 uses scripts/train_adaptive_readout_v81.py with separate preflight and training stages. Preserve its failed original evaluation and partial output. The completed checkpoints are evaluated without optimization using scripts/evaluate_trained_readout_v81.py, first with -stage evaluate in the deterministic GPU0 environment above, then -stage verify with CUDA explicitly hidden. Export all verified images and scores with scripts/export_readout_v81_result.py. The source-bound execution and numerical criteria are in references/adaptive_readout_v81_protocol.md, references/adaptive_readout_v81r2_protocol.md and references/adaptive_readout_v81_evaluation_protocol.md.

The matched decoder-sensitive comparison in Note 39 uses scripts/train_decoder_metric_v82.py with separate preflight, training and CPU verification stages, under references/decoder_metric_v82_protocol.md and references/decoder_metric_v82r2_protocol.md. Preserve the first failed execution and interruption records. Export the verified complete cohort with scripts/export_decoder_metric_v82_result.py. Independently reproduce the spatial failure diagnosis from the public PNGs using scripts/diagnose_writing_failures.py. Standalone execution uses scripts/infer_decoder_metric_v82.py; the fixed one-case restoration check is scripts/check_metric_runtime_replay_v82.py.

The paired-image experiment in Note 40 uses scripts/prepare_paired_image_condition.py for the standalone offline constant, scripts/train_paired_image_language.py for preparation, caching, preflight, optimization and single-image inference, and scripts/infer_paired_image_cohort.py for a fresh input-only cohort. The fixed preprocessing is scripts/trim_paired_input_padding.py; separate CPU recognition is scripts/score_paired_writing.py. Authenticate the complete export with scripts/export_paired_padding_result.py. Protocols and full command examples are in references/paired_image_learning_protocol.md,

references/paired_padding_protocol.md, references/paired_padding_full_cohort_protocol.md and docs/paired-image-learning.md.

The native source and visual-access observation in Note 41 use scripts/acquire_native_unified_source.py, scripts/probe_native_unified_source.py and scripts/train_native_visual_access.py. Complete held-out preparation and matched question interventions use scripts/prepare_native_visual_access_validation.py and scripts/probe_native_visual_access_controls.py. CPU export uses scripts/score_native_visual_access.py. Fixed protocols, the isolated runtime, exact commands and the remaining image-writing requirement are in docs/native-visual-access.md and references/native_visual_access_assessment_protocol.md.

The full-page curriculum in Note 42 uses scripts/prepare_native_visual_alignment.py, scripts/train_native_visual_alignment.py and scripts/score_native_visual_alignment.py. Prospective English pages and source readability checks use scripts/prepare_native_english_transfer.py and scripts/score_native_english_transfer.py. The geometry intervention uses scripts/check_native_reader_geometry.py, scripts/probe_native_unadapted_reader.py and scripts/score_native_reader_geometry.py. All inference indexes exclude answer strings. Exact commands, fresh output paths and sequential GPU stages are specified in docs/native-visual-alignment.md, docs/native-english-transfer.md and references/native_reader_geometry_result.md.

## References


1. Tai, Y., Liao, X., Suglia, A. & Vergari, A. PIXAR: Auto-Regressive Language Modeling in Pixel Space. *arXiv* 2401.03321 (2024). https://arxiv.org/abs/2401.03321.
2. Li, T., Tian, Y., Li, H., Deng, M. & He, K. Autoregressive Image Generation without Vector Quantization. *Advances in Neural Information Processing Systems* (2024). https://arxiv.org/abs/2406.11838.
3. Chen, B. et al. Diffusion Forcing: Next-token Prediction Meets Full-Sequence Diffusion. *arXiv* 2407.01392 (2024). https://arxiv.org/abs/2407.01392.
4. Dhariwal, P. & Nichol, A. Diffusion Models Beat GANs on Image Synthesis. *Advances in Neural Information Processing Systems* (2021). https://arxiv.org/abs/2105.05233.
5. Yu, S. et al. Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think. *International Conference on Learning Representations* (2025). https://arxiv.org/abs/2410.06940.
6. van den Oord, A., Kalchbrenner, N. & Kavukcuoglu, K. Pixel Recurrent Neural Networks. *Proceedings of the 33rd International Conference on Machine Learning* **48**, 1747–1756 (2016). https://proceedings.mlr.press/v48/oord16.html.
7. van den Oord, A. et al. Conditional Image Generation with PixelCNN Decoders. *arXiv* 1606.05328 (2016). https://arxiv.org/abs/1606.05328.
8. Germain, M., Gregor, K., Murray, I. & Larochelle, H. MADE: Masked Autoencoder for Distribution Estimation. *Proceedings of the 32nd International Conference on Machine Learning* **37**, 881–889 (2015). https://arxiv.org/abs/1502.03509.
9. Hu, C., Tai, Y., Vergari, A., Keller, F. & Suglia, A. MIXAR: Scaling Autoregressive Pixel-based Language Models to Multiple Languages and Scripts. *arXiv* 2604.11575v2 (2026). https://arxiv.org/abs/2604.11575v2.
10. Holtzman, A., Buys, J., Du, L., Forbes, M. & Choi, Y. The Curious Case of Neural Text Degeneration. *International Conference on Learning Representations* (2020). https://arxiv.org/abs/1904.09751.
11. See, A., Liu, P. J. & Manning, C. D. Get To The Point: Summarization with Pointer-Generator Networks. *Proceedings of ACL*, 1073–1083 (2017). https://aclanthology.org/P17-1099/.
12. Cui, Y. et al. A Span-Extraction Dataset for Chinese Machine Reading Comprehension. *Proceedings of EMNLP-IJCNLP*, 5883–5889 (2019). https://aclanthology.org/D19-1600/.
13. Zhang, C. et al. Unifying Distributional Training for One-Step Visual Generation. *arXiv* 2609.35763v4 (2026). https://arxiv.org/abs/2609.35763v4.
14. Van Assel, H., De Brouwer, E., Saremi, S., Scalia, G. & Regev, A. Generate in Reconstruction Space, Match in Semantic Space: Transport Geometry for One-Step Generation. *arXiv* 2606.00514v2 (2026). https://arxiv.org/abs/2606.00514v2.
15. Tao, H. et al. Multimodal Flow: Unified Flow Modeling of Language and Vision in Embedding Spaces. *arXiv* 2609.40362v1 (2026). https://arxiv.org/abs/2609.40362v1.
16. Huang, L. & Wu, X. OCG: Optical Character Grounded Visual Reasoning with Flow Matching. Author manuscript (2026). Peer-review status unverified.
17. Bai, S. et al. Qwen3-VL Technical Report. *arXiv* 2511.21631v2 (2025). https://arxiv.org/abs/2511.21631v2.
18. Databricks. Databricks Dolly 15k instruction dataset (2023). https://huggingface.co/datasets/databricks/databricks-dolly-15k.
19. Li, T. & He, K. Back to Basics: Let Denoising Generative Models Denoise. *arXiv* 2511.13720v1 (2025). https://arxiv.org/abs/2511.13720v1.
20. Xie, J., Yang, Z. & Shou, M. Z. Show-o2: Improved Native Unified Multimodal Models. *arXiv* 2506.15564v3 (2025). https://arxiv.org/abs/2506.15564v3.
21. Rajpurkar, P., Zhang, J., Lopyrev, K. & Liang, P. SQuAD: 100,000+ Questions for Machine Comprehension of Text. *Proceedings of EMNLP*, 2383–2392 (2016). https://aclanthology.org/D16-1264/.

22. Wei, H. et al. General OCR Theory: Towards OCR-2.0 via a Unified End-to-end Model. *arXiv* 2409.01704 (2024). https://arxiv.org/abs/2409.01704.
23. Clark, K., Vicol, P., Swersky, K. & Fleet, D. J. Directly Fine-Tuning Diffusion Models on Differentiable Rewards. *International Conference on Learning Representations* (2024). https://arxiv.org/abs/2309.17400.